Math-Booklet

Maths Department

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Welcome to MYP 5 Mathematics

Welcome to your comprehensive revision hub. This isn't just a textbook; it's a **living document** designed to evolve with you as we move toward your final MYP 5 assessments.

> A Growing Resource
>
> This booklet is updated **ongoingly**. New chapters and advanced units will appear as we cover them in class, ensuring your revision material is always synchronized with your learning.

How to use this Booklet

Mastery Tracking

Every unit starts with clear **Learning Targets**. Use these as your personal checklist. If you can't tick it off, you aren't ready for the test!

Dynamic Practice

Our **"Test Your Knowledge"** sections feature a "Try Again" button that generates new numbers for the same problem. Practice the *method*, not the memory.

Formula Cards

No more hunting through notebooks. Quick-access digital cards provide the identities and rules you need at a glance.

Instant Feedback

Every exercise includes a hidden solution. Toggle it only after you've put pen to paper.

The Journey: Tiered Practice

We use a color-coded system to help you scale your skills from foundational to exam-ready:

Level	Focus	Goal
□	**Easy**	Building fluency and basic recall.
□	**Medium**	Multi-step processing and application.
□	**Challenging**	Exam-style logic and complex problem solving.

Get AI Help

Stuck on a problem? Each unit's worksheet includes a **Get Help** button that connects you to an AI tutor trained on your exact exercises. It won't give you the answer — it asks the right questions to guide you to it.

> Pro Tip for Success
>
> Don't just read the solutions. Mathematics is a "doing" subject. Use the **Interactive Testing** feature until you can solve the □ Challenging problems without looking at the hints!

Part I

Algebra

1 Unit 1: Operations

1.1 Learning Targets

LT1

Calculate with Integers

Add, subtract, multiply, and divide positive and negative numbers accurately.

Positive & negative Four operations Number line

LT2

Apply Order of Operations

Evaluate numerical expressions using BODMAS, including brackets and exponents.

BODMAS Brackets Exponents

LT3

Operate with Fractions

Add, subtract, multiply, and divide fractions and mixed numbers fluently.

LCD Mixed numbers Simplify

LT4

Work with Decimals

Perform calculations with decimals and convert between fractions, decimals, and percentages.

Decimal operations FDP conversion Rounding

LT5

Simplify Complex Calculations

Combine integers, fractions, and decimals in multi-step numerical problems.

Multi-step Mixed types Problem solving

1.2 Recall: Formula Card

Essential Operation Rules

Simplifying Fractions

 Pro Tip A fraction is fully simplified when GCF = 1

How to Simplify

1. Find the Greatest Common Factor (GCF) of numerator and denominator
2. Divide both top and bottom by the GCF

$$\frac{a}{b} = \frac{a \div \text{GCF}}{b \div \text{GCF}}$$

Examples: - $\frac{12}{18} = \frac{12 \div 6}{18 \div 6} = \frac{2}{3}$ - $\frac{15}{25} = \frac{15 \div 5}{25 \div 5} = \frac{3}{5}$

Rules of Signs (Multiplication & Division)

 Memory Trick Same signs = Happy (+)
Different signs = Sad (-)

Same Signs → Positive - $(+) \times (+) = (+)$ - $(-) \times (-) = (+)$

Different Signs → Negative - $(+) \times (-) = (-)$ - $(-) \times (+) = (-)$

Adding/Subtracting Fractions

⚠ **Common Mistake** Don't add denominators!
WRONG: $\frac{1}{2} + \frac{1}{3} = \frac{2}{5}$

Same Denominator: Keep denominator, add/subtract numerators

$$\frac{a}{c} + \frac{b}{c} = \frac{a+b}{c}$$

Multiplying & Dividing Fractions

 Quick Method Cancel BEFORE you multiply - saves time!

Multiply: $\frac{a}{b} \times \frac{c}{d} = \frac{a \times c}{b \times d}$

Divide (Keep-Change-Flip): $\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c}$

Order of Operations (PEMDAS/BODMAS)

1. **P**arentheses / **B**rackets
2. **E**xponents / **O**rders
3. **M/D** Multiplication & Division (Left to Right)
4. **A/S** Addition & Subtraction (Left to Right)

1.3 Practice Problems

1.3.1 Easy

Guidance

1. Read each question carefully before you begin answering it.
2. Check your answers seem right.
3. Always show your workings.

Exercise 1.1 (Conversion Table)**.** Complete the following table:

Fraction	Decimal	Percentage
$\frac{1}{2}$	...	...
...	0.75	75%

 View Solution

- **Row 1:** $\frac{1}{2} = 0.5 =$ **50**$\%$
- **Row 2:** $\frac{3}{4} = 0.75 = 75\%$

Topic: Conversion Table Skills: Basic Conversions

∑ × π ÷ √ ◆ ∞ ± ∫ ≠ Δ

Exercise 1.2 (Basic Conversions)**.**

a) Write 20% as a decimal.

b) Write 9% as a fraction.

View Solution

a) $20 \div 100 = \mathbf{0.2}$

b) $\mathbf{\frac{9}{100}}$

Topic: Basic Conversions Skills: Percentage-to-Decimal/Fraction

∑ × π ÷ √ ◆ ∞ ± ∫ ≠ Δ

Level 1: Basic Operations

Calculate

a	b	c	d
$\frac{3}{4}$ of 36	$\frac{1}{3}$ of 24	$\frac{2}{5}$ of 35	$\frac{4}{9}$ of 72

Solutions

a) $(36 \div 4) \times 3 = 27$

b) $(24 \div 3) \times 1 = 8$

c) $(35 \div 5) \times 2 = 14$

d) $(72 \div 9) \times 4 = 32$

Topic: Fractions of Amounts Skills: Multiplicative Reasoning

∑ × π ÷ √ ◆ ∞ ± ∫ ≠ Δ

Level 2: Applied Context

Calculate:
a) Zaid earns 800 QAR a month. He spends $\frac{1}{4}$ on rent and $\frac{2}{5}$ on food. How much money does he have left?
b) Fatima is buying shoes that normally cost 200 QAR. They are $\frac{1}{5}$ off. What is the sale price?
c) Amir has 300 marbles. He gives $\frac{1}{2}$ to his brother and $\frac{1}{3}$ of the remaining marbles to his friend. How many does he keep?

Solutions

a) Rent: $(800 \div 4) \times 1 = 200$
Food: $(800 \div 5) \times 2 = 320$
Total spent: $200 + 320 = 520$
Left: $800 - 520 = 280$ QAR

b) Discount: $(200 \div 5) \times 1 = 40$
Sale Price: $200 - 40 = 160$ QAR

c) To brother: $300 \times \frac{1}{2} = 150$
Remaining: 150
To friend: $150 \times \frac{1}{3} = 50$
Kept: $150 - 50 = 100$

Topic: Fractions of Amounts
Skills: Word Problems

1.3.2 Medium

Exercise 1.3 (Time and Fractions)**.** A train is late. It should arrive at 5 pm but arrives at 5:15 pm.

a) How many minutes late is the train?

b) Write your answer as a fraction of an hour.

View Solution

a) **15 minutes** b) $\frac{15}{60} = \mathbf{\frac{1}{4}}$ of an hour

Topic: Time and Fractions Skills: Fraction Simplification

∑ × π ÷ √ ◆ ∞ ± ∫ ≠ Δ

Exercise 1.4 (Savings and Values)**.** For every £200 Mrs. Joy earns, she saves £34.

a) Work out £34 as a percentage of £200.

b) Last month she earned £1000. How much does she save?

View Solution

a) $\frac{34}{200} \times 100 =$ **17**$\%$

b) 17% of $£1000 = 0.17 \times 1000 = £$**170**

Topic: Savings Skills: Values as Percentages

∑ × π ÷ √ ◆ ∞ ± ∫ ≠ Δ

1.3.3 Challenging

Multi-Step Reasoning

a) Maryam works 8 hours a week and earns 50 QAR per hour. She decides to save $\frac{2}{5}$ of her total weekly earnings. How many weeks will it take her to save 640 QAR?

b) There are 40 students in a class. $\frac{3}{5}$ are girls. $\frac{1}{4}$ of the girls wear glasses. How many girls in the class wear glasses?

Solutions

a) Weekly Pay: $8 \times 50 = 400$
Weekly Savings: $(400 \div 5) \times 2 = 160$
Weeks needed: $640 \div 160 = 4$ weeks

b) Total Girls: $(40 \div 5) \times 3 = 24$
Girls with glasses: $(24 \div 4) \times 1 = 6$

Topic: Fractions of Amounts
Skills: Logic & Savings

1.4 Test Your Knowledge

2 Unit 2: Indices

2.1 Learning Targets

LT1

Apply Multiplication Rules

Simplify expressions by adding powers when multiplying terms with the same base.

Same base Add exponents Index notation

LT2

Apply Division Rules

Simplify expressions by subtracting powers when dividing terms with the same base.

Same base Subtract exponents Fraction form

LT3

Master Power of a Power

Simplify expressions where a power is raised to another power by multiplying the exponents.

Brackets Multiply exponents Coefficients

LT4

Manage Coefficients

Multiply or divide numerical coefficients separately from the algebraic index laws.

Separate coefficients Multiply & divide Combined terms

LT5

Handle Negative Indices

Correctly work with exponents when calculations produce or involve negative indices.

Negative exponents Reciprocal Sign rules

LT6

Simplify Complex Fractions

Combine multiple index laws to simplify multi-step algebraic fractions involving multiplication and division.

Multi-step Mixed laws Algebraic fractions

LT7

Operate with Surds

Add, subtract, multiply, and divide surd expressions, including those with coefficients.

Add & subtract Multiply & divide Like surds

LT8

Simplify Surds

Express surds in simplest form by identifying and extracting perfect square factors.

Perfect squares Simplest form Prime factors

LT9

Apply Surds

Rationalise denominators and solve problems that require exact surd answers.

Rationalise Exact values Problem solving

2.2 Formula Card

Algebraic Indices & Surds

Laws of Indices — The Core Rules

 Pro Tip These rules ONLY apply when the **base** (the big letter) is the same.

Multiplication & Division

1. **Multiplication**: Add the powers $\rightarrow$

$$x^a \times x^b = x^{a+b}$$

2. **Division**: Subtract the powers $\rightarrow$

$$\frac{x^a}{x^b} = x^{a-b}$$

Examples:

$$y^5 \times y^3 = y^{5+3} = y^8$$
$$10w^7 \div 2w^3 = (10 \div 2)\, w^{7-3} = 5w^4$$

Power of a Power & Zero

 Memory Trick Brackets mean multiply!
Anything to the power of 0 is a hero (it becomes 1).

The Bracket Rule:

$$(x^a)^b = x^{a \times b} = x^{ab}$$

Examples:

$$(k^3)^4 = k^{3 \times 4} = k^{12}$$
$$(5x^2)^3 = 5^3 \times x^{2 \times 3} = 125x^6$$

The Zero Rule:

$$x^0 = 1$$

Negative & Fractional Indices

⚠ **Common Mistake** A negative power does NOT make the number negative; it makes it a reciprocal (fraction)!

Negative Powers:

$$x^{-a} = \frac{1}{x^a}$$

Fractional Powers:

$$x^{\frac{1}{n}} = \sqrt[n]{x} \qquad x^{\frac{m}{n}} = \left(\sqrt[n]{x}\right)^m$$

Examples:

$$4^{-2} = \frac{1}{4^2} = \frac{1}{16} \qquad 9^{\frac{1}{2}} = \sqrt{9} = 3 \qquad 8^{\frac{2}{3}} = \left(\sqrt[3]{8}\right)^2 = 2^2 = 4$$

Expanding Brackets with Indices

 Quick Method Multiply the “Big” numbers (coefficients), add the “Small” numbers (indices).

Single Brackets:

$$a(b+c) = ab + ac$$

Examples:

$$3x^2(2x^3+4) = (3\times 2)x^{2+3} + (3\times 4)x^2 = 6x^5 + 12x^2$$

$$x^5(x^2 - x^{-3}) = x^{5+2} - x^{5-3} = x^7 - x^2$$

Algebraic Fractions with Indices

Simplifying Fractions:

1. Simplify the numerator (top) first using multiplication laws.
2. Simplify the denominator (bottom).
3. Use the division law (subtract) to find the final result.

Example:

$$\frac{a^5 \times a^4}{a^2} = \frac{a^9}{a^2} = a^{9-2} = a^7$$

Example with Coefficients:

$$\frac{(2x^3)^2}{4x^4} = \frac{4x^6}{4x^4} = x^{6-4} = x^2$$

Surds — What is a Surd?

 Definition A **surd** is a root that cannot be written as an exact fraction — it is irrational.
$\sqrt{2},\ \sqrt{3},\ \sqrt{5}$ are surds. $\sqrt{4} = 2$ is **not** a surd.
Key fact: $(\sqrt{a})^2 = a$ always.

Operating with Surds

 Key Idea You can only add or subtract surds with the **same radicand**. Think of $\sqrt{3}$ as a unit — just like $3x + 5x = 8x$, we get $3\sqrt{3} + 5\sqrt{3} = 8\sqrt{3}$.

Addition & Subtraction — collect like surds only:

$$a\sqrt{n} + b\sqrt{n} = (a+b)\sqrt{n} \qquad a\sqrt{n} - b\sqrt{n} = (a-b)\sqrt{n}$$

Unlike surds **cannot** be combined: $3\sqrt{2} + 5\sqrt{3}$ stays as is.

Multiplication & Division:

$$\sqrt{a} \times \sqrt{b} = \sqrt{ab} \qquad \frac{\sqrt{a}}{\sqrt{b}} = \sqrt{\frac{a}{b}}$$

$$a\sqrt{m} \times b\sqrt{n} = ab\sqrt{mn}$$

Examples:

$$5\sqrt{7} - 10\sqrt{7} + 2\sqrt{3} = -5\sqrt{7} + 2\sqrt{3}$$
$$2\sqrt{10} \times 7\sqrt{6} = 14\sqrt{60} = 14\sqrt{4 \times 15} = 28\sqrt{15}$$

Simplifying Surds

⚠ **Always** extract the **largest** perfect square factor.
$\sqrt{72} = \sqrt{36 \times 2} = 6\sqrt{2}$ □ not $2\sqrt{18}$ ✗

Method — find largest perfect square k^2 that divides n:

$$\sqrt{n} = \sqrt{k^2 \times r} = k\sqrt{r}$$

Steps: find square factor → split → evaluate $\sqrt{k^2} = k$

Examples:

$$\sqrt{200} = \sqrt{100 \times 2} = 10\sqrt{2}$$
$$3\sqrt{48} = 3 \times \sqrt{16 \times 3} = 3 \times 4\sqrt{3} = 12\sqrt{3}$$

Perfect Squares: $4, 9, 16, 25, 36, 49, 64, 81, 100, 121, 144, 169, 196, 225$

Expanding & Rationalising

□ **Expanding** works exactly like algebra — distribute every term, use FOIL for double brackets, and remember $(\sqrt{a})^2 = a$.

Single Bracket:

$$\sqrt{a}(\sqrt{b} + c) = \sqrt{ab} + c\sqrt{a}$$

Double Bracket (FOIL):

$$(\sqrt{a} + b)(\sqrt{a} + c) = a + (b + c)\sqrt{a} + bc$$

Perfect Square:

$$(\sqrt{a} + b)^2 = a + 2b\sqrt{a} + b^2$$

Conjugate Pair — always gives a rational result:

$$(\sqrt{a} + \sqrt{b})(\sqrt{a} - \sqrt{b}) = a - b$$

Examples:

$$\sqrt{3}(\sqrt{3} + 2) = 3 + 2\sqrt{3}$$
$$(\sqrt{5} + 1)(\sqrt{5} - 1) = 5 - 1 = 4$$

$$(\sqrt{2}+3)^2 = 2 + 6\sqrt{2} + 9 = 11 + 6\sqrt{2}$$

Rationalising the Denominator — remove the surd from the bottom.

Simple: multiply top and bottom by $\sqrt{b}$:

$$\frac{a}{\sqrt{b}} \times \frac{\sqrt{b}}{\sqrt{b}} = \frac{a\sqrt{b}}{b}$$

Binomial: multiply by the **conjugate** of the denominator:

$$\frac{a}{\sqrt{b}+c} \times \frac{\sqrt{b}-c}{\sqrt{b}-c} = \frac{a(\sqrt{b}-c)}{b-c^2}$$

The conjugate of $(\sqrt{b}+c)$ is $(\sqrt{b}-c)$ — flip the sign between the terms.

Examples:

$$\frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{3} \qquad \frac{6}{\sqrt{2}} = \frac{6\sqrt{2}}{2} = 3\sqrt{2}$$

$$\frac{5}{\sqrt{3}+2} \times \frac{\sqrt{3}-2}{\sqrt{3}-2} = \frac{5(\sqrt{3}-2)}{3-4} = \frac{5(\sqrt{3}-2)}{-1} = 10 - 5\sqrt{3}$$

2.3 Practice Problems

2.3.1 Easy

Exercise 2.1. *Simplify. Give your answer in index form.*

a) $w^2 \times w^2$

b) $n^3 \times n^3$

c) $c^7 \times c^2$

d) $w^7 \times w^6$

e) $b^6 \times b^5$

f) $a^2 \times a^2$

View Solutions

a)

$$w^{2+2} = w^4$$

b)

$$n^{3+3} = n^6$$

c)

$$c^{7+2} = c^9$$

d)

$$w^{7+6} = w^{13}$$

e)

$$b^{6+5} = b^{11}$$

f)

$$a^{2+2} = a^4$$

Index notation Same base identification Addition of integers

Exercise 2.2. *Simplify. Give your answer in index form.*

a) $m^3 \times m^6 \times m^6$

b) $a^6 \times a^3 \times a^6$

c) $y^3 \times y^5 \times y^6$

d) $n^2 \times n^3 \times n^5$

e) $x^4 \times x^3 \times x^3$

f) $x^2 \times x^2 \times x^5$

View Solutions

a)

$$m^{3+6+6} = m^{15}$$

b)

$$a^{6+3+6} = a^{15}$$

c)

$$y^{3+5+6} = y^{14}$$

d)

$$n^{2+3+5} = n^{10}$$

e)

$$x^{4+3+3} = x^{10}$$

f)

$$x^{2+2+5} = x^{9}$$

Index notation Addition of integers Multi-step simplification

Exercise 2.3. *Simplify. Give your answer in index form.*

a) $t^6 \div t^2$

b) $y^{10} \div y^5$

c) $n^8 \div n^2$

d) $b^{10} \div b^4$

e) $x^5 \div x^2$

f) $y^7 \div y^3$

View Solutions

a)

$$t^{6-2} = t^4$$

b)

$$y^{10-5} = y^5$$

c)

$$n^{8-2} = n^6$$

d)

$$b^{10-4} = b^6$$

e)

$$x^{5-2} = x^3$$

f)

$$y^{7-3} = y^4$$

Index notation Subtraction of integers

Exercise 2.4. *Simplify. Give your answer in index form.*

a) $(y^3)^2$

b) $(b^5)^4$

c) $(y^5)^5$

d) $(a^2)^2$

e) $(y^4)^2$

f) $(m^3)^3$

View Solutions

a)

$$y^{3\times 2} = y^6$$

b)

$$b^{5\times 4} = b^{20}$$

c)

$$y^{5\times 5} = y^{25}$$

d)

$$a^{2\times 2} = a^4$$

e)

$$y^{4\times 2} = y^8$$

f)

$$m^{3\times 3} = m^9$$

Index notation Multiplication of integers Power of power law

Exercise 2.5. *Evaluate.*

a) 1

b) 1

c) 1

d) 1

e) 1

f) 1

View Solutions

a)

$$n^0 = 1 \quad \text{(any non-zero base to power 0 = 1)}$$

b)

$$a^0 = 1 \quad \text{(any non-zero base to power 0 = 1)}$$

c)

$$8^0 = 1 \quad \text{(any non-zero base to power 0 = 1)}$$

d)

$$3w^0 = 1 \quad \text{(any non-zero base to power 0 = 1)}$$

e)

$$a^0 = 1 \quad \text{(any non-zero base to power 0 = 1)}$$

f)

$$p^0 = 1 \quad \text{(any non-zero base to power 0 = 1)}$$

Zero exponent rule Index notation

Exercise 2.6. *Simplify.*

a) $7\sqrt{6} + 5\sqrt{6}$

b) $5\sqrt{13} + 8\sqrt{13}$

c) $2\sqrt{2} + 4\sqrt{2}$

d) $5\sqrt{6} + 7\sqrt{6}$

e) $4\sqrt{6} + 7\sqrt{6}$

f) $5\sqrt{10} + 6\sqrt{10}$

View Solutions

a)

$$(7+5)\sqrt{6} = 12\sqrt{6}$$

b)

$$(5+8)\sqrt{13} = 13\sqrt{13}$$

c)

$$(2+4)\sqrt{2} = 6\sqrt{2}$$

d)

$$(5+7)\sqrt{6} = 12\sqrt{6}$$

e)

$$(4+7)\sqrt{6} = 11\sqrt{6}$$

f)

$$(5+6)\sqrt{10} = 11\sqrt{10}$$

Like surd identification Integer addition

Exercise 2.7. *Simplify.*

a) $10\sqrt{10} - 5\sqrt{10}$

b) $8\sqrt{10} - 4\sqrt{10}$

c) $7\sqrt{7} - 4\sqrt{7}$

d) $4\sqrt{6} - 2\sqrt{6}$

e) $7\sqrt{10} - 2\sqrt{10}$

f) $5\sqrt{5} - 3\sqrt{5}$

View Solutions

a)

$$(10-5)\sqrt{10} = 5\sqrt{10}$$

b)

$$(8-4)\sqrt{10} = 4\sqrt{10}$$

c)

$$(7-4)\sqrt{7} = 3\sqrt{7}$$

d)

$$(4-2)\sqrt{6} = 2\sqrt{6}$$

e)

$$(7-2)\sqrt{10} = 5\sqrt{10}$$

f)

$$(5-3)\sqrt{5} = 2\sqrt{5}$$

Like surd identification Integer subtraction

Exercise 2.8. *Simplify.*

a) $\sqrt{2} \times \sqrt{5}$

b) $\sqrt{2} \times \sqrt{6}$

c) $\sqrt{3} \times \sqrt{5}$

d) $\sqrt{2} \times \sqrt{6}$

e) $\sqrt{6} \times \sqrt{7}$

f) $\sqrt{2} \times \sqrt{5}$

View Solutions

a)

$$\sqrt{2 \times 5} = \sqrt{10} = \sqrt{10}$$

b)

$$\sqrt{2 \times 6} = \sqrt{12} = 2\sqrt{3}$$

c)

$$\sqrt{3 \times 5} = \sqrt{15} = \sqrt{15}$$

d)

$$\sqrt{2 \times 6} = \sqrt{12} = 2\sqrt{3}$$

e)

$$\sqrt{6 \times 7} = \sqrt{42} = \sqrt{42}$$

f)

$$\sqrt{2 \times 5} = \sqrt{10} = \sqrt{10}$$

Surd multiplication Surd simplification

Exercise 2.9. *Simplify.*

a) $\dfrac{\sqrt{24}}{\sqrt{6}}$

b) $\dfrac{\sqrt{72}}{\sqrt{2}}$

c) $\dfrac{\sqrt{27}}{\sqrt{3}}$

d) $\dfrac{\sqrt{180}}{\sqrt{5}}$

e) $\dfrac{\sqrt{72}}{\sqrt{2}}$

f) $\dfrac{\sqrt{24}}{\sqrt{6}}$

View Solutions

a)

$$\sqrt{\frac{24}{6}} = \sqrt{4} = 2$$

b)

$$\sqrt{\frac{72}{2}} = \sqrt{36} = 6$$

c)

$$\sqrt{\frac{27}{3}} = \sqrt{9} = 3$$

d)

$$\sqrt{\frac{180}{5}} = \sqrt{36} = 6$$

e)

$$\sqrt{\frac{72}{2}} = \sqrt{36} = 6$$

f)

$$\sqrt{\frac{24}{6}} = \sqrt{4} = 2$$

Surd division Surd simplification

Exercise 2.10. *Simplify fully.*

a) $\sqrt{640}$

b) $\sqrt{700}$

c) $\sqrt{28}$

d) $\sqrt{468}$

e) $\sqrt{208}$

f) $\sqrt{1300}$

View Solutions

a)

$$\sqrt{640} = \sqrt{64 \times 10} = \sqrt{64} \times \sqrt{10} = 8\sqrt{10} = 8\sqrt{10}$$

b)

$$\sqrt{700} = \sqrt{100 \times 7} = \sqrt{100} \times \sqrt{7} = 10\sqrt{7} = 10\sqrt{7}$$

c)

$$\sqrt{28} = \sqrt{4 \times 7} = \sqrt{4} \times \sqrt{7} = 2\sqrt{7} = 2\sqrt{7}$$

d)

$$\sqrt{468} = \sqrt{36 \times 13} = \sqrt{36} \times \sqrt{13} = 6\sqrt{13} = 6\sqrt{13}$$

e)

$$\sqrt{208} = \sqrt{16 \times 13} = \sqrt{16} \times \sqrt{13} = 4\sqrt{13} = 4\sqrt{13}$$

f)

$$\sqrt{1300} = \sqrt{100 \times 13} = \sqrt{100} \times \sqrt{13} = 10\sqrt{13} = 10\sqrt{13}$$

Perfect square recognition Surd simplification

∑ × π ÷ √ ◆ ∞ ± ∫ ≠ Δ

2.3.2 Medium

Exercise 2.11. *Simplify. Give your answer in index form.*

a) $4b^6 \times 4b^4$

b) $5a^2 \times 6a^5$

c) $6b^7 \times 4b^6$

d) $6x^7 \times 3x^2$

e) $3a^2 \times 4a^3$

f) $5b^5 \times 4b^7$

View Solutions

a)

$$4 \times 4 = 16, \quad b^{6+4} = b^{10}$$

b)

$$5 \times 6 = 30, \quad a^{2+5} = a^7$$

c)

$$6 \times 4 = 24, \quad b^{7+6} = b^{13}$$

d)

$$6 \times 3 = 18, \quad x^{7+2} = x^9$$

e)

$$3 \times 4 = 12, \quad a^{2+3} = a^5$$

f)

$$5 \times 4 = 20, \quad b^{5+7} = b^{12}$$

Coefficient multiplication Addition of integers Separating coefficients

Exercise 2.12. *Simplify. Give your answer in index form.*

a) $4x^3c \times 4x^4c^2$

b) $3p^5m^2 \times 5p^4m^3$

c) $2a^5n \times 4a^4n^2$

d) $3t^5x^2 \times 5t^5x^3$

e) $4c^5n^2 \times 5c^4n^2$

f) $2p^2k^2 \times 2p^3k^4$

View Solutions

a)

$$4 \times 4 = 16, \quad x^{3+4} = x^7, \quad c^{1+2} = c^3$$

b)

$$3 \times 5 = 15, \quad p^{5+4} = p^9, \quad m^{2+3} = m^5$$

c)

$$2 \times 4 = 8, \quad a^{5+4} = a^9, \quad n^{1+2} = n^3$$

d)

$$3 \times 5 = 15, \quad t^{5+5} = t^{10}, \quad x^{2+3} = x^5$$

e)

$$4 \times 5 = 20, \quad c^{5+4} = c^9, \quad n^{2+2} = n^4$$

f)

$$2 \times 2 = 4, \quad p^{2+3} = p^5, \quad k^{2+4} = k^6$$

Multi-variable expressions Coefficient multiplication Grouping like terms

Exercise 2.13. *Simplify. Give your answer in index form.*

a) $6a^9 \div 3a^4$

b) $18w^9 \div 6w^4$

c) $12c^6 \div 6c^4$

d) $4t^8 \div 2t^4$

e) $6x^6 \div 2x^4$

f) $10t^3 \div 5t^2$

View Solutions

a)

$$\frac{6}{3} = 2, \quad a^{9-4} = a^5$$

b)

$$\frac{18}{6} = 3, \quad w^{9-4} = w^5$$

c)

$$\frac{12}{6} = 2, \quad c^{6-4} = c^2$$

d)

$$\frac{4}{2} = 2, \quad t^{8-4} = t^4$$

e)

$$\frac{6}{2} = 3, \quad x^{6-4} = x^2$$

f)

$$\frac{10}{5} = 2, \quad t^{3-2} = t^1$$

Coefficient division Subtraction of integers

Exercise 2.14. *Simplify. Give your answer in index form.*

a) $\dfrac{12p^6c^4}{3p^2c^3}$

b) $\dfrac{12t^4y^5}{3t^2y^3}$

c) $\dfrac{4w^8x^3}{2w^4x^3}$

d) $\dfrac{4b^3y^4}{2by^3}$

e) $\dfrac{8w^3a^3}{2w^2a}$

f) $\dfrac{6m^6x^4}{2m^4x}$

View Solutions

a)

$$\frac{12}{3} = 4, \quad p^{6-2} = p^4, \quad c^{4-3} = c^1$$

b)

$$\frac{12}{3} = 4, \quad t^{4-2} = t^2, \quad y^{5-3} = y^2$$

c)

$$\frac{4}{2} = 2, \quad w^{8-4} = w^4, \quad x^{3-3} = x^0$$

d)

$$\frac{4}{2} = 2, \quad b^{3-1} = b^2, \quad y^{4-3} = y^1$$

e)

$$\frac{8}{2} = 4, \quad w^{3-2} = w^1, \quad a^{3-1} = a^2$$

f)

$$\frac{6}{2} = 3, \quad m^{6-4} = m^2, \quad x^{4-1} = x^3$$

Multi-variable expressions Coefficient division Subtraction of integers

Exercise 2.15. *Expand and simplify.*

a) $(5p^3)^3$

b) $(4c^5)^2$

c) $(5b^2)^2$

d) $(2w^2)^2$

e) $(5c^5)^3$

f) $(5m^2)^2$

 View Solutions

a)

$$5^3 = 125, \quad p^{3\times 3} = p^9$$

b)

$$4^2 = 16, \quad c^{5\times 2} = c^{10}$$

c)

$$5^2 = 25, \quad b^{2\times 2} = b^4$$

d)

$$2^2 = 4, \quad w^{2\times 2} = w^4$$

e)

$$5^3 = 125, \quad c^{5\times 3} = c^{15}$$

f)

$$5^2 = 25, \quad m^{2\times 2} = m^4$$

Power of power law Coefficient exponentiation Multiplication of integers

Exercise 2.16. *Expand and simplify.*

a) $(3y^3a^4)^3$

b) $(4y^4t^2)^2$

c) $(2nm^3)^2$

d) $(3a^2w)^2$

e) $(4cb^2)^3$

f) $(2bw)^3$

View Solutions

a)

$$3^3 = 27, \quad y^{3\times3} = y^9, \quad a^{4\times3} = a^{12}$$

b)

$$4^2 = 16, \quad y^{4\times2} = y^8, \quad t^{2\times2} = t^4$$

c)

$$2^2 = 4, \quad n^{1\times2} = n^2, \quad m^{3\times2} = m^6$$

d)

$$3^2 = 9, \quad a^{2\times2} = a^4, \quad w^{1\times2} = w^2$$

e)

$$4^3 = 64, \quad c^{1\times3} = c^3, \quad b^{2\times3} = b^6$$

f)

$$2^3 = 8, \quad b^{1\times3} = b^3, \quad w^{1\times3} = w^3$$

Power of power law Coefficient exponentiation Multi-variable expressions

Exercise 2.17. *Write as a single power by expressing each number as a prime power.*

a) 8×343

b) 16×81

c) 25×49

d) 25×49

e) 9×25

f) 8×343

View Solutions

a)

$$8 \times 343 = 2^3 \times 7^3 = (2 \times 7)^3 = 14^3 = 2744$$

b)

$$16 \times 81 = 2^4 \times 3^4 = (2 \times 3)^4 = 6^4 = 1296$$

c)

$$25 \times 49 = 5^2 \times 7^2 = (5 \times 7)^2 = 35^2 = 1225$$

d)

$$25 \times 49 = 5^2 \times 7^2 = (5 \times 7)^2 = 35^2 = 1225$$

e)

$$9 \times 25 = 3^2 \times 5^2 = (3 \times 5)^2 = 15^2 = 225$$

f)

$$8 \times 343 = 2^3 \times 7^3 = (2 \times 7)^3 = 14^3 = 2744$$

Prime factorisation Perfect square recognition Power distribution law

Exercise 2.18. *Evaluate. Write as a fraction.*

a) 4^{-3}

b) 7^{-2}

c) 10^{-3}

d) 3^{-3}

e) 7^{-4}

f) 4^{-1}

View Solutions

a)

$$4^{-3} = \frac{1}{4^3} = \frac{1}{64}$$

b)

$$7^{-2} = \frac{1}{7^2} = \frac{1}{49}$$

c)

$$10^{-3} = \frac{1}{10^3} = \frac{1}{1000}$$

d)

$$3^{-3} = \frac{1}{3^3} = \frac{1}{27}$$

e)

$$7^{-4} = \frac{1}{7^4} = \frac{1}{2401}$$

f)

$$4^{-1} = \frac{1}{4^1} = \frac{1}{4}$$

Negative index as reciprocal Integer exponentiation Fraction notation

Exercise 2.19. *Evaluate. Simplify your answer.*

a) $\left(\frac{2}{3}\right)^{-3}$

b) $\left(\frac{3}{4}\right)^{-2}$

c) $\left(\frac{2}{3}\right)^{-1}$

d) $\left(\frac{2}{5}\right)^{-1}$

e) $\left(\frac{2}{5}\right)^{-1}$

f) $\left(\frac{3}{4}\right)^{-3}$

View Solutions

a)

$$\left(\frac{3}{2}\right)^{3} = \frac{3^3}{2^3} = \frac{27}{8}$$

b)

$$\left(\frac{4}{3}\right)^{2} = \frac{4^2}{3^2} = \frac{16}{9}$$

c)

$$\text{Flip: } \left(\frac{3}{2}\right)^{1} = \frac{3}{2}$$

d)

$$\text{Flip: } \left(\frac{5}{2}\right)^{1} = \frac{5}{2}$$

e)

$$\text{Flip: } \left(\frac{5}{2}\right)^{1} = \frac{5}{2}$$

f)

$$\left(\frac{4}{3}\right)^{3} = \frac{4^3}{3^3} = \frac{64}{27}$$

Negative index as reciprocal Fraction reciprocal

Exercise 2.20. *Simplify. Give your answer in index form.*

a) $y^6 \times y^{-5}$

b) $b^2 \times b^{-4}$

c) $p^2 \times p^{-3}$

d) $t^6 \times t^{-4}$

e) $y^3 \times y^{-5}$

f) $n^4 \times n^{-5}$

View Solutions

a)

$$y^{6+(-5)} = y$$

b)

$$b^{2+(-4)} = b^{-2}$$

c)

$$p^{2+(-3)} = p^{-1}$$

d)

$$t^{6+(-4)} = t^2$$

e)

$$y^{3+(-5)} = y^{-2}$$

f)

$$n^{4+(-5)} = n^{-1}$$

Addition of integers Negative integer addition Index notation

Exercise 2.21. *Simplify. Give your answer in index form.*

a) $x^3 \div x^5$

b) $w^2 \div w^5$

c) $p^5 \div p^7$

d) $m^3 \div m^5$

e) $y^2 \div y^4$

f) $x^5 \div x^7$

View Solutions

a)

$$x^{3-5} = x^{-2}$$

b)

$$w^{2-5} = w^{-3}$$

c)

$$p^{5-7} = p^{-2}$$

d)

$$m^{3-5} = m^{-2}$$

e)

$$y^{2-4} = y^{-2}$$

f)

$$x^{5-7} = x^{-2}$$

Subtraction of integers Negative index as reciprocal Index notation

Exercise 2.22. *Evaluate.*

a) $256^{\frac{1}{4}}$

b) $64^{\frac{1}{2}}$

c) $81^{\frac{1}{2}}$

d) $125^{\frac{1}{3}}$

e) $625^{\frac{1}{4}}$

f) $8^{\frac{1}{3}}$

View Solutions

a)

$$\sqrt[4]{256} = 4$$

b)

$$\sqrt[2]{64} = 8$$

c)

$$\sqrt[2]{81} = 9$$

d)

$$\sqrt[3]{125} = 5$$

e)

$$\sqrt[4]{625} = 5$$

f)

$$\sqrt[3]{8} = 2$$

Fractional index as root Perfect square recognition nth root evaluation

Exercise 2.23. *Simplify. Give your answer in index form.*

a) $\dfrac{b^5 \times b^3}{b^7}$

b) $\dfrac{k^6 \times k^4}{k^8}$

c) $\dfrac{t^6 \times t^5}{t^9}$

d) $\dfrac{k^3 \times k^5}{k^4}$

e) $\dfrac{n^3 \times n^4}{n^5}$

f) $\dfrac{k^3 \times k^4}{k^4}$

View Solutions

a)

$$b^{5+3-7} = b^1$$

b)

$$k^{6+4-8} = k^2$$

c)

$$t^{6+5-9} = t^2$$

d)

$$k^{3+5-4} = k^4$$

e)

$$n^{3+4-5} = n^2$$

f)

$$k^{3+4-4} = k^3$$

Combining multiplication and division laws Multi-step simplification

Exercise 2.24. *Write as a single power of the given base.*

a) Write 81 as a power of 3

b) Write 9 as a power of 3

c) Write 32 as a power of 2

d) Write 100 as a power of 10

e) Write $\frac{1}{5}$ as a power of 5

f) Write 4 as a power of 2

View Solutions

a)

$$81 = 3^4$$

b)

$$9 = 3^2$$

c)

$$32 = 2^5$$

d)

$$100 = 10^2$$

e)

$$\frac{1}{5} = 5^{-1}$$

f)

$$4 = 2^2$$

Prime factorisation Index notation Negative index as reciprocal Zero exponent rule

Exercise 2.25. *Find the missing index.*

a) $y^2 \times y^{\square} = y^7$

b) $y^7 \times y^{\square} = y^{11}$

c) $k^3 \times k^{\square} = k^{10}$

d) $y^7 \times y^{\square} = y^{10}$

e) $p^4 \times p^{\square} = p^{11}$

f) $t^5 \times t^{\square} = t^{11}$

View Solutions

a)

$$\square = 7 - 2 = 5$$

b)

$$\square = 11 - 7 = 4$$

c)

$$\square = 10 - 3 = 7$$

d)

$$\square = 10 - 7 = 3$$

e)

$$\square = 11 - 4 = 7$$

f)

$$\square = 11 - 5 = 6$$

Reverse index law Subtraction of integers Algebraic reasoning

Exercise 2.26. *Find the missing index.*

a) $\dfrac{k^{14}}{k^{\square}} = k^{10}$

b) $\dfrac{t^{9}}{t^{\square}} = t^{6}$

c) $\dfrac{m^{5}}{m^{\square}} = m^{4}$

d) $\dfrac{k^{8}}{k^{\square}} = k^{6}$

e) $\dfrac{t^{16}}{t^{\square}} = t^{11}$

f) $\dfrac{y^{7}}{y^{\square}} = y^{4}$

View Solutions

a)

$$\square = 14 - 10 = 4$$

b)

$$\square = 9 - 6 = 3$$

c)

$$\square = 5 - 4 = 1$$

d)

$$\square = 8 - 6 = 2$$

e)

$$\square = 16 - 11 = 5$$

f)

$$\square = 7 - 4 = 3$$

Reverse index law Subtraction of integers Algebraic reasoning

Exercise 2.27. *Simplify by collecting like surds.*

a) $7\sqrt{7} + 6\sqrt{5} + 5\sqrt{7}$

b) $3\sqrt{5} + 4\sqrt{2} + 2\sqrt{5}$

c) $4\sqrt{5} + 2\sqrt{3} + 5\sqrt{5}$

d) $2\sqrt{3} + 2\sqrt{5} + 5\sqrt{3}$

e) $7\sqrt{5} + 5\sqrt{3} + \sqrt{5}$

f) $5\sqrt{2} + 5\sqrt{5} + 4\sqrt{2}$

View Solutions

a)

$$(7 + 5)\sqrt{7} + 6\sqrt{5} = 12\sqrt{7} + 6\sqrt{5}$$

b)

$$(3+2)\sqrt{5}+4\sqrt{2}=5\sqrt{5}+4\sqrt{2}$$

c)

$$(4+5)\sqrt{5}+2\sqrt{3}=9\sqrt{5}+2\sqrt{3}$$

d)

$$(2+5)\sqrt{3}+2\sqrt{5}=7\sqrt{3}+2\sqrt{5}$$

e)

$$(7+1)\sqrt{5}+5\sqrt{3}=8\sqrt{5}+5\sqrt{3}$$

f)

$$(5+4)\sqrt{2}+5\sqrt{5}=9\sqrt{2}+5\sqrt{5}$$

Like surd identification Collecting like terms Integer addition

Exercise 2.28. *Simplify by collecting like surds.*

a) $3\sqrt{5}+5\sqrt{3}-2\sqrt{5}-\sqrt{3}$

b) $6\sqrt{2}+3\sqrt{7}-5\sqrt{2}-\sqrt{7}$

c) $3\sqrt{3}+4\sqrt{5}-\sqrt{3}-3\sqrt{5}$

d) $7\sqrt{7}+6\sqrt{3}-4\sqrt{7}-4\sqrt{3}$

e) $5\sqrt{5}+7\sqrt{7}-\sqrt{5}-5\sqrt{7}$

f) $8\sqrt{2}+4\sqrt{3}-3\sqrt{2}-3\sqrt{3}$

View Solutions

a)

$$(3-2)\sqrt{5}+(5-1)\sqrt{3}=\sqrt{5}+4\sqrt{3}$$

b)

$$(6-5)\sqrt{2}+(3-1)\sqrt{7}=\sqrt{2}+2\sqrt{7}$$

c)

$$(3-1)\sqrt{3}+(4-3)\sqrt{5}=2\sqrt{3}+\sqrt{5}$$

d)

$$(7-4)\sqrt{7}+(6-4)\sqrt{3}=3\sqrt{7}+2\sqrt{3}$$

e)

$$(5-1)\sqrt{5}+(7-5)\sqrt{7}=4\sqrt{5}+2\sqrt{7}$$

f)

$$(8-3)\sqrt{2}+(4-3)\sqrt{3}=5\sqrt{2}+\sqrt{3}$$

Like surd identification Collecting like terms

Exercise 2.29. *Simplify.*

a) $3\sqrt{2}\times 5\sqrt{6}$

b) $5\sqrt{5}\times 2\sqrt{3}$

c) $4\sqrt{5}\times 5\sqrt{2}$

d) $3\sqrt{5}\times 3\sqrt{6}$

e) $2\sqrt{7}\times 3\sqrt{3}$

f) $4\sqrt{3}\times 2\sqrt{2}$

View Solutions

a)

$$3\times 5=15,\quad \sqrt{2}\times\sqrt{6}=\sqrt{12}=2\sqrt{3}\quad \Rightarrow 30\sqrt{3}$$

b)

$$5\times 2=10,\quad \sqrt{5}\times\sqrt{3}=\sqrt{15}=\sqrt{15}\quad \Rightarrow 10\sqrt{15}$$

c)

$$4\times 5=20,\quad \sqrt{5}\times\sqrt{2}=\sqrt{10}=\sqrt{10}\quad \Rightarrow 20\sqrt{10}$$

d)

e)

$$3 \times 3 = 9, \quad \sqrt{5} \times \sqrt{6} = \sqrt{30} = \sqrt{30} \quad \Rightarrow 9\sqrt{30}$$

f)

$$2 \times 3 = 6, \quad \sqrt{7} \times \sqrt{3} = \sqrt{21} = \sqrt{21} \quad \Rightarrow 6\sqrt{21}$$

$$4 \times 2 = 8, \quad \sqrt{3} \times \sqrt{2} = \sqrt{6} = \sqrt{6} \quad \Rightarrow 8\sqrt{6}$$

Surd multiplication Coefficient multiplication Surd simplification

Exercise 2.30. *Simplify.*

a) $2\sqrt{3} \times 3\sqrt{3} \times 2\sqrt{3}$

b) $3\sqrt{2} \times 2\sqrt{2} \times 2\sqrt{5}$

c) $3\sqrt{5} \times 2\sqrt{2} \times 3\sqrt{3}$

d) $2\sqrt{3} \times 3\sqrt{3} \times 2\sqrt{3}$

e) $2\sqrt{5} \times 3\sqrt{3} \times 2\sqrt{2}$

f) $2\sqrt{3} \times 2\sqrt{3} \times 3\sqrt{5}$

View Solutions

a)

$$2 \times 3 \times 2 = 12, \quad \sqrt{27} = 3\sqrt{3} \quad \Rightarrow 36\sqrt{3}$$

b)

$$3 \times 2 \times 2 = 12, \quad \sqrt{20} = 2\sqrt{5} \quad \Rightarrow 24\sqrt{5}$$

c)

$$3 \times 2 \times 3 = 18, \quad \sqrt{30} = \sqrt{30} \quad \Rightarrow 18\sqrt{30}$$

d)

$$2 \times 3 \times 2 = 12, \quad \sqrt{27} = 3\sqrt{3} \quad \Rightarrow 36\sqrt{3}$$

e)

$$2 \times 3 \times 2 = 12, \quad \sqrt{30} = \sqrt{30} \quad \Rightarrow 12\sqrt{30}$$

f)

$$2 \times 2 \times 3 = 12, \quad \sqrt{45} = 3\sqrt{5} \quad \Rightarrow 36\sqrt{5}$$

Surd multiplication Multi-step simplification

Exercise 2.31. *Simplify.*

a) $\dfrac{6\sqrt{2}}{3\sqrt{2}}$

b) $\dfrac{12\sqrt{20}}{3\sqrt{5}}$

c) $\dfrac{9\sqrt{27}}{3\sqrt{3}}$

d) $\dfrac{12\sqrt{12}}{3\sqrt{3}}$

e) $\dfrac{8\sqrt{32}}{2\sqrt{2}}$

f) $\dfrac{6\sqrt{20}}{2\sqrt{5}}$

View Solutions

a)

$$\frac{6}{3} = 2, \quad \frac{\sqrt{2}}{\sqrt{2}} = \sqrt{1} = 1 \quad \Rightarrow 2$$

b)

$$\frac{12}{3} = 4, \quad \frac{\sqrt{20}}{\sqrt{5}} = \sqrt{4} = 2 \quad \Rightarrow 8$$

c)

$$\frac{9}{3} = 3, \quad \frac{\sqrt{27}}{\sqrt{3}} = \sqrt{9} = 3 \quad \Rightarrow 9$$

d)

$$\frac{12}{3} = 4, \quad \frac{\sqrt{12}}{\sqrt{3}} = \sqrt{4} = 2 \quad \Rightarrow 8$$

e)

$$\frac{8}{2} = 4, \quad \frac{\sqrt{32}}{\sqrt{2}} = \sqrt{16} = 4 \quad \Rightarrow 16$$

f)

$$\frac{6}{2} = 3, \quad \frac{\sqrt{20}}{\sqrt{5}} = \sqrt{4} = 2 \quad \Rightarrow 6$$

Surd division Coefficient division Surd simplification

Exercise 2.32. *Simplify fully.*

a) $4\sqrt{112}$

b) $3\sqrt{63}$

c) $5\sqrt{54}$

d) $2\sqrt{96}$

e) $5\sqrt{12}$

f) $4\sqrt{24}$

View Solutions

a)

$$4\sqrt{112} = 4 \times 4\sqrt{7} = 16\sqrt{7} = 16\sqrt{7}$$

b)

$$3\sqrt{63} = 3 \times 3\sqrt{7} = 9\sqrt{7} = 9\sqrt{7}$$

c)

$$5\sqrt{54} = 5 \times 3\sqrt{6} = 15\sqrt{6} = 15\sqrt{6}$$

d)

$$2\sqrt{96} = 2 \times 4\sqrt{6} = 8\sqrt{6} = 8\sqrt{6}$$

e)

$$5\sqrt{12} = 5 \times 2\sqrt{3} = 10\sqrt{3} = 10\sqrt{3}$$

f)

$$4\sqrt{24} = 4 \times 2\sqrt{6} = 8\sqrt{6} = 8\sqrt{6}$$

Perfect square recognition Surd simplification Coefficient multiplication

Exercise 2.33. *Rationalise the denominator and simplify.*

a) $\dfrac{10}{\sqrt{2}}$

b) $\dfrac{6}{\sqrt{6}}$

c) $\dfrac{12}{\sqrt{3}}$

d) $\dfrac{3}{\sqrt{3}}$

e) $\dfrac{18}{\sqrt{6}}$

f) $\dfrac{25}{\sqrt{5}}$

View Solutions

a)

$$\frac{10}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}} = \frac{10\sqrt{2}}{2} = 5\sqrt{2}$$

b)

$$\frac{6}{\sqrt{6}} \times \frac{\sqrt{6}}{\sqrt{6}} = \frac{6\sqrt{6}}{6} = \sqrt{6}$$

c)

$$\frac{12}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} = \frac{12\sqrt{3}}{3} = 4\sqrt{3}$$

d)

$$\frac{3}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} = \frac{3\sqrt{3}}{3} = \sqrt{3}$$

e)

$$\frac{18}{\sqrt{6}} \times \frac{\sqrt{6}}{\sqrt{6}} = \frac{18\sqrt{6}}{6} = 3\sqrt{6}$$

f)

$$\frac{25}{\sqrt{5}} \times \frac{\sqrt{5}}{\sqrt{5}} = \frac{25\sqrt{5}}{5} = 5\sqrt{5}$$

Rationalise denominator Surd multiplication

Exercise 2.34. *Simplify fully.*

a) $\sqrt{10} \times \sqrt{6}$

b) $\sqrt{10} \times \sqrt{2}$

c) $\sqrt{5} \times \sqrt{10}$

d) $\sqrt{5} \times \sqrt{10}$

e) $\sqrt{10} \times \sqrt{6}$

f) $\sqrt{10} \times \sqrt{5}$

View Solutions

a)

$$\sqrt{60} = 2\sqrt{15}$$

b)

$$\sqrt{20} = 2\sqrt{5}$$

c)

$$\sqrt{50} = 5\sqrt{2}$$

d)

$$\sqrt{50} = 5\sqrt{2}$$

e)

$$\sqrt{60} = 2\sqrt{15}$$

f)

$$\sqrt{50} = 5\sqrt{2}$$

Surd multiplication Surd simplification

Exercise 2.35. *Expand and simplify.*

a) $\sqrt{5}(\sqrt{7} + 5)$

b) $\sqrt{2}(2\sqrt{2} + 5)$

c) $\sqrt{6}(\sqrt{3} + 2)$

d) $\sqrt{3}(1\sqrt{3} + 4)$

e) $\sqrt{5}(\sqrt{3} + 5)$

f) $\sqrt{5}(2\sqrt{5} + 5)$

View Solutions

a)

$$\sqrt{35} + 5\sqrt{5} = \sqrt{35} + 5\sqrt{5}$$

b)

$$\sqrt{2} \times 2\sqrt{2} + 5\sqrt{2} = 2 \times 2 + 5\sqrt{2} = 4 + 5\sqrt{2}$$

c)

$$\sqrt{18} + 2\sqrt{6} = 3\sqrt{2} + 2\sqrt{6}$$

d)

$$\sqrt{3} \times 1\sqrt{3} + 4\sqrt{3} = 1 \times 3 + 4\sqrt{3} = 3 + 4\sqrt{3}$$

e)

$$\sqrt{15} + 5\sqrt{5} = \sqrt{15} + 5\sqrt{5}$$

f)

$$\sqrt{5} \times 2\sqrt{5} + 5\sqrt{5} = 2 \times 5 + 5\sqrt{5} = 10 + 5\sqrt{5}$$

Distributive law Surd multiplication Surd simplification

Exercise 2.36. *Expand and simplify.*

a) $(\sqrt{3} + 3)(\sqrt{3} + 4)$

b) $(\sqrt{5} + 5)(\sqrt{5} - 4)$

c) $(\sqrt{5} + 4)(\sqrt{5} - 5)$

d) $(\sqrt{3} + 2)(\sqrt{3} + 5)$

e) $(\sqrt{5} + 1)(\sqrt{5} - 4)$

f) $(\sqrt{2} + 5)(\sqrt{2} - 1)$

View Solutions

a)

$$m + (a + b)\sqrt{m} + ab = 3 + 7\sqrt{3} + 12 = 15 + 7\sqrt{3}$$

b)

$$m + (a + b)\sqrt{m} + ab = 5 + 1\sqrt{5} + -20 = -15 + 1\sqrt{5}$$

c)

$$m + (a+b)\sqrt{m} + ab = 5 + -1\sqrt{5} + -20 = -15 - 1\sqrt{5}$$

d)

$$m + (a+b)\sqrt{m} + ab = 3 + 7\sqrt{3} + 10 = 13 + 7\sqrt{3}$$

e)

$$m + (a+b)\sqrt{m} + ab = 5 + -3\sqrt{5} + -4 = 1 - 3\sqrt{5}$$

f)

$$m + (a+b)\sqrt{m} + ab = 2 + 4\sqrt{2} + -5 = -3 + 4\sqrt{2}$$

FOIL method Surd multiplication Collecting like terms

Exercise 2.37. *Expand and simplify.*

a) $(\sqrt{7}+2)(\sqrt{7}-2)$

b) $(\sqrt{11}+1)(\sqrt{11}-1)$

c) $(\sqrt{6}+\sqrt{5})(\sqrt{6}-\sqrt{5})$

d) $(\sqrt{6}+4)(\sqrt{6}-4)$

e) $(\sqrt{13}+1)(\sqrt{13}-1)$

f) $(3\sqrt{2}+\sqrt{11})(3\sqrt{2}-\sqrt{11})$

View Solutions

a)

$$(\sqrt{7})^2 - 2^2 = 7 - 4 = 3$$

b)

$$(\sqrt{11})^2 - 1^2 = 11 - 1 = 10$$

c)

$$(\sqrt{6})^2 - (\sqrt{5})^2 = 1 \times 6 - 5 = 6 - 5 = 1$$

d)

$$(\sqrt{6})^2 - 4^2 = 6 - 16 = -10$$

e)

$$(\sqrt{13})^2 - 1^2 = 13 - 1 = 12$$

f)

$$(3\sqrt{2})^2 - (\sqrt{11})^2 = 9 \times 2 - 11 = 18 - 11 = 7$$

Difference of squares Conjugate pairs Surd simplification

Exercise 2.38. *Expand and simplify.*

a) $(\sqrt{3} + 2)^2$

b) $(\sqrt{5} + 4)^2$

c) $(\sqrt{2} + 5)^2$

d) $(\sqrt{5} + 2)^2$

e) $(\sqrt{7} + 2)^2$

f) $(\sqrt{2} + 5)^2$

View Solutions

a)

$$(\sqrt{3})^2 + 2 \times 2 \times \sqrt{3} + 2^2 = 3 + 4\sqrt{3} + 4 = 7 + 4\sqrt{3}$$

b)

$$(\sqrt{5})^2 + 2 \times 4 \times \sqrt{5} + 4^2 = 5 + 8\sqrt{5} + 16 = 21 + 8\sqrt{5}$$

c)

$$(\sqrt{2})^2 + 2 \times 5 \times \sqrt{2} + 5^2 = 2 + 10\sqrt{2} + 25 = 27 + 10\sqrt{2}$$

d)

$$(\sqrt{5})^2 + 2 \times 2 \times \sqrt{5} + 2^2 = 5 + 4\sqrt{5} + 4 = 9 + 4\sqrt{5}$$

e)

f)

$$(\sqrt{7})^2 + 2 \times 2 \times \sqrt{7} + 2^2 = 7 + 4\sqrt{7} + 4 = 11 + 4\sqrt{7}$$

$$(\sqrt{2})^2 + 2 \times 5 \times \sqrt{2} + 5^2 = 2 + 10\sqrt{2} + 25 = 27 + 10\sqrt{2}$$

Squaring binomials Surd multiplication Collecting like terms

Exercise 2.39. *Rationalise the denominator and simplify.*

a) $\dfrac{2}{\sqrt{5}}$

b) $\dfrac{1}{\sqrt{7}}$

c) $\dfrac{8}{\sqrt{6}}$

d) $\dfrac{7}{\sqrt{2}}$

e) $\dfrac{5}{\sqrt{5}}$

f) $\dfrac{6}{\sqrt{2}}$

View Solutions

a)

$$\frac{2}{\sqrt{5}} \times \frac{\sqrt{5}}{\sqrt{5}} = \frac{2\sqrt{5}}{5} = \frac{2\sqrt{5}}{5}$$

b)

$$\frac{1}{\sqrt{7}} \times \frac{\sqrt{7}}{\sqrt{7}} = \frac{1\sqrt{7}}{7} = \frac{1\sqrt{7}}{7}$$

c)

d)

$$\frac{8}{\sqrt{6}} \times \frac{\sqrt{6}}{\sqrt{6}} = \frac{8\sqrt{6}}{6} = \frac{4\sqrt{6}}{3}$$

e)

$$\frac{7}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}} = \frac{7\sqrt{2}}{2} = \frac{7\sqrt{2}}{2}$$

f)

$$\frac{5}{\sqrt{5}} \times \frac{\sqrt{5}}{\sqrt{5}} = \frac{5\sqrt{5}}{5} = \sqrt{5}$$

$$\frac{6}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}} = \frac{6\sqrt{2}}{2} = 3\sqrt{2}$$

Rationalise denominator Surd multiplication Fraction simplification

Exercise 2.40. *Rationalise the denominator and simplify.*

a) $\frac{16}{4\sqrt{2}}$

b) $\frac{16}{4\sqrt{2}}$

c) $\frac{24}{4\sqrt{2}}$

d) $\frac{60}{5\sqrt{3}}$

e) $\frac{9}{3\sqrt{3}}$

f) $\frac{36}{2\sqrt{6}}$

 View Solutions

a)

$$\frac{16}{4\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}} = \frac{16\sqrt{2}}{8} = 2\sqrt{2}$$

b)

$$\frac{16}{4\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}} = \frac{16\sqrt{2}}{8} = 2\sqrt{2}$$

c)

$$\frac{24}{4\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}} = \frac{24\sqrt{2}}{8} = 3\sqrt{2}$$

d)

$$\frac{60}{5\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} = \frac{60\sqrt{3}}{15} = 4\sqrt{3}$$

e)

$$\frac{9}{3\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} = \frac{9\sqrt{3}}{9} = \sqrt{3}$$

f)

$$\frac{36}{2\sqrt{6}} \times \frac{\sqrt{6}}{\sqrt{6}} = \frac{36\sqrt{6}}{12} = 3\sqrt{6}$$

Rationalise denominator Surd multiplication Fraction simplification

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2.3.3 Challenging

Exercise 2.41. *Simplify. Give your answer with positive exponents.*

a) $(4w^4)^{-3}$

b) $(3x^2)^{-3}$

c) $(2x^3)^{-2}$

d) $(4c^4)^{-2}$

e) $(3k^2)^{-1}$

f) $(4k^4)^{-1}$

View Solutions

a)

$$\frac{1}{4^3 w^{4\times 3}} = \frac{1}{64w^{12}}$$

b)

$$\frac{1}{3^3 x^{2\times 3}} = \frac{1}{27x^6}$$

c)

$$\frac{1}{2^2 x^{3\times 2}} = \frac{1}{4x^6}$$

d)

$$\frac{1}{4^2 c^{4\times 2}} = \frac{1}{16c^8}$$

e)

$$\frac{1}{3^1 k^{2\times 1}} = \frac{1}{3k^2}$$

f)

$$\frac{1}{4^1 k^{4\times 1}} = \frac{1}{4k^4}$$

Power of power law Negative index as reciprocal Coefficient exponentiation

Exercise 2.42. *Evaluate.*

a) $16^{\frac{2}{4}}$

b) $16^{\frac{2}{4}}$

c) $25^{\frac{4}{2}}$

d) $64^{\frac{2}{3}}$

e) $32^{\frac{4}{5}}$

f) $49^{\frac{3}{2}}$

View Solutions

a)

$$\left(\sqrt[4]{16}\right)^2 = 2^2 = 4$$

b)

$$\left(\sqrt[4]{16}\right)^2 = 2^2 = 4$$

c)

$$\left(\sqrt[2]{25}\right)^4 = 5^4 = 625$$

d)

$$\left(\sqrt[3]{64}\right)^2 = 4^2 = 16$$

e)

$$\left(\sqrt[5]{32}\right)^4 = 2^4 = 16$$

f)

$$\left(\sqrt[2]{49}\right)^3 = 7^3 = 343$$

Fractional index as root and power nth root evaluation Integer exponentiation

Exercise 2.43. *Evaluate.*

a) $256^{-\frac{3}{4}}$

b) $49^{-\frac{2}{2}}$

c) $256^{-\frac{3}{4}}$

d) $64^{-\frac{2}{3}}$

e) $256^{-\frac{3}{4}}$

f) $9^{-\frac{2}{2}}$

View Solutions

a)

$$256^{-\frac{3}{4}} = \frac{1}{256^{\frac{3}{4}}} = \frac{1}{\left(\sqrt[4]{256}\right)^3} = \frac{1}{4^3} = \frac{1}{64}$$

b)

$$49^{-\frac{2}{2}} = \frac{1}{49^{\frac{2}{2}}} = \frac{1}{\left(\sqrt[2]{49}\right)^2} = \frac{1}{7^2} = \frac{1}{49}$$

c)

$$256^{-\frac{3}{4}} = \frac{1}{256^{\frac{3}{4}}} = \frac{1}{\left(\sqrt[4]{256}\right)^3} = \frac{1}{4^3} = \frac{1}{64}$$

d)

$$64^{-\frac{2}{3}} = \frac{1}{64^{\frac{2}{3}}} = \frac{1}{\left(\sqrt[3]{64}\right)^2} = \frac{1}{4^2} = \frac{1}{16}$$

e)

$$256^{-\frac{3}{4}} = \frac{1}{256^{\frac{3}{4}}} = \frac{1}{\left(\sqrt[4]{256}\right)^3} = \frac{1}{4^3} = \frac{1}{64}$$

f)

$$9^{-\frac{2}{2}} = \frac{1}{9^{\frac{2}{2}}} = \frac{1}{\left(\sqrt[2]{9}\right)^2} = \frac{1}{3^2} = \frac{1}{9}$$

Fractional index as root and power Negative index as reciprocal

Exercise 2.44. *Evaluate. Simplify your answer.*

a) $\left(\frac{27}{216}\right)^{\frac{3}{3}}$

b) $\left(\frac{125}{216}\right)^{\frac{1}{3}}$

c) $\left(\frac{64}{27}\right)^{\frac{2}{3}}$

d) $\left(\frac{27}{216}\right)^{\frac{2}{3}}$

e) $\left(\frac{4}{16}\right)^{\frac{3}{2}}$

f) $\left(\frac{8}{27}\right)^{\frac{3}{3}}$

View Solutions

a)

$$\left(\sqrt[3]{\frac{27}{216}}\right)^{3} = \left(\frac{3}{6}\right)^{3} = \frac{3^3}{6^3} = \frac{1}{8}$$

b)

$$\left(\sqrt[3]{\frac{125}{216}}\right)^{1} = \left(\frac{5}{6}\right)^{1} = \frac{5^1}{6^1} = \frac{5}{6}$$

c)

$$\left(\sqrt[3]{\frac{64}{27}}\right)^{2} = \left(\frac{4}{3}\right)^{2} = \frac{4^2}{3^2} = \frac{16}{9}$$

d)

$$\left(\sqrt[3]{\frac{27}{216}}\right)^{2} = \left(\frac{3}{6}\right)^{2} = \frac{3^2}{6^2} = \frac{1}{4}$$

e)

$$\left(\sqrt[2]{\frac{4}{16}}\right)^3 = \left(\frac{2}{4}\right)^3 = \frac{2^3}{4^3} = \frac{1}{8}$$

f)

$$\left(\sqrt[3]{\frac{8}{27}}\right)^3 = \left(\frac{2}{3}\right)^3 = \frac{2^3}{3^3} = \frac{8}{27}$$

Fractional index as root and power nth root of fraction

Exercise 2.45. *Simplify fully.*

a) $\dfrac{3b^4 \times 3b^2}{3b^3}$

b) $\dfrac{4c^3 \times 2c^2}{8c^4}$

c) $\dfrac{3m^2 \times 4m^3}{6m^4}$

d) $\dfrac{3y^2 \times 4y^2}{2y^2}$

e) $\dfrac{3t^5 \times 3t^3}{9t}$

f) $\dfrac{3y^3 \times 3y^3}{9y^2}$

 View Solutions

a)

$$3 \times 3 = 9,\ b^{4+2} = b^6 \quad \frac{9}{3} = 3,\ b^{6-3} = b^3$$

b)

$$4 \times 2 = 8,\ c^{3+2} = c^5 \quad \frac{8}{8} = 1,\ c^{5-4} = c^1$$

c)

$$3 \times 4 = 12,\ m^{2+3} = m^5 \quad \frac{12}{6} = 2,\ m^{5-4} = m^1$$

d)

$$3 \times 4 = 12,\ y^{2+2} = y^4 \quad \frac{12}{2} = 6,\ y^{4-2} = y^2$$

e)

$$3 \times 3 = 9,\ t^{5+3} = t^8 \quad \frac{9}{9} = 1,\ t^{8-1} = t^7$$

f)

$$3 \times 3 = 9,\ y^{3+3} = y^6 \quad \frac{9}{9} = 1,\ y^{6-2} = y^4$$

Coefficient division Combining multiplication and division laws

Exercise 2.46. *Simplify fully.*

a) $\dfrac{6a^7k^6}{2a^3k^3}$

b) $\dfrac{12c^2w^3}{4cw^2}$

c) $\dfrac{6c^5p^3}{3c^2p}$

d) $\dfrac{6k^6x^4}{3k^3x^2}$

e) $\dfrac{8k^3a^5}{2ka^2}$

f) $\dfrac{4a^3b^4}{2a^2b}$

View Solutions

a)

$$\frac{6}{2} = 3, \quad a^{7-3} = a^4, \quad k^{6-3} = k^3$$

b)

$$\frac{12}{4} = 3, \quad c^{2-1} = c^1, \quad w^{3-2} = w^1$$

c)

$$\frac{6}{3} = 2, \quad c^{5-2} = c^3, \quad p^{3-1} = p^2$$

d)

$$\frac{6}{3} = 2, \quad k^{6-3} = k^3, \quad x^{4-2} = x^2$$

e)

$$\frac{8}{2} = 4, \quad k^{3-1} = k^2, \quad a^{5-2} = a^3$$

f)

$$\frac{4}{2} = 2, \quad a^{3-2} = a^1, \quad b^{4-1} = b^3$$

Multi-variable expressions Coefficient division Multi-step simplification

Exercise 2.47. *Simplify fully.*

a) $\dfrac{c^5 \times (c^4)^2}{c^{10}}$

b) $\dfrac{m^5 \times (m^4)^3}{m^{12}}$

c) $\dfrac{c^2 \times (c^2)^3}{c}$

d) $\dfrac{t^2 \times (t^3)^3}{t^7}$

e) $\dfrac{m^2 \times (m^4)^2}{m^2}$

f) $\dfrac{n^2 \times (n^4)^2}{n^6}$

View Solutions

a)

$$(c^4)^2 = c^8, \quad c^5 \times c^8 = c^{13}, \quad c^{13-10} = c^3$$

b)

$$(m^4)^3 = m^{12}, \quad m^5 \times m^{12} = m^{17}, \quad m^{17-12} = m^5$$

c)

$$(c^2)^3 = c^6, \quad c^2 \times c^6 = c^8, \quad c^{8-1} = c^7$$

d)

$$(t^3)^3 = t^9, \quad t^2 \times t^9 = t^{11}, \quad t^{11-7} = t^4$$

e)

$$(m^4)^2 = m^8, \quad m^2 \times m^8 = m^{10}, \quad m^{10-2} = m^8$$

f)

$$(n^4)^2 = n^8, \quad n^2 \times n^8 = n^{10}, \quad n^{10-6} = n^4$$

Power of power law Combining multiplication and division laws

Exercise 2.48. *Find the missing power.*

a) $(k^4)^{\square} = k^8$

b) $(k^3)^{\square} = k^9$

c) $(n^4)^{\square} = n^{12}$

d) $(t^4)^{\square} = t^{16}$

e) $(a^3)^{\square} = a^6$

f) $(m^5)^{\square} = m^{15}$

View Solutions

a)

$$\square = 8 \div 4 = 2$$

b)

$$\square = 9 \div 3 = 3$$

c)

$$\square = 12 \div 4 = 3$$

d)

$$\square = 16 \div 4 = 4$$

e)

$$\square = 6 \div 3 = 2$$

f)

$$\square = 15 \div 5 = 3$$

Reverse power of power law Division of integers Algebraic reasoning

Exercise 2.49. *Identify the mistake and write the correct answer.*

A student wrote: $27^{\frac{1}{3}} = 9$ (claiming $\frac{1}{3} \times 27 = 9$)

(a) Identify the mistake.

(b) Write the correct answer.

View Solution

a)

$$a^{\frac{1}{n}} = \sqrt[n]{a}, \text{ not } \frac{1}{n} \times a$$

b)

$$\sqrt[3]{27} = 3$$

Mathematical reasoning Error analysis Verbal mathematical explanation

Exercise 2.50. *Simplify fully.*

a) $5\sqrt{3} + 1\sqrt{3}$

b) $2\sqrt{2} + 4\sqrt{2}$

c) $2\sqrt{2} + 2\sqrt{2}$

d) $3\sqrt{3} + 4\sqrt{3}$

e) $5\sqrt{3} + 4\sqrt{3}$

f) $4\sqrt{3} + 1\sqrt{3}$

View Solutions

a)

$$5 \times 1\sqrt{3} + 1 \times 1\sqrt{3} = (5+1)\sqrt{3} = 6\sqrt{3}$$

b)

$$2 \times 1\sqrt{2} + 4 \times 1\sqrt{2} = (2+4)\sqrt{2} = 6\sqrt{2}$$

c)

$$2 \times 1\sqrt{2} + 2 \times 1\sqrt{2} = (2+2)\sqrt{2} = 4\sqrt{2}$$

d)

$$3 \times 1\sqrt{3} + 4 \times 1\sqrt{3} = (3+4)\sqrt{3} = 7\sqrt{3}$$

e)

$$5 \times 1\sqrt{3} + 4 \times 1\sqrt{3} = (5+4)\sqrt{3} = 9\sqrt{3}$$

f)

$$4 \times 1\sqrt{3} + 1 \times 1\sqrt{3} = (4+1)\sqrt{3} = 5\sqrt{3}$$

Surd simplification Collecting like terms Multi-step simplification

Exercise 2.51. *Simplify fully.*

a) $\sqrt{27} + \sqrt{27}$

b) $\sqrt{24} + \sqrt{96}$

c) $\sqrt{18} + \sqrt{18}$

d) $\sqrt{12} + \sqrt{27}$

e) $\sqrt{24} + \sqrt{24}$

f) $\sqrt{12} + \sqrt{12}$

View Solutions

a)

$$3\sqrt{3} + 3\sqrt{3} = 6\sqrt{3}$$

b)

$$2\sqrt{6} + 4\sqrt{6} = 6\sqrt{6}$$

c)

$$3\sqrt{2} + 3\sqrt{2} = 6\sqrt{2}$$

d)

$$2\sqrt{3} + 3\sqrt{3} = 5\sqrt{3}$$

e)

$$2\sqrt{6} + 2\sqrt{6} = 4\sqrt{6}$$

f)

$$2\sqrt{3} + 2\sqrt{3} = 4\sqrt{3}$$

Surd simplification Multi-step simplification Collecting like terms

Exercise 2.52. *Expand and simplify.*

a) $(\sqrt{2} + 3)(2\sqrt{2} + 4)$

b) $(\sqrt{2} + 4)(2\sqrt{2} + 4)$

c) $(2\sqrt{2} + 3)(\sqrt{2} - 1)$

d) $(\sqrt{3} + 1)(2\sqrt{3} + 3)$

e) $(\sqrt{2}+1)(2\sqrt{2}-2)$

f) $(2\sqrt{5}+4)(3\sqrt{5}+1)$

View Solutions

a)

$$1 \times 2 \times 2 = 4, \quad (1 \times 4 + 3 \times 2)\sqrt{2} = 10\sqrt{2}, \quad \text{const} = 12 \Rightarrow 16 + 10\sqrt{2}$$

b)

$$1 \times 2 \times 2 = 4, \quad (1 \times 4 + 4 \times 2)\sqrt{2} = 12\sqrt{2}, \quad \text{const} = 16 \Rightarrow 20 + 12\sqrt{2}$$

c)

$$2 \times 1 \times 2 = 4, \quad (2 \times -1 + 3 \times 1)\sqrt{2} = 1\sqrt{2}, \quad \text{const} = -3 \Rightarrow 1 + 1\sqrt{2}$$

d)

$$1 \times 2 \times 3 = 6, \quad (1 \times 3 + 1 \times 2)\sqrt{3} = 5\sqrt{3}, \quad \text{const} = 3 \Rightarrow 9 + 5\sqrt{3}$$

e)

$$1 \times 2 \times 2 = 4, \quad (1 \times -2 + 1 \times 2)\sqrt{2} = 0\sqrt{2}, \quad \text{const} = -2 \Rightarrow 2$$

f)

$$2 \times 3 \times 5 = 30, \quad (2 \times 1 + 4 \times 3)\sqrt{5} = 14\sqrt{5}, \quad \text{const} = 4 \Rightarrow 34 + 14\sqrt{5}$$

FOIL method Surd multiplication Collecting like terms

Exercise 2.53. *Rationalise the denominator and simplify.*

a) $\dfrac{2}{\sqrt{3}+1}$

b) $\dfrac{3}{\sqrt{2}+2}$

c) $\dfrac{4}{\sqrt{7}+2}$

d) $\dfrac{2}{\sqrt{5}+1}$

e) $\dfrac{1}{\sqrt{5}+2}$

f) $\dfrac{1}{\sqrt{5}+2}$

View Solutions

a)

$$\frac{2}{\sqrt{3}+1} \times \frac{\sqrt{3}-1}{\sqrt{3}-1} = \frac{2(\sqrt{3}-1)}{3-1^2} = \frac{2(\sqrt{3}-1)}{2} = \sqrt{3}-1$$

b)

$$\frac{3}{\sqrt{2}+2} \times \frac{\sqrt{2}-2}{\sqrt{2}-2} = \frac{3(\sqrt{2}-2)}{2-2^2} = \frac{3(\sqrt{2}-2)}{-2} = -\frac{3(\sqrt{2}-2)}{2}$$

c)

$$\frac{4}{\sqrt{7}+2} \times \frac{\sqrt{7}-2}{\sqrt{7}-2} = \frac{4(\sqrt{7}-2)}{7-2^2} = \frac{4(\sqrt{7}-2)}{3} = \frac{4(\sqrt{7}-2)}{3}$$

d)

$$\frac{2}{\sqrt{5}+1} \times \frac{\sqrt{5}-1}{\sqrt{5}-1} = \frac{2(\sqrt{5}-1)}{5-1^2} = \frac{2(\sqrt{5}-1)}{4} = \frac{1(\sqrt{5}-1)}{2}$$

e)

$$\frac{1}{\sqrt{5}+2} \times \frac{\sqrt{5}-2}{\sqrt{5}-2} = \frac{1(\sqrt{5}-2)}{5-2^2} = \frac{1(\sqrt{5}-2)}{1} = \sqrt{5}-2$$

f)

$$\frac{1}{\sqrt{5}+2} \times \frac{\sqrt{5}-2}{\sqrt{5}-2} = \frac{1(\sqrt{5}-2)}{5-2^2} = \frac{1(\sqrt{5}-2)}{1} = \sqrt{5}-2$$

Conjugate pairs Difference of squares Rationalise denominator

Exercise 2.54. *Rationalise the denominator fully and simplify.*

Rationalise the denominator: $\dfrac{4+\sqrt{5}}{\sqrt{7}-3}$

(a) State the conjugate of the denominator.

(b) Multiply numerator and denominator by the conjugate and simplify.

View Solution

a)

$$\text{Conjugate of } (\sqrt{7}-3) \text{ is } (\sqrt{7}+3)$$

b)

$$\frac{(4+\sqrt{5})(\sqrt{7}+3)}{(\sqrt{7})^2-3^2} = \frac{4\sqrt{7}+12+\sqrt{35}+3\sqrt{5}}{-2}$$

Conjugate pairs Difference of squares Multi-step rationalisation

Exercise 2.55. *Rationalise each denominator and add.*

Rationalise and simplify: $\dfrac{2}{\sqrt{3}-1}+\dfrac{2}{3-\sqrt{3}}$

(a) Rationalise each fraction separately.

(b) Add the simplified fractions together.

View Solution

a)

$$\frac{2}{\sqrt{3}-1}\times\frac{\sqrt{3}+1}{\sqrt{3}+1} = \frac{2(\sqrt{3}+1)}{2} = \sqrt{3}+1$$
$$\frac{2}{3-\sqrt{3}}\times\frac{3+\sqrt{3}}{3+\sqrt{3}} = \frac{2(3+\sqrt{3})}{6} = \frac{3+\sqrt{3}}{3}$$

b)

$$(\sqrt{3}+1)+\frac{3+\sqrt{3}}{3}=\frac{3\sqrt{3}+3+3+\sqrt{3}}{3}=\frac{4\sqrt{3}+6}{3}$$

Rationalise denominator Adding fractions Multi-step simplification

Exercise 2.56. *Rationalise and simplify.*

a) $\dfrac{\sqrt{12}+6}{\sqrt{3}}$

b) $\dfrac{\sqrt{12}+6}{\sqrt{3}}$

c) $\dfrac{\sqrt{18}+10}{\sqrt{2}}$

d) $\dfrac{\sqrt{12}+6}{\sqrt{3}}$

e) $\dfrac{\sqrt{50}+10}{\sqrt{2}}$

f) $\dfrac{\sqrt{12}+6}{\sqrt{3}}$

View Solutions

a)

$$\frac{\sqrt{12}}{\sqrt{3}}+\frac{6}{\sqrt{3}}=\sqrt{4}+\frac{6\sqrt{3}}{3}=2+2\sqrt{3}$$

b)

$$\frac{\sqrt{12}}{\sqrt{3}}+\frac{6}{\sqrt{3}}=\sqrt{4}+\frac{6\sqrt{3}}{3}=2+2\sqrt{3}$$

c)

$$\frac{\sqrt{18}}{\sqrt{2}}+\frac{10}{\sqrt{2}}=\sqrt{9}+\frac{10\sqrt{2}}{2}=3+5\sqrt{2}$$

d)

$$\frac{\sqrt{12}}{\sqrt{3}} + \frac{6}{\sqrt{3}} = \sqrt{4} + \frac{6\sqrt{3}}{3} = 2 + 2\sqrt{3}$$

e)

$$\frac{\sqrt{50}}{\sqrt{2}} + \frac{10}{\sqrt{2}} = \sqrt{25} + \frac{10\sqrt{2}}{2} = 5 + 5\sqrt{2}$$

f)

$$\frac{\sqrt{12}}{\sqrt{3}} + \frac{6}{\sqrt{3}} = \sqrt{4} + \frac{6\sqrt{3}}{3} = 2 + 2\sqrt{3}$$

Splitting fractions Rationalise denominator Surd simplification

Exercise 2.57. *Find the values of x and y.*

a) $(3 - \sqrt{x})^2 = y - 6\sqrt{2}$

b) $(7 - \sqrt{x})^2 = y - 14\sqrt{2}$

c) $(3 - \sqrt{x})^2 = y - 6\sqrt{2}$

d) $(5 - \sqrt{x})^2 = y - 10\sqrt{2}$

e) $(6 - \sqrt{x})^2 = y - 12\sqrt{2}$

f) $(5 - \sqrt{x})^2 = y - 10\sqrt{2}$

View Solutions

a)

$$(3)^2 - 6\sqrt{x} + x \Rightarrow x = 2,\ y = 9 + 2 = 11$$

b)

$$(7)^2 - 14\sqrt{x} + x \Rightarrow x = 2,\ y = 49 + 2 = 51$$

c)

$$(3)^2 - 6\sqrt{x} + x \Rightarrow x = 2,\ y = 9 + 2 = 11$$

d)

$$(5)^2 - 10\sqrt{x} + x \Rightarrow x = 2,\ y = 25 + 2 = 27$$

e)

$$(6)^2 - 12\sqrt{x} + x \Rightarrow x = 2,\ y = 36 + 2 = 38$$

f)

$$(5)^2 - 10\sqrt{x} + x \Rightarrow x = 2,\ y = 25 + 2 = 27$$

Coefficient matching Surd expansion Algebraic reasoning

Exercise 2.58. *Use Pythagoras' theorem. Give your answer as a simplified surd.*

The diagram shows a right-angled triangle with hypotenuse $\sqrt{31}$ and one leg $\sqrt{22}$.

(a) Find the exact value of x. Give your answer in simplest surd form.

View Solution

a)

$$\begin{aligned} x^2 = (\sqrt{31})^2 - (\sqrt{22})^2 = 31 - 22 = 9 \\ x = \sqrt{9} = 3 \end{aligned}$$

Pythagoras theorem Surd simplification Surd arithmetic

Exercise 2.59. *Show your working clearly.*

A trapezium has parallel sides of length 4 cm and k cm, a perpendicular height of $\sqrt{3}$ cm, and an area of $5\sqrt{6}$ cm².

(a) Show that $k = 10\sqrt{2} - 4$.

View Solution

a)

$$\frac{(4+k)\sqrt{3}}{2} = 5\sqrt{6}$$
$$4 + k = 10\sqrt{2}$$
$$k = 10\sqrt{2} - 4$$

Area formula Surd division Rationalise denominator Multi-step simplification

∑ × π ÷ √ ◆ ∞ ± ∫ ≠ Δ

2.4 Test Your Knowledge

3 Unit 3: Mensuration

3.1 Learning Targets

Volume

LT1

Volume — Prisms

Calculate the volume of right prisms using $V = A_{\text{base}} \times l$, including rectangular, triangular, trapezoidal, and parallelogram cross-sections.

Cross-section area Prism formula Mixed cross-sections

LT2

Volume — Cylinders

Calculate the volume of a full cylinder using $V = \pi r^2 h$, given either radius or diameter.

Circle area Radius vs diameter Cylinder formula

LT3

Volume — Portions

Find volumes of partial cylinders: half $\frac{1}{2}\pi r^2 h$, quarter, and sector $\frac{\theta}{360}\pi r^2 h$.

Half cylinder Sector fraction Angle proportion

LT4

Volume — Reverse

Rearrange volume formulas to find unknown dimensions such as length, height, or radius.

Rearrange formula Unknown dimension Algebraic substitution

LT5

Volume — Context

Solve real-world problems involving volume, including converting m^3 to litres.

Word problems Unit conversion m^3 to litres

LT6

Composite Volume

Break complex solids into simpler shapes and add or subtract their volumes.

Decompose solids Add & subtract Multi-step

Surface Area

LT7

SA — Prisms

Calculate surface area of cuboids, triangular and trapezoidal prisms, and open boxes.

Net of prism Face areas Open box

LT8

SA — Cylinders

Calculate surface area of closed cylinders, open-top cylinders, tubes, and half-cylinders.

Curved surface Open vs closed Half cylinder

LT9

SA — Composite

Find surface area of composite solids, including faces that are added or removed at joins.

Composite solids Remove shared faces Multi-step

Advanced & Algebra

LT10

Unit Conversions

Convert between length, area, and volume units using the correct powers of conversion factors.

Length units Area units $\times^2$ Volume units $\times^3$

LT11

Pyramids, Cones & Spheres

Apply $V = \frac{1}{3}Ah$, $V = \frac{1}{3}\pi r^2 h$, $SA = \pi r^2 + \pi rl$, $V = \frac{4}{3}\pi r^3$, and $SA = 4\pi r^2$, including composite solids.

Pyramid Cone Sphere

LT12

Algebraic Volume

Work with algebraic dimensions — expand and simplify expressions, or solve for x given a volume.

Expand & simplify Solve for x Algebraic dimensions

3.2 Formula Cards

Unit 3: Mensuration — Formula Reference

Unit Conversions

 Key Rule When converting units:
• **Length** → multiply or divide by the factor
• **Area** → use the factor **squared**
• **Volume** → use the factor **cubed**

Length chain: km $\xrightarrow{\times 1000}$ m $\xrightarrow{\times 100}$ cm $\xrightarrow{\times 10}$ mm

Area chain: km^2 $\xrightarrow{\times 10^6}$ m^2 $\xrightarrow{\times 10^4}$ cm^2 $\xrightarrow{\times 100}$ mm^2

Volume chain: km^3 $\xrightarrow{\times 10^9}$ m^3 $\xrightarrow{\times 10^6}$ cm^3 $\xrightarrow{\times 10^3}$ mm^3

Capacity: 1 m$^3 = 1000$ L $\quad 1$ cm$^3 = 1$ mL

⚠ **Common Error** 1 m$^2 = 100$ cm^2 ✗ — should be **10 000** cm².
1 m$^3 = 100$ cm^3 ✗ — should be **1 000 000** cm³.

Volume Formulae

 Core Idea For ANY prism or cylinder:

$$V = A_{\text{cross-section}} \times \text{length}$$

Solid	Cross-section	Formula
Rectangular prism	Rectangle	$V = l \times w \times h$
Triangular prism	Triangle	$V = \frac{1}{2}\, b\, h\, l$
Trapezoidal prism	Trapezoid	$V = \frac{1}{2}(a+b)\, h\, l$
Cylinder	Circle	$V = \pi r^2 h$
Half-cylinder	Semicircle	$V = \frac{1}{2}\pi r^2 h$
Quarter-cylinder	Quarter circle	$V = \frac{1}{4}\pi r^2 h$
Cylindrical sector	Sector (θ°)	$V = \dfrac{\theta}{360}\,\pi r^2 h$

Composite solids: $V_{\text{total}} = V_1 + V_2$ (add) or $V_1 - V_2$ (subtract for holes)

⚠ **Common Errors**
- Using diameter instead of radius: $r = d \div 2$ first!

- Missing the $\frac{1}{2}$ for triangular cross-sections.
- Adding volumes when you should subtract (hollow solids).

Surface Area Formulae

 Strategy Surface Area = **sum of areas of ALL outer faces**.
Sketch the solid, label every face, calculate each, then add.

Cuboid (all 6 faces):

$$SA = 2(lw + lh + wh)$$

Open-top box (5 faces — no lid):

$$SA = lw + 2lh + 2wh$$

Triangular prism (5 faces):

$$SA = 2 \times A_{\triangle} + (b + s_1 + s_2) \times l$$

Trapezoidal prism (6 faces):

$$SA = 2 \times \tfrac{1}{2}(a + b)h + (a + b + s_1 + s_2) \times l$$

Closed cylinder (curved + 2 circles):

$$SA = 2\pi rh + 2\pi r^2$$

Open-top cylinder (curved + 1 circle):

$$SA = 2\pi rh + \pi r^2$$

Tube/pipe (curved only, both ends open):

$$SA = 2\pi rh$$

Half-cylinder (3 regions):

$$SA = \underbrace{\pi rh}_{\text{half-curved}} + \underbrace{2rh}_{\text{flat rect}} + \underbrace{\pi r^2}_{\text{2 semicircles}}$$

 Cylinder Unrolled The curved surface of a cylinder unrolls into a rectangle: width $= 2\pi r$ (the circumference), height $= h$.
So curved $SA = 2\pi r \times h = 2\pi rh$.

⚠ **Common Errors — Surface Area**
- Using the full cylinder SA formula when the cylinder is open at one or both ends.
- Forgetting to include **both** triangular ends of a triangular prism.
- Counting an internal join face that is **not** on the outside surface.
- Confusing SA (cm²) with Volume (cm³).

Pyramids, Cones and Spheres

 Key Pattern Pyramids and cones are exactly **⅓** the volume of the prism or cylinder with the same base and height.

Solid	Volume	Surface Area
Pyramid (any base)	$V = \frac{1}{3}A_{\text{base}} \times h$	sum of all faces
Rectangular pyramid	$V = \frac{1}{3}lwh$	
Square pyramid	$V = \frac{1}{3}s^2h$	
Cone	$V = \frac{1}{3}\pi r^2 h$	$SA = \pi r^2 + \pi rl$
Sphere	$V = \frac{4}{3}\pi r^3$	$SA = 4\pi r^2$
Hemisphere	$V = \frac{2}{3}\pi r^3$	$SA = 3\pi r^2$

Cone slant height: $l = \sqrt{r^2 + h^2}$ (Pythagoras — slant is the hypotenuse)
Remember: $r = d \div 2$ always before substituting.

⚠ **Common Errors — Curved Solids**
- Forgetting the $\frac{1}{3}$ in pyramid and cone formulas.
- Squaring r when you should cube it (sphere formula uses r^3).
- Using $\frac{4}{3}$ instead of $\frac{2}{3}$ for hemisphere volume.
- Using the perpendicular height h instead of slant l in cone SA.

Algebraic Volume

 Method Substitute the algebraic expressions into the volume formula, then expand brackets and collect like terms using index laws.

Steps:

1. Write the formula: e.g. $V = \frac{1}{2} \times \text{base} \times h \times l$
2. Substitute each expression: e.g. base $= 2x$, $h = x$, $l = (x+4)$
3. Multiply step by step: $V = \frac{1}{2} \times 2x \times x \times (x+4) = x^2(x+4)$
4. Expand: $V = x^3 + 4x^2$

Index law reminder: $(ax)^2 = a^2x^2$ — square both the coefficient AND the variable.

Example — solve for x: If $V = 3x^3 = 375$, then $x^3 = 125$, so $x = \sqrt[3]{125} = 5$.

⚠ **Common Errors — Algebraic Volume**
- $(2x)^2 = 4x^2$ ✓ not $2x^2$ ✗
- Stopping before expanding all brackets.
- Forgetting $V =$ on the left-hand side of the formula.
- Square rooting when the equation requires a **cube root**.

3.3 Part A — Unit Conversions

 Key Conversion Facts

Dimension	Conversion
Length	$\times 10$: cm→mm · $\times 100$: m→cm · $\times 1000$: km→m
Area	$\times 100$: cm²→mm² · $\times 10{,}000$: m²→cm² · $\times 10^6$: km²→m²
Volume	$\times 1000$: cm³→mm³ · $\times 10^6$: m³→cm³ · $1\ \text{m}^3 = 1000$ L

To go the other way: divide by the same factor.

3.3.1 Easy

Exercise 3.1 (Converting Units of Length)**.** Convert the measurement.

a) $5.1\,\text{cm} = \square\,\text{mm}$

b) $3.2\,\text{cm} = \square\,\text{mm}$

c) $4373\,\text{mm} = \square\,\text{m}$

d) $6.7\,\text{km} = \square\,\text{m}$

e) $1.3\,\text{m} = \square\,\text{cm}$

f) $2342\,\text{mm} = \square\,\text{m}$

View Solution

a) $51\,\text{mm}$

b) $32\,\text{mm}$

c) $4.373\,\text{m}$

d) $6700\,\text{m}$

e) $130\,\text{cm}$

f) $2.342\,\text{m}$

Unit conversion length Multiply by powers of 10

∑ × π ÷ √ ◆ ∞ ± ∫ ≠ Δ

3.3.2 Medium

Exercise 3.2 (Converting Units of Area)**.** Convert the measurement.

a) $3.6\,\text{cm}^2 = \square\,\text{mm}^2$

b) $4029618\,\text{m}^2 = \square\,\text{km}^2$

c) $5.0\,\text{cm}^2 = \square\,\text{mm}^2$

d) $717\,\text{mm}^2 = \square\,\text{cm}^2$

e) $4.5\,\text{m}^2 = \square\,\text{cm}^2$

f) $7.4\,\text{m}^2 = \square\,\text{cm}^2$

View Solution

a) $360\,\text{mm}^2$

b) $4.029618\,\text{km}^2$

c) $500\,\text{mm}^2$

d) $7.17\,\text{cm}^2$

e) $45{,}000\,\text{cm}^2$

f) $74{,}000\,\text{cm}^2$

Unit conversion area Squaring conversion factors

Exercise 3.3 (Converting Units of Volume)**.** Convert the measurement.

a) $3.1\,\text{m}^3 = \square\,\text{cm}^3$

b) $3483\,\text{L} = \square\,\text{m}^3$

c) $3970\,\text{L} = \square\,\text{m}^3$

d) $4743\,\text{mm}^3 = \square\,\text{cm}^3$

e) $295\,\text{cm}^3 = \square\,\text{mL}$

f) $6144958\,\text{cm}^3 = \square\,\text{m}^3$

View Solution

a) $3{,}100{,}000\,\text{cm}^3$

b) $3.483\,\text{m}^3$

c) $3.97\,\text{m}^3$

d) $4.743\,\text{cm}^3$

e) $295\,\text{mL}$

f) $6.144958\,\text{m}^3$

Unit conversion volume Cubing conversion factors

Exercise 3.4 (Mixed Unit Conversions)**.** Convert the measurement.

a) $6.4\,\text{cm}^2 = \square\,\text{mm}^2$

b) $2.7\,\text{m} = \square\,\text{cm}$

c) $3.3\,\text{km}^2 = \square\,\text{m}^2$

d) $7461\,\text{m} = \square\,\text{km}$

e) $4657692\,\text{cm}^3 = \square\,\text{m}^3$

f) $7.0\,\text{cm}^2 = \square\,\text{mm}^2$

 View Solution

a) $640\,\text{mm}^2$

b) $270\,\text{cm}$

c) $3{,}300{,}000\,\text{m}^2$

d) $7.461\,\text{km}$

e) $4.657692\,\text{m}^3$

f) $700\,\text{mm}^2$

Unit conversion length Unit conversion area Unit conversion volume

∑ × π ÷ √ ◆ ∞ ± ∫ ≠ Δ

3.4 Part B — Volume of Prisms and Cylinders

3.4.1 Easy

Exercise 3.5 (Volume of Rectangular Prism)**.** Find the volume of each solid.

a) Cuboid: $l = 14$ cm, $w = 4$ cm, $h = 4$ cm

b) Cuboid: $l = 13$ cm, $w = 3$ cm, $h = 8$ cm

c) Cuboid: $l = 8$ cm, $w = 4$ cm, $h = 2$ cm

d) Cuboid: $l = 9$ cm, $w = 8$ cm, $h = 2$ cm

e) Cuboid: $l = 4$ cm, $w = 4$ cm, $h = 4$ cm

f) Cuboid: $l = 10$ cm, $w = 9$ cm, $h = 5$ cm

View Solution

a) 224 cm^3

b) 312 cm^3

c) 64 cm^3

d) 144 cm^3

e) 64 cm^3

f) 450 cm^3

Volume formula prism Multiplication of integers Unit recognition

Exercise 3.6 (Volume from Cross-Section Area)**.** Find the volume. The cross-section area is given.

a) Prism: $A = 21$ cm^2, $l = 6$ cm

b) Prism: $A = 28$ cm^2, $l = 13$ cm

c) Prism: $A = 29$ cm^2, $l = 8$ cm

d) Prism: $A = 35$ cm^2, $l = 4$ cm

e) Prism: $A = 48$ cm^2, $l = 11$ cm

f) Prism: $A = 39$ cm^2, $l = 9$ cm

View Solution

a) 126 cm^3

b) 364 cm^3

c) 232 cm^3

d) 140 cm^3

e) $528\,\text{cm}^3$

f) $351\,\text{cm}^3$

Volume formula prism Area identification

Exercise 3.7 (Volume of Triangular Prism)**.** Find the volume of each triangular prism.

a) Triangular prism: $b = 4,\ h = 4,\ l = 14$ cm

b) Triangular prism: $b = 6,\ h = 3,\ l = 8$ cm

c) Triangular prism: $b = 4,\ h = 3,\ l = 8$ cm

d) Triangular prism: $b = 7,\ h = 8,\ l = 9$ cm

e) Triangular prism: $b = 7,\ h = 8,\ l = 9$ cm

f) Triangular prism: $b = 5,\ h = 7,\ l = 6$ cm

View Solution

a) $112.0\,\text{cm}^3$

b) $72.0\,\text{cm}^3$

c) $48.0\,\text{cm}^3$

d) $252.0\,\text{cm}^3$

e) $252.0\,\text{cm}^3$

f) $105.0\,\text{cm}^3$

Area triangle Volume formula prism Multi step calculation

Exercise 3.8 (Volume of Cylinder (radius))**.** Find the volume. Round to 1 decimal place.

a) Cylinder: $r = 3$ cm, $h = 4$ cm

b) Cylinder: $r = 7$ cm, $h = 7$ cm

c) Cylinder: $r = 3$ cm, $h = 13$ cm

d) Cylinder: $r = 6$ cm, $h = 13$ cm

e) Cylinder: $r = 2$ cm, $h = 15$ cm

f) Cylinder: $r = 2$ cm, $h = 6$ cm

View Solution

a) $113.1\,\text{cm}^3$

b) $1077.6\,\text{cm}^3$

c) $367.6\,\text{cm}^3$

d) $1470.3\,\text{cm}^3$

e) $188.5\,\text{cm}^3$

f) $75.4\,\text{cm}^3$

Cylinder formula Squaring numbers Use of pi

∑ × π ÷ √ ◆ ∞ ± ∫ ≠ Δ

3.4.2 Medium

Exercise 3.9 (Volume of Trapezoidal Prism)**.** Find the volume of each trapezoidal prism.

a) Trapezoidal prism: $a = 2,\ b = 6,\ h = 3,\ l = 8$ cm

b) Trapezoidal prism: $a = 2,\ b = 4,\ h = 4,\ l = 10$ cm

c) Trapezoidal prism: $a = 5,\ b = 8,\ h = 4,\ l = 7$ cm

d) Trapezoidal prism: $a = 3,\ b = 8,\ h = 2,\ l = 11$ cm

e) Trapezoidal prism: $a = 4,\ b = 7,\ h = 2,\ l = 8$ cm

f) Trapezoidal prism: $a = 5,\ b = 8,\ h = 3,\ l = 9$ cm

View Solution

a) $96.0\,\text{cm}^3$

b) $120.0\,\text{cm}^3$

c) $182.0\,\text{cm}^3$

d) $121.0\,\text{cm}^3$

e) $88.0\,\text{cm}^3$

f) $175.5\,\text{cm}^3$

Area trapezoid Volume formula prism Multi step calculation

Exercise 3.10 (Volume of Parallelogram Prism)**.** Find the volume of each parallelogram prism.

a) Parallelogram prism: $b = 4,\ h = 8,\ l = 10$ cm

b) Parallelogram prism: $b = 9,\ h = 8,\ l = 11$ cm

c) Parallelogram prism: $b = 7,\ h = 7,\ l = 5$ cm

d) Parallelogram prism: $b = 10,\ h = 8,\ l = 7$ cm

e) Parallelogram prism: $b = 4,\ h = 6,\ l = 13$ cm

f) Parallelogram prism: $b = 6,\ h = 3,\ l = 6$ cm

View Solution

a) $320\,\text{cm}^3$

b) $792\,\text{cm}^3$

c) $245\,\text{cm}^3$

d) $560\,\text{cm}^3$

e) $312\,\text{cm}^3$

f) $108\,\text{cm}^3$

Area parallelogram Volume formula prism

Exercise 3.11 (Volume of Cylinder (diameter))**.** Find the volume. Round to 1 decimal place.

a) Cylinder: $d = 7$ cm, $h = 11$ cm

b) Cylinder: $d = 12$ cm, $h = 14$ cm

c) Cylinder: $d = 4$ cm, $h = 5$ cm

d) Cylinder: $d = 4$ cm, $h = 13$ cm

e) Cylinder: $d = 15$ cm, $h = 11$ cm

f) Cylinder: $d = 18$ cm, $h = 5$ cm

View Solution

a) $423.3\,\text{cm}^3$

b) $1583.4\,\text{cm}^3$

c) $62.8\,\text{cm}^3$

d) $163.4\,\text{cm}^3$

e) $1943.9\,\text{cm}^3$

f) $1272.3\,\text{cm}^3$

Cylinder formula Diameter to radius Use of pi

Exercise 3.12 (Volume of Half-Cylinder)**.** Find the volume. Round to 1 decimal place.

a) Half-cylinder: $r = 6$ cm, $l = 6$ cm

b) Half-cylinder: $r = 8$ cm, $l = 8$ cm

c) Half-cylinder: $r = 3$ cm, $l = 10$ cm

d) Half-cylinder: $r = 3$ cm, $l = 7$ cm

e) Half-cylinder: $r = 5$ cm, $l = 7$ cm

f) Half-cylinder: $r = 5$ cm, $l = 5$ cm

View Solution

a) $339.3\,\text{cm}^3$

b) $804.2\,\text{cm}^3$

c) $141.4\,\text{cm}^3$

d) $99.0\,\text{cm}^3$

e) $274.9\,\text{cm}^3$

f) $196.3\,\text{cm}^3$

Cylinder formula Fraction of solid Use of pi

Exercise 3.13 (Volume of Quarter-Cylinder)**.** Find the volume. Round to 1 decimal place.

a) Quarter-cylinder: $r = 6$ cm, $l = 6$ cm

b) Quarter-cylinder: $r = 8$ cm, $l = 6$ cm

c) Quarter-cylinder: $r = 3$ cm, $l = 11$ cm

d) Quarter-cylinder: $r = 9$ cm, $l = 4$ cm

e) Quarter-cylinder: $r = 4$ cm, $l = 7$ cm

f) Quarter-cylinder: $r = 6$ cm, $l = 10$ cm

 View Solution

a) $169.6\,\text{cm}^3$

b) $301.6\,\text{cm}^3$

c) $77.8\,\text{cm}^3$

d) $254.5\,\text{cm}^3$

e) $88.0\,\text{cm}^3$

f) $282.7\,\text{cm}^3$

Cylinder formula Fraction of solid Use of pi

∑ × π ÷ √ ◆ ∞ ± ∫ ≠ Δ

3.4.3 Challenging

Exercise 3.14 (Volume of Cylindrical Sector)**.** Find the volume. Round to 1 decimal place.

a) Sector cylinder: $r = 4,\ l = 8$ cm, $\theta = 90^\circ$

b) Sector cylinder: $r = 8,\ l = 13$ cm, $\theta = 120^\circ$

c) Sector cylinder: $r = 5,\ l = 14$ cm, $\theta = 270^\circ$

d) Sector cylinder: $r = 4,\ l = 4$ cm, $\theta = 120^\circ$

e) Sector cylinder: $r = 3,\ l = 10$ cm, $\theta = 120^\circ$

f) Sector cylinder: $r = 7,\ l = 10$ cm, $\theta = 300^\circ$

View Solution

a) $100.5\,\text{cm}^3$

b) $871.3\,\text{cm}^3$

c) $824.7\,\text{cm}^3$

d) $67.0\,\text{cm}^3$

e) $94.2\,\text{cm}^3$

f) $1282.8\,\text{cm}^3$

Cylinder formula Sector fraction Use of pi Proportional reasoning

∑ × π ÷ √ ◆ ∞ ± ∫ ≠ Δ

3.5 Part C — Composite Solids & Missing Dimensions

3.5.1 Medium

Exercise 3.15 (Composite: L-Shaped Prism)**.** Find the volume of the L-shaped prism.

a)

$$\text{L-prism: } (4 \times 4) \text{ and } (2 \times 4) \text{ cm}, \; l = 11 \text{ cm}$$

b)

$$\text{L-prism: } (4 \times 3) \text{ and } (3 \times 4) \text{ cm}, \; l = 14 \text{ cm}$$

c)

$$\text{L-prism: } (6 \times 4) \text{ and } (4 \times 4) \text{ cm}, \; l = 8 \text{ cm}$$

d)

$$\text{L-prism: } (7 \times 5) \text{ and } (4 \times 3) \text{ cm}, \; l = 11 \text{ cm}$$

e)

$$\text{L-prism: } (8 \times 5) \text{ and } (2 \times 2) \text{ cm}, \; l = 7 \text{ cm}$$

f)

$$\text{L-prism: } (7 \times 6) \text{ and } (3 \times 3) \text{ cm}, \ l = 10 \text{ cm}$$

View Solution

a)

$$264 \text{ cm}^3$$

b)

$$336 \text{ cm}^3$$

c)

$$320 \text{ cm}^3$$

d)

$$517 \text{ cm}^3$$

e)

$$308 \text{ cm}^3$$

f)

$$510 \text{ cm}^3$$

Composite volume Area l shape Multi step calculation

Exercise 3.16 (Find Missing Prism Dimension)**.** Find the missing dimension.

a)

$$\text{Cuboid: } V = 80 \text{ cm}^3, \ w = 4, \ h = 2 \text{ cm. Find } l.$$

b)

$$\text{Cuboid: } V = 160 \text{ cm}^3, \ w = 8, \ h = 5 \text{ cm. Find } l.$$

c)

$$\text{Cuboid: } V = 200 \text{ cm}^3, \ w = 4, \ h = 5 \text{ cm. Find } l.$$

d)

$$\text{Cuboid: } V = 210 \text{ cm}^3, \ w = 6, \ h = 7 \text{ cm. Find } l.$$

e)

$$\text{Cuboid: } V = 48 \text{ cm}^3, \ w = 2, \ h = 3 \text{ cm. Find } l.$$

f)

$$\text{Cuboid: } V = 100 \text{ cm}^3, \ w = 5, \ h = 5 \text{ cm. Find } l.$$

View Solution

a)

$$l = 10 \text{ cm}$$

b)

$$l = 4 \text{ cm}$$

c)

$$l = 10 \text{ cm}$$

d)

$$l = 5 \text{ cm}$$

e)

$$l = 8 \text{ cm}$$

f)

$$l = 4 \text{ cm}$$

Rearranging formula Division of integers Algebraic reasoning

Exercise 3.17 (Find Missing Height — Triangular Prism)**.** Find the value of x.

a)

Triangular prism: $V = 125.0\,\text{cm}^3,\ b = 5$ cm, $l = 10$ cm. Find x.

b)

Triangular prism: $V = 100.0\,\text{cm}^3,\ b = 5$ cm, $l = 10$ cm. Find x.

c)

Triangular prism: $V = 192.5\,\text{cm}^3,\ b = 5$ cm, $l = 11$ cm. Find x.

d)

Triangular prism: $V = 192.0\,\text{cm}^3,\ b = 8$ cm, $l = 12$ cm. Find x.

e)

Triangular prism: $V = 67.5\,\text{cm}^3,\ b = 5$ cm, $l = 9$ cm. Find x.

f)

Triangular prism: $V = 140.0\,\text{cm}^3,\ b = 8$ cm, $l = 7$ cm. Find x.

View Solution

a)
$$x = 5 \text{ cm}$$

b)
$$x = 4 \text{ cm}$$

c)
$$x = 7 \text{ cm}$$

d)
$$x = 4 \text{ cm}$$

e)
$$x = 3 \text{ cm}$$

f)
$$x = 5 \text{ cm}$$

Rearranging formula Division of integers

Exercise 3.18 (Real-World: Water Tank)**.** Solve the word problem. Show all working.

a) Cylindrical tank: $r = 1.8$ m, $h = 1.6$ m. How many litres?

b) Cylindrical tank: $r = 0.8$ m, $h = 2.7$ m. How many litres?

c) Cylindrical tank: $r = 1.1$ m, $h = 3.9$ m. How many litres?

d) Cylindrical tank: $r = 1.3$ m, $h = 2.2$ m. How many litres?

e) Cylindrical tank: $r = 2.1$ m, $h = 4.3$ m. How many litres?

f) Cylindrical tank: $r = 2.0$ m, $h = 1.5$ m. How many litres?

View Solution

a)

$$16{,}290 \text{ L}$$

b)

$$5430 \text{ L}$$

c)

$$14{,}830 \text{ L}$$

d)

$$11{,}680 \text{ L}$$

e)

$$59{,}570 \text{ L}$$

f)

$$18{,}850 \text{ L}$$

Cylinder formula Unit conversion m3 to litres Real world application

Exercise 3.19 (Real-World: Shed Roof)**.** Solve the word problem. Show all working.

a)

Triangular prism roof: $b = 5$ m, $h = 2$ m, $l = 12$ m. Find volume.

b)

Triangular prism roof: $b = 6$ m, $h = 2$ m, $l = 13$ m. Find volume.

c)

Triangular prism roof: $b = 3$ m, $h = 3$ m, $l = 6$ m. Find volume.

d)

Triangular prism roof: $b = 7$ m, $h = 5$ m, $l = 9$ m. Find volume.

e)

Triangular prism roof: $b = 7$ m, $h = 2$ m, $l = 8$ m. Find volume.

f)

Triangular prism roof: $b = 3$ m, $h = 5$ m, $l = 10$ m. Find volume.

 View Solution

a)
$$60.0\,\text{m}^3$$

b)
$$78.0\,\text{m}^3$$

c)
$$27.0\,\text{m}^3$$

d)
$$157.5\,\text{m}^3$$

e)
$$56.0\,\text{m}^3$$

f)
$$75.0\,\text{m}^3$$

Prism formula Real world application

∑ × π ÷ √ ◆ ∞ ± ∫ ≠ Δ

3.5.2 Challenging

Exercise 3.20 (Composite: Prism + Half-Cylinder)**.** Find the total volume.

a)
$$\text{Prism+half-cyl: } l = 14, w = 6, h = 7 \text{ cm}, \ r = 3.0 \text{ cm}$$

b)
$$\text{Prism+half-cyl: } l = 6, w = 8, h = 8 \text{ cm}, \ r = 4.0 \text{ cm}$$

c)
$$\text{Prism+half-cyl: } l = 9, w = 6, h = 3 \text{ cm}, \ r = 3.0 \text{ cm}$$

d)
$$\text{Prism+half-cyl: } l = 12, w = 7, h = 8 \text{ cm}, \ r = 3.5 \text{ cm}$$

e)
$$\text{Prism+half-cyl: } l = 8, w = 4, h = 6 \text{ cm}, \ r = 2.0 \text{ cm}$$

f)

$$\text{Prism+half-cyl: } l = 6, w = 8, h = 4 \text{ cm}, \ r = 4.0 \text{ cm}$$

View Solution

a)

$$785.9 \text{ cm}^3$$

b)

$$534.8 \text{ cm}^3$$

c)

$$289.2 \text{ cm}^3$$

d)

$$902.9 \text{ cm}^3$$

e)

$$242.3 \text{ cm}^3$$

f)

$$342.8 \text{ cm}^3$$

Composite volume Multi step calculation Combining shapes

Exercise 3.21 (Composite: L-Prism)**.** Find the total volume.

a)

$$\text{L-prism: } (4 \times 3) \text{ and } (2 \times 4) \text{ cm}, \ l = 14 \text{ cm}$$

b)

$$\text{L-prism: } (7 \times 5) \text{ and } (4 \times 2) \text{ cm}, \ l = 12 \text{ cm}$$

c)

$$\text{L-prism: } (4 \times 4) \text{ and } (4 \times 4) \text{ cm}, \ l = 10 \text{ cm}$$

d)

$$\text{L-prism: } (10 \times 6) \text{ and } (4 \times 2) \text{ cm}, \ l = 10 \text{ cm}$$

e)

$$\text{L-prism: } (9 \times 3) \text{ and } (4 \times 3) \text{ cm}, \ l = 10 \text{ cm}$$

f)

$$\text{L-prism: } (5 \times 5) \text{ and } (3 \times 3) \text{ cm}, \ l = 14 \text{ cm}$$

View Solution

a)

$$280\,\text{cm}^3$$

b)

$$516\,\text{cm}^3$$

c)

$$320\,\text{cm}^3$$

d)

$$680\,\text{cm}^3$$

e)

$$390\,\text{cm}^3$$

f)

$$476\,\text{cm}^3$$

Composite volume Decomposing shapes Multi step calculation

Exercise 3.22 (Composite: Subtract Cylinder)**.** Find the volume of the solid.

a)

Box$(l = 10, w = 9, h = 8$ cm), hole $r = 2$ cm

b)

Box$(l = 8, w = 10, h = 4$ cm), hole $r = 2$ cm

c)

Box$(l = 12, w = 9, h = 4$ cm), hole $r = 1$ cm

d)

Box$(l = 13, w = 6, h = 6$ cm), hole $r = 3$ cm

e)

Box$(l = 10, w = 8, h = 6$ cm), hole $r = 3$ cm

f)

Box$(l = 12, w = 6, h = 8$ cm), hole $r = 3$ cm

View Solution

a)

$$594.3\,\text{cm}^3$$

b)

$$219.5\,\text{cm}^3$$

c)

$$394.3\,\text{cm}^3$$

d)

$$100.4\,\text{cm}^3$$

e)

$$197.3\,\text{cm}^3$$

f)

$$236.7\,\text{cm}^3$$

Composite volume Subtraction of volumes Use of pi

Exercise 3.23 (Composite: Hollow Cylinder)**.** Find the volume of material.

a)

Hollow pipe: $R = 7,\ r = 3,\ l = 10$ cm

b)

Hollow pipe: $R = 8,\ r = 2,\ l = 17$ cm

c)

Hollow pipe: $R = 5,\ r = 2,\ l = 14$ cm

d)

Hollow pipe: $R = 6,\ r = 1,\ l = 17$ cm

e)

Hollow pipe: $R = 5,\ r = 1,\ l = 18$ cm

f)

Hollow pipe: $R = 6,\ r = 1,\ l = 15$ cm

View Solution

a)

$$1256.7\,\text{cm}^3$$

b)

$$3204.5\,\text{cm}^3$$

c)

$$923.7\,\text{cm}^3$$

d)

$$1869.3\,\text{cm}^3$$

e)

$$1357.2\,\text{cm}^3$$

f)

$$1649.4\,\text{cm}^3$$

Composite volume Subtraction of volumes Use of pi

Exercise 3.24 (Composite: T-Shape Cuboids)**.** Find the total volume of the compound solid.

a)

T-shape: base $(9 \times 7 \times 2)$ cm, top $(7 \times 5 \times 9)$ cm. Find total volume.

b)

T-shape: base $(5 \times 5 \times 5)$ cm, top $(4 \times 3 \times 6)$ cm. Find total volume.

c)

T-shape: base $(12 \times 8 \times 3)$ cm, top $(4 \times 5 \times 8)$ cm. Find total volume.

d)

T-shape: base $(6 \times 9 \times 4)$ cm, top $(2 \times 3 \times 7)$ cm. Find total volume.

e)

T-shape: base $(12 \times 4 \times 3)$ cm, top $(8 \times 2 \times 7)$ cm. Find total volume.

f)

T-shape: base $(8 \times 6 \times 4)$ cm, top $(6 \times 2 \times 6)$ cm. Find total volume.

View Solution

a)

$$441\,\text{cm}^3$$

b)

$$197\,\text{cm}^3$$

c)

$$448\,\text{cm}^3$$

d)

$$258\,\text{cm}^3$$

e)

$$256\,\text{cm}^3$$

f)

$$264\,\text{cm}^3$$

Composite volume Decomposing shapes Multi step calculation

Exercise 3.25 (Find Missing Cylinder Dimension)**.** Find the missing dimension. Round to 1 d.p.

a)

Cylinder: $V = 169.6\,\text{cm}^3,\ h = 6$ cm. Find r.

b)

Cylinder: $V = 615.8\,\text{cm}^3,\ r = 7$ cm. Find h.

c)

Cylinder: $V = 314.2\,\text{cm}^3,\ r = 5$ cm. Find h.

d)

Cylinder: $V = 311.0\,\text{cm}^3,\ h = 11$ cm. Find r.

e)

Cylinder: $V = 150.8\,\text{cm}^3,\ r = 2$ cm. Find h.

f)

$$\text{Cylinder: } V = 1608.5\,\text{cm}^3,\ h = 8\text{ cm. Find } r.$$

View Solution

a)

$$r = 3\text{ cm}$$

b)

$$h = 4\text{ cm}$$

c)

$$h = 4\text{ cm}$$

d)

$$r = 3\text{ cm}$$

e)

$$h = 12\text{ cm}$$

f)

$$r = 8\text{ cm}$$

Rearranging formula Square root Use of pi Algebraic reasoning

Exercise 3.26 (Similar Cylinders — Find Volume)**.** Use the scale factor to find the volume.

a)

$$\text{Similar cylinders A and B: } l_A = 3\text{ cm},\ l_B = 8\text{ cm. } V_A = 58\,\text{cm}^3\text{. Find } V_B.$$

b)

$$\text{Similar cylinders A and B: } l_A = 5\text{ cm},\ l_B = 7\text{ cm. } V_A = 171\,\text{cm}^3\text{. Find } V_B.$$

c)

$$\text{Similar cylinders A and B: } l_A = 6\text{ cm},\ l_B = 9\text{ cm. } V_A = 163\,\text{cm}^3\text{. Find } V_B.$$

d)

$$\text{Similar cylinders A and B: } l_A = 8\text{ cm},\ l_B = 12\text{ cm. } V_A = 144\,\text{cm}^3\text{. Find } V_B.$$

e)

Similar cylinders A and B: $l_A = 3$ cm, $l_B = 5$ cm. $V_A = 57\,\text{cm}^3$. Find V_B.

f)

Similar cylinders A and B: $l_A = 4$ cm, $l_B = 5$ cm. $V_A = 194\,\text{cm}^3$. Find V_B.

View Solution

a)

$$1099.9\,\text{cm}^3$$

b)

$$469.2\,\text{cm}^3$$

c)

$$550.1\,\text{cm}^3$$

d)

$$486.0\,\text{cm}^3$$

e)

$$263.9\,\text{cm}^3$$

f)

$$378.9\,\text{cm}^3$$

Similar solids Scale factor Cubing scale factor

∑ × π ÷ √ ◆ ∞ ± ∫ ≠ Δ

3.6 Part D — Surface Area

3.6.1 Easy

Exercise 3.27 (Surface Area of a Cuboid)**.** Find the total surface area.

a) Cuboid: $l = 10,\ w = 6,\ h = 6$ cm

b) Cuboid: $l = 3,\ w = 3,\ h = 4$ cm

c) Cuboid: $l = 6,\ w = 3,\ h = 4$ cm

d) Cuboid: $l = 7,\ w = 7,\ h = 5$ cm

e) Cuboid: $l = 3,\ w = 6,\ h = 7$ cm

f) Cuboid: $l = 5,\ w = 2,\ h = 7$ cm

View Solution

a) $312\,\text{cm}^2$

b) $66\,\text{cm}^2$

c) $108\,\text{cm}^2$

d) $238\,\text{cm}^2$

e) $162\,\text{cm}^2$

f) $118\,\text{cm}^2$

Sa formula cuboid Counting faces Multiplication of integers

Exercise 3.28 (Surface Area of Open-Top Box)**.** Find the surface area (no lid).

a) Open box: $l = 7,\ w = 7,\ h = 5$ cm

b) Open box: $l = 4,\ w = 4,\ h = 6$ cm

c) Open box: $l = 9,\ w = 3,\ h = 4$ cm

d) Open box: $l = 9,\ w = 4,\ h = 6$ cm

e) Open box: $l = 12,\ w = 3,\ h = 5$ cm

f) Open box: $l = 4,\ w = 5,\ h = 4$ cm

View Solution

a) $189\,\text{cm}^2$

b) $112\,\text{cm}^2$

c) $123\,\text{cm}^2$

d) $192\,\text{cm}^2$

e) $186\,\text{cm}^2$

f) $92\,\text{cm}^2$

Sa formula cuboid Counting faces Omit one face

Exercise 3.29 (Surface Area: Open Tube)**.** Find the curved surface area. Round to 1 d.p.

a) Open tube: $r = 6$ cm, $l = 5$ cm

b) Open tube: $r = 2$ cm, $l = 16$ cm

c) Open tube: $r = 8$ cm, $l = 11$ cm

d) Open tube: $r = 4$ cm, $l = 18$ cm

e) Open tube: $r = 5$ cm, $l = 6$ cm

f) Open tube: $r = 7$ cm, $l = 10$ cm

View Solution

a) $188.5\,\text{cm}^2$

b) $201.1\,\text{cm}^2$

c) $552.9\,\text{cm}^2$

d) $452.4\,\text{cm}^2$

e) $188.5\,\text{cm}^2$

f) $439.8\,\text{cm}^2$

Curved surface area Use of pi

∑ × π ÷ √ ◆ ∞ ± ∫ ≠ Δ

3.6.2 Medium

Exercise 3.30 (Surface Area of a Triangular Prism)**.** Find the total surface area.

a) Tri-prism: $b = 4,\ h = 4,\ s_1 = 6,\ s_2 = 5,\ l = 13$ cm

b) Tri-prism: $b = 5,\ h = 5,\ s_1 = 7,\ s_2 = 5,\ l = 14$ cm

c) Tri-prism: $b = 4,\ h = 6,\ s_1 = 7,\ s_2 = 8,\ l = 6$ cm
d) Tri-prism: $b = 4,\ h = 7,\ s_1 = 5,\ s_2 = 8,\ l = 14$ cm
e) Tri-prism: $b = 3,\ h = 3,\ s_1 = 7,\ s_2 = 8,\ l = 9$ cm
f) Tri-prism: $b = 3,\ h = 4,\ s_1 = 5,\ s_2 = 7,\ l = 12$ cm

View Solution

a) $211.0\,\text{cm}^2$
b) $263.0\,\text{cm}^2$
c) $138.0\,\text{cm}^2$
d) $266.0\,\text{cm}^2$
e) $171.0\,\text{cm}^2$
f) $192.0\,\text{cm}^2$

Area triangle Sa formula prism Counting faces

Exercise 3.31 (Surface Area of a Trapezoidal Prism)**.** Find the total surface area.

a) Trap-prism: $a = 2,\ b = 7,\ h = 4,\ s_1 = 7,\ s_2 = 6,\ l = 9$ cm
b) Trap-prism: $a = 3,\ b = 5,\ h = 2,\ s_1 = 4,\ s_2 = 7,\ l = 11$ cm
c) Trap-prism: $a = 3,\ b = 8,\ h = 2,\ s_1 = 6,\ s_2 = 4,\ l = 5$ cm
d) Trap-prism: $a = 5,\ b = 10,\ h = 3,\ s_1 = 4,\ s_2 = 6,\ l = 8$ cm
e) Trap-prism: $a = 3,\ b = 6,\ h = 4,\ s_1 = 5,\ s_2 = 3,\ l = 7$ cm
f) Trap-prism: $a = 4,\ b = 7,\ h = 3,\ s_1 = 3,\ s_2 = 4,\ l = 7$ cm

View Solution

a) $234.0\,\text{cm}^2$
b) $225.0\,\text{cm}^2$
c) $127.0\,\text{cm}^2$
d) $245.0\,\text{cm}^2$
e) $155.0\,\text{cm}^2$

f) $159.0\,\text{cm}^2$

Area trapezoid Sa formula prism Counting faces

Exercise 3.32 (Surface Area of Closed Cylinder)**.** Find the total surface area. Round to 1 d.p.

a) Closed cylinder: $r = 2$ cm, $h = 11$ cm

b) Closed cylinder: $r = 10$ cm, $h = 7$ cm

c) Closed cylinder: $r = 9$ cm, $h = 7$ cm

d) Closed cylinder: $r = 10$ cm, $h = 3$ cm

e) Closed cylinder: $r = 10$ cm, $h = 6$ cm

f) Closed cylinder: $r = 2$ cm, $h = 12$ cm

View Solution

a) $163.3\,\text{cm}^2$

b) $1068.1\,\text{cm}^2$

c) $904.7\,\text{cm}^2$

d) $816.8\,\text{cm}^2$

e) $1005.3\,\text{cm}^2$

f) $175.9\,\text{cm}^2$

Sa formula cylinder Use of pi Curved surface area

Exercise 3.33 (Surface Area: Open-Top Cylinder)**.** Find the surface area (no lid). Round to 1 d.p.

a) Open-top cylinder: $r = 7$ cm, $h = 8$ cm

b) Open-top cylinder: $r = 3$ cm, $h = 4$ cm

c) Open-top cylinder: $r = 2$ cm, $h = 8$ cm

d) Open-top cylinder: $r = 2$ cm, $h = 8$ cm

e) Open-top cylinder: $r = 3$ cm, $h = 6$ cm

f) Open-top cylinder: $r = 4$ cm, $h = 13$ cm

View Solution

a) $505.8\,\text{cm}^2$

b) $103.7\,\text{cm}^2$

c) $113.1\,\text{cm}^2$

d) $113.1\,\text{cm}^2$

e) $141.4\,\text{cm}^2$

f) $377.0\,\text{cm}^2$

Sa formula cylinder Use of pi Omit one face

∑ × π ÷ √ ◆ ∞ ± ∫ ≠ Δ

3.6.3 Challenging

Exercise 3.34 (Surface Area of L-Shaped Prism)**.** Find the total surface area of the L-prism.

a)
$$\text{L-prism: outer } a = 4, b = 9, \text{ notch } c = 3, d = 2, \ l = 12 \text{ cm}$$

b)
$$\text{L-prism: outer } a = 9, b = 8, \text{ notch } c = 5, d = 3, \ l = 10 \text{ cm}$$

c)
$$\text{L-prism: outer } a = 8, b = 6, \text{ notch } c = 2, d = 4, \ l = 12 \text{ cm}$$

d)
$$\text{L-prism: outer } a = 10, b = 5, \text{ notch } c = 3, d = 4, \ l = 8 \text{ cm}$$

e)
$$\text{L-prism: outer } a = 6, b = 5, \text{ notch } c = 4, d = 3, \ l = 12 \text{ cm}$$

f)
$$\text{L-prism: outer } a = 9, b = 10, \text{ notch } c = 5, d = 3, \ l = 14 \text{ cm}$$

View Solution

a)
$$370\,\text{cm}^2$$

b)
$$444\,\text{cm}^2$$

c)
$$408\,\text{cm}^2$$

d)
$$326\,\text{cm}^2$$

e)
$$316\,\text{cm}^2$$

f)
$$656\,\text{cm}^2$$

Area l shape Sa formula prism Multi step calculation

Exercise 3.35 (Surface Area of Half-Cylinder)**.** Find the total surface area. Round to 1 d.p.

a) Half-cylinder: $r = 4$ cm, $l = 5$ cm
b) Half-cylinder: $r = 5$ cm, $l = 6$ cm
c) Half-cylinder: $r = 3$ cm, $l = 12$ cm
d) Half-cylinder: $r = 5$ cm, $l = 9$ cm
e) Half-cylinder: $r = 7$ cm, $l = 12$ cm
f) Half-cylinder: $r = 3$ cm, $l = 12$ cm

View Solution

a) $153.1\,\text{cm}^2$
b) $232.7\,\text{cm}^2$
c) $213.4\,\text{cm}^2$
d) $309.9\,\text{cm}^2$
e) $585.8\,\text{cm}^2$

f) $213.4\,\text{cm}^2$

Fraction of solid Curved surface area Area rectangle Area semicircle

Exercise 3.36 (SA: Prism + Half-Cylinder)**.** Find the total surface area.

a)
$$\text{House prism: } l = 12,\ w = 6,\ h_{\text{wall}} = 7,\ r = 3.0 \text{ cm}$$

b)
$$\text{House prism: } l = 13,\ w = 7,\ h_{\text{wall}} = 3,\ r = 3.5 \text{ cm}$$

c)
$$\text{House prism: } l = 9,\ w = 7,\ h_{\text{wall}} = 5,\ r = 3.5 \text{ cm}$$

d)
$$\text{House prism: } l = 8,\ w = 7,\ h_{\text{wall}} = 6,\ r = 3.5 \text{ cm}$$

e)
$$\text{House prism: } l = 6,\ w = 8,\ h_{\text{wall}} = 4,\ r = 4.0 \text{ cm}$$

f)
$$\text{House prism: } l = 6,\ w = 4,\ h_{\text{wall}} = 4,\ r = 2.0 \text{ cm}$$

View Solution

a)
$$465.3\,\text{cm}^2$$

b)
$$392.3\,\text{cm}^2$$

c)
$$360.4\,\text{cm}^2$$

d)
$$362.4\,\text{cm}^2$$

e)
$$285.6\,\text{cm}^2$$

f)
$$154.3\,\text{cm}^2$$

Composite sa Multi step calculation Combining shapes

Exercise 3.37 (SA: Cuboid with Hole)**.** Find the total outer surface area.

a)
$$\text{Cuboid}(l = 11, w = 8, h = 5 \text{ cm}) \text{ with hole } r = 1 \text{ cm}$$

b)
$$\text{Cuboid}(l = 7, w = 8, h = 5 \text{ cm}) \text{ with hole } r = 2 \text{ cm}$$

c)
$$\text{Cuboid}(l = 10, w = 6, h = 3 \text{ cm}) \text{ with hole } r = 1 \text{ cm}$$

d)
$$\text{Cuboid}(l = 10, w = 4, h = 4 \text{ cm}) \text{ with hole } r = 1 \text{ cm}$$

e)
$$\text{Cuboid}(l = 10, w = 7, h = 7 \text{ cm}) \text{ with hole } r = 1 \text{ cm}$$

f)
$$\text{Cuboid}(l = 11, w = 4, h = 4 \text{ cm}) \text{ with hole } r = 1 \text{ cm}$$

 View Solution

a)
$$362.9 \text{ cm}^2$$

b)
$$249.4 \text{ cm}^2$$

c)
$$212.9 \text{ cm}^2$$

d)
$$188.9 \text{ cm}^2$$

e)
$$374.9 \text{ cm}^2$$

f)
$$204.9 \text{ cm}^2$$

Composite sa Subtraction of areas Use of pi

Exercise 3.38 (SA: Cylinder + Cone)**.** Find the total surface area. Round to 1 d.p.

a)

$$\text{Cylinder+cone: } d = 12 \text{ cm},\ h_{\text{cyl}} = 7,\ h_{\text{cone}} = 4 \text{ cm}$$

b)

$$\text{Cylinder+cone: } d = 6 \text{ cm},\ h_{\text{cyl}} = 10,\ h_{\text{cone}} = 6 \text{ cm}$$

c)

$$\text{Cylinder+cone: } d = 6 \text{ cm},\ h_{\text{cyl}} = 9,\ h_{\text{cone}} = 7 \text{ cm}$$

d)

$$\text{Cylinder+cone: } d = 10 \text{ cm},\ h_{\text{cyl}} = 14,\ h_{\text{cone}} = 5 \text{ cm}$$

e)

$$\text{Cylinder+cone: } d = 12 \text{ cm},\ h_{\text{cyl}} = 9,\ h_{\text{cone}} = 8 \text{ cm}$$

f)

$$\text{Cylinder+cone: } d = 6 \text{ cm},\ h_{\text{cyl}} = 12,\ h_{\text{cone}} = 5 \text{ cm}$$

View Solution

a)

$$512.9\,\text{cm}^2$$

b)

$$280.0\,\text{cm}^2$$

c)

$$269.7\,\text{cm}^2$$

d)

$$629.4\,\text{cm}^2$$

e)

$$640.9\,\text{cm}^2$$

f)

$$309.4\,\text{cm}^2$$

Composite sa Cone sa formula Curved surface area

∑ × π ÷ √ ◆ ∞ ± ∫ ≠ Δ

3.7 Part E — Pyramids, Cones and Spheres

3.7.1 Easy

Exercise 3.39 (Volume of Rectangular Pyramid)**.** Find the volume. Round to 1 d.p. where needed.

a) Rect pyramid: $l = 10,\ w = 7,\ h = 9$ cm

b) Rect pyramid: $l = 6,\ w = 10,\ h = 9$ cm

c) Rect pyramid: $l = 4,\ w = 8,\ h = 7$ cm

d) Rect pyramid: $l = 11,\ w = 4,\ h = 6$ cm

e) Rect pyramid: $l = 8,\ w = 4,\ h = 9$ cm

f) Rect pyramid: $l = 8,\ w = 6,\ h = 7$ cm

View Solution

a) $210.0\,\text{cm}^3$

b) $180.0\,\text{cm}^3$

c) $74.7\,\text{cm}^3$

d) $88.0\,\text{cm}^3$

e) $96.0\,\text{cm}^3$

f) $112.0\,\text{cm}^3$

Pyramid formula One third factor

Exercise 3.40 (Volume of Square Pyramid)**.** Find the volume.

a) Square pyramid: $s = 4$ cm, $h = 9$ cm

b) Square pyramid: $s = 9$ cm, $h = 14$ cm

c) Square pyramid: $s = 4$ cm, $h = 5$ cm

d) Square pyramid: $s = 8$ cm, $h = 14$ cm

e) Square pyramid: $s = 9$ cm, $h = 4$ cm

f) Square pyramid: $s = 6$ cm, $h = 9$ cm

View Solution

a) $48.0\,\text{cm}^3$

b) $378.0\,\text{cm}^3$

c) $26.7\,\text{cm}^3$

d) $298.7\,\text{cm}^3$

e) $108.0\,\text{cm}^3$

f) $108.0\,\text{cm}^3$

Pyramid formula Squaring numbers One third factor

Exercise 3.41 (Volume of Pyramid (area given))**.** Find the volume.

a) Pyramid: $A = 25\,\text{cm}^2,\ h = 10$ cm

b) Pyramid: $A = 14\,\text{cm}^2,\ h = 7$ cm

c) Pyramid: $A = 36\,\text{cm}^2,\ h = 8$ cm

d) Pyramid: $A = 45\,\text{cm}^2,\ h = 11$ cm

e) Pyramid: $A = 16\,\text{cm}^2,\ h = 8$ cm

f) Pyramid: $A = 47\,\text{cm}^2,\ h = 7$ cm

View Solution

a) $83.3\,\text{cm}^3$

b) $32.7\,\text{cm}^3$

c) $96.0\,\text{cm}^3$

d) $165.0\,\text{cm}^3$

e) $42.7\,\text{cm}^3$

f) $109.7\,\text{cm}^3$

Pyramid formula One third factor

∑ × π ÷ √ ◆ ∞ ± ∫ ≠ Δ

3.7.2 Medium

Exercise 3.42 (Find Missing Pyramid Dimension)**.** Find the missing dimension.

a)
$$\text{Pyramid: } V = 50.0\,\text{cm}^3,\ A = 15\,\text{cm}^2.\ \text{Find } h.$$

b)
$$\text{Pyramid: } V = 60.0\,\text{cm}^3,\ A = 15\,\text{cm}^2.\ \text{Find } h.$$

c)
$$\text{Pyramid: } V = 130.0\,\text{cm}^3,\ A = 39\,\text{cm}^2.\ \text{Find } h.$$

d)
$$\text{Pyramid: } V = 58.7\,\text{cm}^3,\ A = 16\,\text{cm}^2.\ \text{Find } h.$$

e)
$$\text{Pyramid: } V = 45.0\,\text{cm}^3,\ A = 27\,\text{cm}^2.\ \text{Find } h.$$

f)
$$\text{Pyramid: } V = 124.0\,\text{cm}^3,\ A = 31\,\text{cm}^2.\ \text{Find } h.$$

View Solution

a)
$$h = 10 \text{ cm}$$

b)
$$h = 12 \text{ cm}$$

c)
$$h = 10 \text{ cm}$$

d)
$$h = 11 \text{ cm}$$

e)
$$h = 5 \text{ cm}$$

f)
$$h = 12 \text{ cm}$$

Rearranging formula Pyramid formula

Exercise 3.43 (Volume of a Cone)**.** Find the volume. Round to 3 s.f.

a) Cone: $r = 4$ cm, $h = 4$ cm

b) Cone: $r = 7$ cm, $h = 4$ cm

c) Cone: $r = 8$ cm, $h = 11$ cm

d) Cone: $r = 10$ cm, $h = 16$ cm

e) Cone: $r = 5$ cm, $h = 6$ cm

f) Cone: $r = 3$ cm, $h = 6$ cm

View Solution

a) $67.0\,\text{cm}^3$

b) $205.3\,\text{cm}^3$

c) $737.2\,\text{cm}^3$

d) $1675.5\,\text{cm}^3$

e) $157.1\,\text{cm}^3$

f) $56.5\,\text{cm}^3$

Cone formula Use of pi One third factor

Exercise 3.44 (Volume of a Sphere)**.** Find the volume. Round to 3 s.f.

a) Sphere: $r = 6$ cm

b) Sphere: $r = 10$ cm

c) Sphere: $r = 8$ cm

d) Sphere: $r = 9$ cm

e) Sphere: $r = 8$ cm

f) Sphere: $r = 8$ cm

View Solution

a) $904.8\,\text{cm}^3$

b) $4188.8\,\text{cm}^3$

c) $2144.7\,\text{cm}^3$

d) $3053.6\,\text{cm}^3$

e) $2144.7\,\text{cm}^3$

f) $2144.7\,\text{cm}^3$

Sphere formula Use of pi Cubing numbers

Exercise 3.45 (Sphere Volume (diameter given))**.** Find the volume. Round to 1 d.p.

a) Sphere: $d = 11$ cm

b) Sphere: $d = 5$ cm

c) Sphere: $d = 10$ cm

d) Sphere: $d = 5$ cm

e) Sphere: $d = 9$ cm

f) Sphere: $d = 4$ cm

View Solution

a) $696.9\,\text{cm}^3$

b) $65.4\,\text{cm}^3$

c) $523.6\,\text{cm}^3$

d) $65.4\,\text{cm}^3$

e) $381.7\,\text{cm}^3$

f) $33.5\,\text{cm}^3$

Sphere formula Diameter to radius

Exercise 3.46 (Volume of a Hemisphere)**.** Find the volume. Round to 1 d.p.

a) Hemisphere: $r = 3$ cm

b) Hemisphere: $r = 4$ cm

c) Hemisphere: $r = 10$ cm

d) Hemisphere: $r = 3$ cm

e) Hemisphere: $r = 7$ cm

f) Hemisphere: $r = 8$ cm

View Solution

a) $56.5\,\text{cm}^3$

b) $134.0\,\text{cm}^3$

c) $2094.4\,\text{cm}^3$

d) $56.5\,\text{cm}^3$

e) $718.4\,\text{cm}^3$

f) $1072.3\,\text{cm}^3$

Sphere formula Fraction of solid

Exercise 3.47 (Surface Area of a Sphere)**.** Find the surface area. Round to 1 d.p.

a) Sphere: $r = 4$ cm. Find surface area.

b) Sphere: $r = 4$ cm. Find surface area.

c) Sphere: $r = 2$ cm. Find surface area.

d) Sphere: $r = 4$ cm. Find surface area.

e) Sphere: $r = 2$ cm. Find surface area.

f) Sphere: $r = 6$ cm. Find surface area.

View Solution

a) $201.1\,\text{cm}^2$

b) $201.1\,\text{cm}^2$

c) $50.3\,\text{cm}^2$

d) $201.1\,\text{cm}^2$

e) $50.3\,\text{cm}^2$

f) $452.4\,\text{cm}^2$

Sphere sa formula Use of pi

Exercise 3.48 (Surface Area of Sphere (diameter given))**.** Find the surface area. Round to 1 d.p.

a) Sphere: diameter $d = 15$ cm. Find surface area.

b) Sphere: diameter $d = 4$ cm. Find surface area.

c) Sphere: diameter $d = 9$ cm. Find surface area.

d) Sphere: diameter $d = 20$ cm. Find surface area.

e) Sphere: diameter $d = 5$ cm. Find surface area.

f) Sphere: diameter $d = 11$ cm. Find surface area.

View Solution

a) $706.9\,\text{cm}^2$

b) $50.3\,\text{cm}^2$

c) $254.5\,\text{cm}^2$

d) $1256.6\,\text{cm}^2$

e) $78.5\,\text{cm}^2$

f) $380.1\,\text{cm}^2$

Sphere sa formula Diameter to radius Use of pi

Exercise 3.49 (Surface Area of a Hemisphere)**.** Find the total surface area. Round to 1 d.p.

a) Hemisphere: $r = 12$ cm. Find total surface area.

b) Hemisphere: $r = 7$ cm. Find total surface area.

c) Hemisphere: $r = 12$ cm. Find total surface area.

d) Hemisphere: $r = 3$ cm. Find total surface area.

e) Hemisphere: $r = 5$ cm. Find total surface area.

f) Hemisphere: $r = 10$ cm. Find total surface area.

View Solution

a) $1357.2\,\text{cm}^2$

b) $461.8\,\text{cm}^2$

c) $1357.2\,\text{cm}^2$

d) $84.8\,\text{cm}^2$

e) $235.6\,\text{cm}^2$

f) $942.5\,\text{cm}^2$

Sphere sa formula Fraction of solid Area circle

Exercise 3.50 (SA of Cone (slant height given))**.** Find the total surface area. Round to 1 d.p.

a)
$$\text{Cone: base radius } r = 5 \text{ cm, slant height } l = 11 \text{ cm. Find total SA.}$$

b)
$$\text{Cone: base radius } r = 5 \text{ cm, slant height } l = 6 \text{ cm. Find total SA.}$$

c)
$$\text{Cone: base radius } r = 4 \text{ cm, slant height } l = 13 \text{ cm. Find total SA.}$$

d)
$$\text{Cone: base radius } r = 9 \text{ cm, slant height } l = 12 \text{ cm. Find total SA.}$$

e)
$$\text{Cone: base radius } r = 5 \text{ cm, slant height } l = 7 \text{ cm. Find total SA.}$$

f)
$$\text{Cone: base radius } r = 9 \text{ cm, slant height } l = 10 \text{ cm. Find total SA.}$$

View Solution

a)
$$251.3\,\text{cm}^2$$

b)
$$172.8\,\text{cm}^2$$

c)

$$213.6\,\text{cm}^2$$

d)

$$593.8\,\text{cm}^2$$

e)

$$188.5\,\text{cm}^2$$

f)

$$537.2\,\text{cm}^2$$

Cone sa formula Use of pi

∑ × π ÷ √ ◆ ∞ ± ∫ ≠ Δ

3.7.3 Challenging

Exercise 3.51 (Cone Volume using Pythagoras)**.** Find the volume. Round to 3 s.f.

a)

$$\text{Cone: } r = 5 \text{ cm}, \ l = 11.18 \text{ cm. Find volume.}$$

b)

$$\text{Cone: } r = 4 \text{ cm}, \ l = 9.849 \text{ cm. Find volume.}$$

c)

$$\text{Cone: } r = 3 \text{ cm}, \ l = 14.318 \text{ cm. Find volume.}$$

d)

$$\text{Cone: } r = 7 \text{ cm}, \ l = 8.062 \text{ cm. Find volume.}$$

e)

$$\text{Cone: } r = 3 \text{ cm}, \ l = 14.318 \text{ cm. Find volume.}$$

f)

$$\text{Cone: } r = 4 \text{ cm}, \ l = 14.56 \text{ cm. Find volume.}$$

View Solution

a)

$$261.8\,\text{cm}^3$$

b)

$$150.8\,\text{cm}^3$$

c)

$$131.9\,\text{cm}^3$$

d)

$$205.3\,\text{cm}^3$$

e)

$$131.9\,\text{cm}^3$$

f)

$$234.6\,\text{cm}^3$$

Cone formula Pythagoras theorem

Exercise 3.52 (Composite: Cone + Hemisphere)**.** Find the total volume. Round to 3 s.f.

a)

$$\text{Hemisphere}(r = 5) + \text{Cone}(r = 5, h = 13)\ \text{cm}$$

b)

$$\text{Hemisphere}(r = 7) + \text{Cone}(r = 7, h = 12)\ \text{cm}$$

c)

$$\text{Hemisphere}(r = 3) + \text{Cone}(r = 3, h = 9)\ \text{cm}$$

d)

$$\text{Hemisphere}(r = 4) + \text{Cone}(r = 4, h = 11)\ \text{cm}$$

e)

$$\text{Hemisphere}(r = 4) + \text{Cone}(r = 4, h = 6)\ \text{cm}$$

f)

$$\text{Hemisphere}(r = 3) + \text{Cone}(r = 3, h = 5)\ \text{cm}$$

View Solution

a)

$$602.1\,\text{cm}^3$$

b)

$$1334.2\,\text{cm}^3$$

c)

$$141.3\,\text{cm}^3$$

d)

$$318.3\,\text{cm}^3$$

e)

$$234.5\,\text{cm}^3$$

f)

$$103.6\,\text{cm}^3$$

Composite volume Cone formula Sphere formula

Exercise 3.53 (Surface Area of a Cone)**.** Find the total surface area. Round to 1 d.p.

a)

Cone: $r = 6$ cm, $h = 10$ cm. Find total SA.

b)

Cone: $r = 5$ cm, $h = 12$ cm. Find total SA.

c)

Cone: $r = 4$ cm, $h = 7$ cm. Find total SA.

d)

Cone: $r = 9$ cm, $h = 4$ cm. Find total SA.

e)

Cone: $r = 7$ cm, $h = 13$ cm. Find total SA.

f)

Cone: $r = 5$ cm, $h = 14$ cm. Find total SA.

View Solution

a)

$$332.9\,\text{cm}^2$$

b)

$$282.7\,\text{cm}^2$$

c)

$$151.6\,\text{cm}^2$$

d)

$$533.0\,\text{cm}^2$$

e)

$$478.5\,\text{cm}^2$$

f)

$$312.1\,\text{cm}^2$$

Cone sa formula Pythagoras theorem

∑ × π ÷ √ ◆ ∞ ± ∫ ≠ Δ

3.8 Part F — Algebraic Volume

 What to expect

These questions give dimensions as algebraic expressions (e.g. $2x$, $x+3$). Apply the volume formula, expand brackets, and simplify using index laws.

3.8.1 Challenging

Exercise 3.54 (Algebraic Volume — Rectangular Prism)**.** Express V in terms of x.

a)

Rect prism: $l = 4x,\ w = 3x,\ h = 7$ cm. Express V.

b)

Rect prism: $l = 3x,\ w = 3x,\ h = 4$ cm. Express V.

c)

$$\text{Rect prism: } l = 5x,\ w = 3x,\ h = 5 \text{ cm. Express } V.$$

d)

$$\text{Rect prism: } l = 2x,\ w = 2x,\ h = 8 \text{ cm. Express } V.$$

e)

$$\text{Rect prism: } l = 3x,\ w = 2x,\ h = 6 \text{ cm. Express } V.$$

f)

$$\text{Rect prism: } l = 2x,\ w = 2x,\ h = 8 \text{ cm. Express } V.$$

View Solution

a)

$$V = 84x^2 \text{ cm}^3$$

b)

$$V = 36x^2 \text{ cm}^3$$

c)

$$V = 75x^2 \text{ cm}^3$$

d)

$$V = 32x^2 \text{ cm}^3$$

e)

$$V = 36x^2 \text{ cm}^3$$

f)

$$V = 32x^2 \text{ cm}^3$$

Algebraic expressions Index laws

Exercise 3.55 (Algebraic Volume — Triangular Prism)**.** Find V in terms of x. Simplify.

a)

$$\text{Tri-prism: base} = 2x,\ h = x,\ \text{length} = (x+8) \text{ cm. Find } V.$$

b)

$$\text{Tri-prism: base} = 2x,\ h = x,\ \text{length} = (x+2) \text{ cm. Find } V.$$

c)

$$\text{Tri-prism: base} = 2x,\ h = x,\ \text{length} = (x+5) \text{ cm. Find } V.$$

d)

$$\text{Tri-prism: base} = 2x,\ h = x,\ \text{length} = (x+8)\ \text{cm. Find } V.$$

e)

$$\text{Tri-prism: base} = 2x,\ h = x,\ \text{length} = (x+7)\ \text{cm. Find } V.$$

f)

$$\text{Tri-prism: base} = 2x,\ h = x,\ \text{length} = (x+3)\ \text{cm. Find } V.$$

View Solution

a)

$$V = x^3 + 8x^2$$

b)

$$V = x^3 + 2x^2$$

c)

$$V = x^3 + 5x^2$$

d)

$$V = x^3 + 8x^2$$

e)

$$V = x^3 + 7x^2$$

f)

$$V = x^3 + 3x^2$$

Expanding brackets Volume formula prism

Exercise 3.56 (Algebraic Volume — Cylinder)**.** Express V in terms of x. Leave π in answer.

a)

$$\text{Cylinder: } r = 1x \text{ cm},\ h = 4x \text{ cm. Express } V.$$

b)

$$\text{Cylinder: } r = 3x \text{ cm},\ h = 3x \text{ cm. Express } V.$$

c)

$$\text{Cylinder: } r = 3x \text{ cm},\ h = 2x \text{ cm. Express } V.$$

d)

$$\text{Cylinder: } r = 3x \text{ cm}, \ h = 2x \text{ cm. Express } V.$$

e)

$$\text{Cylinder: } r = 2x \text{ cm}, \ h = 3x \text{ cm. Express } V.$$

f)

$$\text{Cylinder: } r = 1x \text{ cm}, \ h = 3x \text{ cm. Express } V.$$

View Solution

a)

$$V = 4\pi x^3 \text{ cm}^3$$

b)

$$V = 27\pi x^3 \text{ cm}^3$$

c)

$$V = 18\pi x^3 \text{ cm}^3$$

d)

$$V = 18\pi x^3 \text{ cm}^3$$

e)

$$V = 12\pi x^3 \text{ cm}^3$$

f)

$$V = 3\pi x^3 \text{ cm}^3$$

Algebraic expressions Cylinder formula

Exercise 3.57 (Solve for x given Volume)**.** Find the value of x.

a)

$$\text{Prism: } V = 192 \text{ cm}^3, \ l = x, \ w = x, \ h = 3x. \text{ Find } x.$$

b)

$$\text{Prism: } V = 128 \text{ cm}^3, \ l = x, \ w = x, \ h = 2x. \text{ Find } x.$$

c)

$$\text{Prism: } V = 81 \text{ cm}^3, \ l = x, \ w = x, \ h = 3x. \text{ Find } x.$$

d)

$$\text{Prism: } V = 192 \text{ cm}^3, \ l = x, \ w = x, \ h = 3x. \text{ Find } x.$$

e)

$$\text{Prism: } V = 162\,\text{cm}^3,\ l = x,\ w = x,\ h = 6x. \text{ Find } x.$$

f)

$$\text{Prism: } V = 48\,\text{cm}^3,\ l = x,\ w = x,\ h = 6x. \text{ Find } x.$$

View Solution

a)

$$x = 4 \text{ cm}$$

b)

$$x = 4 \text{ cm}$$

c)

$$x = 3 \text{ cm}$$

d)

$$x = 4 \text{ cm}$$

e)

$$x = 3 \text{ cm}$$

f)

$$x = 2 \text{ cm}$$

Cube root Rearranging formula

∑ × π ÷ √ ◆ ∞ ± ∫ ≠ Δ

3.9 Test Your Knowledge

4 Unit 4: Statistics

4.1 Learning Targets

LT1

Averages and Range from a List

Calculate mean, median, mode and range from a list. Find missing values given the mean, interpret stem-and-leaf diagrams, and work with unknown values given their ratio and overall mean.

Mean, median, mode Missing value Stem & leaf Ratio problems

LT2

Frequency Tables — Ungrouped

Calculate mean using $\bar{x} = \dfrac{\sum fx}{\sum f}$, find the median using cumulative frequencies, identify the mode, and determine the range of an unknown frequency x that produces a target median.

Mean from table Cumulative frequency Mode Unknown frequency

LT3

Grouped Frequency Tables

Identify modal and median class, estimate the mean using midpoints $\bar{x} \approx \dfrac{\sum fx}{\sum f}$, calculate combined means, and find missing frequencies a and b given total frequency and estimated mean.

Modal class Midpoint estimate Combined mean Missing frequencies

LT4

Cumulative Frequency

Build cumulative frequency tables, read values above or below a threshold, use cumulative frequency graphs to estimate Q_1, Q_2, Q_3 and IQR, and find missing values in a cumulative frequency table.

CF table CF graph Q_1, Q_2, Q_3 IQR

LT5

Box Plots and IQR

Calculate IQR from raw data, draw box-and-whisker plots from a five-number summary, identify outliers using $Q_3 + 1.5 \times$ IQR and $Q_1 - 1.5 \times$ IQR, and find unknowns in ordered lists given conditions on median, mode and range.

IQR Box plot Outliers Five-number summary

4.2 Formula Cards

Statistics — Key Formulas & Rules

Averages and Range

 Pro Tip Always **sort your data first** before finding the median or quartiles.

Mean:

$$\bar{x} = \frac{\sum x}{n} = \frac{\text{sum of all values}}{\text{number of values}}$$

Median: Middle value when data is ordered.
If n is even: mean of the two middle values.

Mode: Most frequently occurring value.

Range:

$$\text{Range} = \text{maximum} - \text{minimum}$$

Example: Data: $3, 7, 7, 9, 14$

$$\bar{x} = \frac{3+7+7+9+14}{5} = 8 \qquad \text{Median} = 7 \qquad \text{Mode} = 7 \qquad \text{Range} = 11$$

Mean from a Frequency Table

 Memory Trick Multiply each value by its frequency first — then divide by the **total frequency**, not the number of rows!

$$\bar{x} = \frac{\sum fx}{\sum f}$$

x	f	fx
2	3	6
4	7	28
6	5	30
Total	**15**	**64**

$$\bar{x} = \frac{64}{15} \approx 4.3$$

Grouped Frequency Tables

⚠ **Common Mistake** Use the **midpoint** of each class — not the lower or upper boundary!

Estimated mean:

$$\bar{x} \approx \frac{\sum fx}{\sum f} \quad \text{where } x = \text{midpoint of class}$$

Modal class: Class with the **highest frequency**.

Median class: Class containing the $\frac{n}{2}$th value (use cumulative frequency).

Example: Class $10 \leq x < 20$, frequency $= 8$ → midpoint $x = 15$, so $fx = 120$

Cumulative Frequency

 Quick Method Read across from the CF axis to the curve, then **down** to the x-axis.

Building CF table: Each row = previous cumulative total $+$ current frequency.

Reading quartiles from a CF graph (total = n):

Measure	Read at CF value
Lower quartile Q_1	$\frac{n}{4}$
Median Q_2	$\frac{n}{2}$
Upper quartile Q_3	$\frac{3n}{4}$

$$\text{IQR} = Q_3 - Q_1$$

Box Plots and Outliers

Five-number summary: Min, Q_1, Median, Q_3, Max

Box plot: Box spans Q_1 to Q_3. Median line inside. Whiskers extend to min and max.

Outlier rule:

$$\text{Outlier if: } x > Q_3 + 1.5 \times \text{IQR} \quad \text{or} \quad x < Q_1 - 1.5 \times \text{IQR}$$

Example: $Q_1 = 35,\ Q_3 = 55,\ \text{IQR} = 20$
Upper fence $= 55 + 1.5 \times 20 = 85$
Any value > 85 is an outlier.

Combined Mean

 Pro Tip You **cannot** just average two means — you must use the totals!

$$\bar{x}_{\text{combined}} = \frac{n_1\bar{x}_1 + n_2\bar{x}_2}{n_1 + n_2}$$

Example: Group 1: $n = 100$, mean $= 18.2$. Group 2: $n = 50$, mean $= 17.3$

$$\bar{x} = \frac{100 \times 18.2 + 50 \times 17.3}{150} = \frac{1820 + 865}{150} = \frac{2685}{150} = 17.9$$

4.3 Part A — Averages and Range

 Key Formulas

$$\bar{x} = \frac{\sum x}{n} \qquad \text{Median} = \text{middle value (sorted)} \qquad \text{Range} = \text{max} - \text{min}$$

4.3.1 Easy

Exercise 4.1 (Mean from a List)**.** Calculate the mean.

a)

$$15,\ 17,\ 20,\ 2,\ 8$$

b)

$$17,\ 10,\ 7,\ 3,\ 18,\ 17,\ 12,\ 4$$

c)

$$13,\ 3,\ 15,\ 6,\ 13,\ 14$$

d)

$$11,\ 10,\ 16,\ 7,\ 11,\ 13,\ 6,\ 16$$

View Solution

a)

$$\bar{x} = 12.4$$

$$\frac{15 + 17 + 20 + 2 + 8}{5} = 12.4$$

b)

$$\bar{x} = 11$$

$$\frac{17 + 10 + 7 + 3 + 18 + 17 + 12 + 4}{8} = 11$$

c)

$$\bar{x} = 10.67$$

$$\frac{13 + 3 + 15 + 6 + 13 + 14}{6} = 10.67$$

d)

$$\bar{x} = 11.25$$

$$\frac{11 + 10 + 16 + 7 + 11 + 13 + 6 + 16}{8} = 11.25$$

Sum of values Division Mean formula

Exercise 4.2 (Median from a List)**.** Find the median.

a)

$$1,\ 2,\ 13,\ 15,\ 19,\ 20$$

b)

$$5,\ 7,\ 8,\ 10,\ 12,\ 18$$

c)

$$3,\ 11,\ 14,\ 15,\ 16,\ 18,\ 19$$

d)

$$2,\ 2,\ 6,\ 8,\ 8,\ 10,\ 14,\ 16,\ 20$$

View Solution

a)

$$\text{Median} = 14$$

$$\text{Mean of positions 3 and 4: } \frac{13+15}{2} = 14$$

b)

$$\text{Median} = 9$$

$$\text{Mean of positions 3 and 4: } \frac{8+10}{2} = 9$$

c)

$$\text{Median} = 15$$

$$\text{Ordered. Position } \frac{n+1}{2} = 4\text{th value} = 15$$

d)

$$\text{Median} = 8$$

$$\text{Ordered. Position } \frac{n+1}{2} = 5\text{th value} = 8$$

Ordering data Locating middle

Exercise 4.3 (Mode from a List)**.** Find the mode.

a)

$$2,\ 11,\ 2,\ 9,\ 2,\ 2$$

b)

$$2,\ 2,\ 2,\ 2,\ 3,\ 8$$

c)

$$5,\ 5,\ 5,\ 10,\ 3,\ 11$$

d)

$$11,\ 3,\ 11,\ 3,\ 3,\ 3$$

View Solution

a)

$$\text{Mode} = 2$$

$$\text{Most frequent value(s): } 2$$

b)

$$\text{Mode} = 2$$

$$\text{Most frequent value(s): } 2$$

c)

$$\text{Mode} = 5$$

$$\text{Most frequent value(s): } 5$$

d)

$$\text{Mode} = 3$$

$$\text{Most frequent value(s): } 3$$

Frequency counting

Exercise 4.4 (Range from a List)**.** Find the range.

a)

$$24,\ 3,\ 12,\ 30,\ 12,\ 9,\ 4,\ 10,\ 16$$

b)

$$20,\ 7,\ 14,\ 29,\ 17,\ 23,\ 30,\ 9$$

c)

$$10,\ 18,\ 17,\ 21,\ 17,\ 4,\ 7,\ 17,\ 23$$

d)

$$30,\ 14,\ 6,\ 15,\ 19,\ 20,\ 13,\ 19$$

View Solution

a)

$$\text{Range} = 27$$

$$30 - 3 = 27$$

b)

$$\text{Range} = 23$$

$$30 - 7 = 23$$

c)

$$\text{Range} = 19$$

$$23 - 4 = 19$$

d)

$$\text{Range} = 24$$

$$30 - 6 = 24$$

Max min Subtraction

∑ × π ÷ √ ◆ ∞ ± ∫ ≠ Δ

4.3.2 Medium

Exercise 4.5 (Mean, Median, Mode and Range)**.** Find the mean, median, mode and range.

a)

$$4,\ 8,\ 13,\ 3,\ 8,\ 9,\ 10,\ 8,\ 9$$

b)

$$3,\ 9,\ 15,\ 14,\ 15,\ 15,\ 11$$

c)

$$13,\ 12,\ 7,\ 10,\ 10,\ 7,\ 7,\ 8,\ 4$$

d)

$$5,\ 15,\ 6,\ 15,\ 8,\ 8,\ 13,\ 8$$

View Solution

a)

$$\text{Mean} = 8,\ \text{Median} = 8,\ \text{Mode} = 8,\ \text{Range} = 10$$

$$\bar{x} = \frac{72}{9} = 8 \quad \text{Median} = 8 \quad \text{Mode} = 8 \quad \text{Range} = 10$$

b)

$$\text{Mean} = 11.71,\ \text{Median} = 14,\ \text{Mode} = 15,\ \text{Range} = 12$$

$$\bar{x} = \frac{82}{7} = 11.71 \quad \text{Median} = 14 \quad \text{Mode} = 15 \quad \text{Range} = 12$$

c)

$$\text{Mean} = 8.67,\ \text{Median} = 8,\ \text{Mode} = 7,\ \text{Range} = 9$$

$$\bar{x} = \frac{78}{9} = 8.67 \quad \text{Median} = 8 \quad \text{Mode} = 7 \quad \text{Range} = 9$$

d)

$$\text{Mean} = 9.75,\ \text{Median} = 8,\ \text{Mode} = 8,\ \text{Range} = 10$$

$$\bar{x} = \frac{78}{8} = 9.75 \quad \text{Median} = 8 \quad \text{Mode} = 8 \quad \text{Range} = 10$$

All averages Ordering data

∑ × π ÷ √ ◆ ∞ ± ∫ ≠ Δ

4.3.3 Challenging

Exercise 4.6 (Find Missing Value / Updated Mean)**.** Find the missing value or new mean.

a)

$$\text{Mean} = 14, \quad n = 6 \text{Numbers: } 14,\ 8,\ 13,\ 7,\ 4,\ x$$

b)

$$\text{Mean} = 26, \quad \text{three values: } 30,\ 35,\ 30 \text{Other two in ratio } 1 : 4$$

c)

$$\text{Mean} = 11, \quad n = 5 \text{Numbers: } 15,\ 14,\ 8,\ 4,\ x$$

d)

$$\text{Mean of } 6 \text{ numbers} = 3, \quad \text{new value} = 12$$

View Solution

a)

$$x = 38$$

$$\text{Total} = 14 \times 6 = 84 \quad x = 84 - (14 + 8 + 13 + 7 + 4) = 38$$

b)

$$7 \text{ and } 28$$

Step 1: Total $= 26 \times 5 = 130$Step 2: Remaining $= 130 - (30 + 35 + 30) = 35$Step 3: Ratio 1 :

c)

$$x = 14$$

$$\text{Total} = 11 \times 5 = 55 \quad x = 55 - (15 + 14 + 8 + 4) = 14$$

d)

$$\text{New mean} = 4.29$$

$$\text{Old total} = 3 \times 6 = 18 \quad \text{New total} = 18 + 12 = 30 \quad \bar{x}_{\text{new}} = \frac{30}{7} = 4.29$$

Rearranging formula Sum from mean

Exercise 4.7 (Averages from Stem & Leaf)**.** Use the stem and leaf diagram to find the median, range and modal class.

a) Find the median.

b) Find the range.

c) Find the mode.

View Solution

a) Median $= 31$

$$\text{Ordered: } 27 \text{ values, middle} = 31$$

b) Range $= 47$

$$59 - 12 = 47$$

c) Mode $= 17$

$$\text{Most frequent value: } 17$$

Reading stem leaf Ordering data

Exercise 4.8 (Averages from Stem & Leaf)**.** Use the stem and leaf diagram to find the median, range and modal class.

a) Find the median.

b) Find the range.

c) Find the mode.

View Solution

a) Median $= 37$

$$\text{Ordered: } 15 \text{ values, middle} = 37$$

b) Range $= 35$

$$58 - 23 = 35$$

c) Mode $= 23$

$$\text{Most frequent value: } 23$$

Reading stem leaf Ordering data

Exercise 4.9 (Averages from Stem & Leaf)**.** Use the stem and leaf diagram to find the median, range and modal class.

a) Find the median.

b) Find the range.

c) Find the mode.

View Solution

a) Median $= 60$

$$\text{Ordered: } 11 \text{ values, middle} = 60$$

b) Range $= 36$

$$77 - 41 = 36$$

c) Mode $= 60$

Most frequent value: 60

Reading stem leaf Ordering data

∑ × π ÷ √ ◆ ∞ ± ∫ ≠ Δ

4.4 Part B — Frequency Tables

 Key Formula

$$\bar{x} = \frac{\sum fx}{\sum f}$$

4.4.1 Easy

Exercise 4.10 (Mean from Frequency Table)**.** Calculate the mean from the frequency table.

a)

x	2	3	5	7	9	10
f	3	1	7	7	2	2

b)

x	4	6	8	9
f	4	4	6	6

c)

x	1	2	4	7	8
f	4	10	4	1	4

d)

x	2	4	5	6
f	5	5	5	3

View Solution

a)

$$\bar{x} = 5.95$$

$$\frac{2 \times 3 + 3 \times 1 + 5 \times 7 + 7 \times 7 + 9 \times 2 + 10 \times 2}{22} = \frac{131}{22} = 5.95$$

b)

$$\bar{x} = 7.1$$

$$\frac{4 \times 4 + 6 \times 4 + 8 \times 6 + 9 \times 6}{20} = \frac{142}{20} = 7.1$$

c)

$$\bar{x} = 3.43$$

$$\frac{1 \times 4 + 2 \times 10 + 4 \times 4 + 7 \times 1 + 8 \times 4}{23} = \frac{79}{23} = 3.43$$

d)

$$\bar{x} = 4.06$$

$$\frac{2 \times 5 + 4 \times 5 + 5 \times 5 + 6 \times 3}{18} = \frac{73}{18} = 4.06$$

Fx column Sum of frequencies

Exercise 4.11 (Mode from Frequency Table)**.** Find the mode from the frequency table.

a)

x	1	2	3	4	5	9
f	1	2	10	2	2	6

b)

x	1	2	5	7	8
f	4	4	9	9	1

c)

x	1	3	8	9
f	3	7	5	2

d)

x	2	4	7	9	10
f	2	5	3	9	7

View Solution

a)

$$\text{Mode} = 3$$

$$\text{Highest frequency is } 10\text{, giving mode(s): } 3$$

b)

$$\text{Mode} = 5, 7$$

$$\text{Highest frequency is } 9\text{, giving mode(s): } 5, 7$$

c)

$$\text{Mode} = 3$$

d)

Highest frequency is 7, giving mode(s): 3

Mode $= 9$

Highest frequency is 9, giving mode(s): 9

Reading frequency table

∑ × π ÷ √ ◆ ∞ ± ∫ ≠ Δ

4.4.2 Medium

Exercise 4.12 (Median from Frequency Table)**.** Find the median from the frequency table.

a)

x	1	2	5	6	7	10
f	10	4	1	1	10	5

b)

x	2	3	6	8	9
f	5	9	7	3	3

c)

x	2	3	7	9	10
f	2	10	1	9	4

d)

x	1	2	3	9	10
f	1	6	7	8	9

View Solution

a)

$$\text{Median} = 6$$

$$n = 31 \quad \text{Middle position } \frac{n+1}{2} = 16.0 \quad \text{Median} = 6$$

b)

$$\text{Median} = 3$$

$$n = 27 \quad \text{Middle position } \frac{n+1}{2} = 14.0 \quad \text{Median} = 3$$

c)

$$\text{Median} = 8$$

$$n = 26 \quad \text{Middle position } \frac{n+1}{2} = 13.5 \quad \text{Median} = 8$$

d)

$$\text{Median} = 9$$

$$n = 31 \quad \text{Middle position } \frac{n+1}{2} = 16.0 \quad \text{Median} = 9$$

Cumulative frequency Locating middle

Exercise 4.13 (Mean, Median & Mode from Table)**.** Find the mean, median and mode from the frequency table.

a)

Calculate the mean.

b)

Find the median.

c)

(c) Find the mode.

View Solution

a)

$$\bar{x} = 4.58$$

$$\bar{x} = \frac{\sum fx}{\sum f} = \frac{1 \times 3 + 2 \times 4 + 4 \times 5 + 8 \times 7}{19} = \frac{87}{19} = 4.58$$

b)

$$\text{Median} = 4$$

$$n = 19 \quad \text{Position } \frac{n+1}{2} = 10.0 \quad \text{Build CF: } 1 : CF = 3,\ 2 : CF = 7,\ 4 : CF = 12,\ 8 : CF$$

c)

$$\text{Mode} = 8$$

Highest frequency = 7 at $x = 8$

Fx column Cumulative frequency Reading frequency table

∑ × π ÷ √ ◆ ∞ ± ∫ ≠ Δ

4.4.3 Challenging

Exercise 4.14 (Find x Given Median (Frequency Table))**.** Find the largest and smallest possible values of x such that the median is as given.

a)

$$\text{Median} = 6$$

Score	5	6	7	8
Freq	3	7	x	5

b)

$$\text{Median} = 6$$

Score	5	6	7	8
Freq	5	9	x	6

c)

$$\text{Median} = 6$$

Score	5	6	7	8
Freq	4	5	x	4

d)

$$\text{Median} = 6$$

Score	5	6	7	8
Freq	5	5	x	7

View Solution

a)

$$x_{\text{min}} = 0,\ x_{\text{max}} = 4$$

For median = 6: cumulative freq up to 6 $\geq \dfrac{n+1}{2}$ Largest: $3+7-5-1 = 4$ Smallest: 0

b)

$$x_{\text{min}} = 0,\ x_{\text{max}} = 7$$

For median = 6: cumulative freq up to 6 $\geq \dfrac{n+1}{2}$ Largest: $5+9-6-1 = 7$ Smallest: 0

c)

$$x_{\text{min}} = 0,\ x_{\text{max}} = 4$$

For median = 6: cumulative freq up to 6 $\geq \dfrac{n+1}{2}$ Largest: $4+5-4-1 = 4$ Smallest: 0

d)

$$x_{\text{min}} = 0,\ x_{\text{max}} = 2$$

For median = 6: cumulative freq up to 6 $\geq \dfrac{n+1}{2}$ Largest: $5+5-7-1 = 2$ Smallest: 0

Cumulative frequency Inequality reasoning

∑ × π ÷ √ ◆ ∞ ± ∫ ≠ Δ

4.5 Part C — Grouped Frequency Tables

 Key Points

Use the **midpoint** of each class interval for estimated mean: $\bar{x} \approx \dfrac{\sum fx}{\sum f}$

Modal class = class with highest frequency. **Median class** = class containing the $\frac{n}{2}$th value.

4.5.1 Easy

Exercise 4.15 (Estimated Mean (Grouped))**.** Estimate the mean from the grouped frequency table.

a)

Class	Midpoint	Frequency
$19 \leq x < 27$	23	3
$27 \leq x < 35$	31	9
$35 \leq x < 43$	39	11
$43 \leq x < 51$	47	4
$51 \leq x < 59$	55	15
$59 \leq x < 67$	63	15

b)

Class	Midpoint	Frequency
$14 \leq x < 18$	16	17
$18 \leq x < 22$	20	3
$22 \leq x < 26$	24	19
$26 \leq x < 30$	28	10
$30 \leq x < 34$	32	3
$34 \leq x < 38$	36	5

c)

Class	Midpoint	Frequency
$29 \leq x < 37$	33	15
$37 \leq x < 45$	41	14
$45 \leq x < 53$	49	20
$53 \leq x < 61$	57	5

d)

Class	Midpoint	Frequency
$10 \leq x < 15$	12.5	11
$15 \leq x < 20$	17.5	16
$20 \leq x < 25$	22.5	6
$25 \leq x < 30$	27.5	11
$30 \leq x < 35$	32.5	20

View Solution

a)

Estimated mean ≈ 48

$$\frac{23 \times 3 + 31 \times 9 + 39 \times 11 + 47 \times 4 + 55 \times 15 + 63 \times 15}{57} = \frac{2735}{57} \approx 48$$

b)

Estimated mean ≈ 23.6

$$\frac{16 \times 17 + 20 \times 3 + 24 \times 19 + 28 \times 10 + 32 \times 3 + 36 \times 5}{57} = \frac{1344}{57} \approx 23.6$$

c)

$$\text{Estimated mean} \approx 43.2$$

$$\frac{33 \times 15 + 41 \times 14 + 49 \times 20 + 57 \times 5}{54} = \frac{2334}{54} \approx 43.2$$

d)

$$\text{Estimated mean} \approx 23.5$$

$$\frac{12.5 \times 11 + 17.5 \times 16 + 22.5 \times 6 + 27.5 \times 11 + 32.5 \times 20}{64} = \frac{1505}{64} \approx 23.5$$

Midpoints Fx column

Exercise 4.16 (Modal Class (Grouped))**.** Write down the modal class.

a)

Class	Frequency
$20 \leq x < 27$	7
$27 \leq x < 34$	12
$34 \leq x < 41$	12
$41 \leq x < 48$	11

b)

Class	Frequency
$7 \leq x < 13$	10
$13 \leq x < 19$	13
$19 \leq x < 25$	6
$25 \leq x < 31$	3
$31 \leq x < 37$	16
$37 \leq x < 43$	19

c)

Class	Frequency
$10 \leq x < 19$	8
$19 \leq x < 28$	18
$28 \leq x < 37$	13
$37 \leq x < 46$	19
$46 \leq x < 55$	17

d)

Class	Frequency
$15 \leq x < 24$	13
$24 \leq x < 33$	5
$33 \leq x < 42$	3
$42 \leq x < 51$	6
$51 \leq x < 60$	18
$60 \leq x < 69$	18

 View Solution

a)

$$\text{Modal class: } 27 \leq x < 34$$

$$\text{Highest frequency} = 12\text{, class: } 27 \leq x < 34$$

b)

$$\text{Modal class: } 37 \leq x < 43$$

$$\text{Highest frequency} = 19\text{, class: } 37 \leq x < 43$$

c)

$$\text{Modal class: } 37 \leq x < 46$$

$$\text{Highest frequency} = 19\text{, class: } 37 \leq x < 46$$

d)

$$\text{Modal class: } 51 \leq x < 60$$

$$\text{Highest frequency} = 18\text{, class: } 51 \leq x < 60$$

Reading grouped table

∑ × π ÷ √ ◆ ∞ ± ∫ ≠ Δ

4.5.2 Medium

Exercise 4.17 (Median Class (Grouped))**.** Identify the median class.

a)

Class	Frequency
$20 \leq x < 27$	11
$27 \leq x < 34$	8
$34 \leq x < 41$	8
$41 \leq x < 48$	8
$48 \leq x < 55$	19

b)

Class	Frequency
$18 \leq x < 28$	10
$28 \leq x < 38$	9
$38 \leq x < 48$	18
$48 \leq x < 58$	14
$58 \leq x < 68$	16

c)

Class	Frequency
$14 \leq x < 23$	15
$23 \leq x < 32$	15
$32 \leq x < 41$	12
$41 \leq x < 50$	18
$50 \leq x < 59$	14
$59 \leq x < 68$	12

d)

Class	Frequency
$9 \leq x < 19$	8
$19 \leq x < 29$	12
$29 \leq x < 39$	19
$39 \leq x < 49$	13
$49 \leq x < 59$	3
$59 \leq x < 69$	9

View Solution

a)

$$\text{Median class: } 34 \leq x < 41$$

$$n = 54 \quad \frac{n}{2} = 27 \quad \text{Median class: } 34 \leq x < 41$$

b)

$$\text{Median class: } 38 \leq x < 48$$

c)

$$n = 67 \quad \frac{n}{2} = 33.5 \quad \text{Median class: } 38 \leq x < 48$$

Median class: $41 \leq x < 50$

d)

$$n = 86 \quad \frac{n}{2} = 43 \quad \text{Median class: } 41 \leq x < 50$$

Median class: $29 \leq x < 39$

$$n = 64 \quad \frac{n}{2} = 32 \quad \text{Median class: } 29 \leq x < 39$$

Cumulative frequency Locating middle

∑ × π ÷ √ ◆ ∞ ± ∫ ≠ Δ

4.5.3 Challenging

Exercise 4.18 (Modal, Median & Mean (Grouped))**.** Find the modal class, median class, and estimated mean.

a)

Write down the modal class.

b)

Identify the median class.

c)

Estimate the mean.

View Solution

a)

Modal class: $24 \leq x < 33$

Highest frequency = 20 → modal class: $24 \leq x < 33$

b)

Median class: $33 \leq x < 42$

$n = 50 \quad \frac{n}{2} = 25$ Build CF to find class containing position 25 Median class: $33 \leq x < 42$

c)

$$\bar{x} \approx 35.2$$

$$\bar{x} = \frac{\sum fx}{\sum f} = \frac{19.5 \times 4 + 28.5 \times 20 + 37.5 \times 11 + 46.5 \times 15}{50} = \frac{1758}{50} \approx 35.2$$

Midpoints Cumulative frequency Reading grouped table

Exercise 4.19 (Combined Mean (Two Groups))**.** Find the combined mean of the two groups.

a)

$$n_1 = 50,\ \bar{x}_1 = 17.8 n_2 = 50,\ \bar{x}_2 = 10.1$$

b)

$$n_1 = 120,\ \bar{x}_1 = 23.1 n_2 = 50,\ \bar{x}_2 = 13.7$$

c)

$$n_1 = 80,\ \bar{x}_1 = 23.3 n_2 = 20,\ \bar{x}_2 = 15.5$$

d)

$$n_1 = 50,\ \bar{x}_1 = 23.9 n_2 = 50,\ \bar{x}_2 = 21.7$$

View Solution

a)

$$\text{Combined mean} = 13.95$$

$$\frac{50 \times 17.8 + 50 \times 10.1}{50 + 50} = \frac{890.0 + 505.0}{100} = \frac{1395.0}{100} = 13.95$$

b)

$$\text{Combined mean} = 20.34$$

$$\frac{120 \times 23.1 + 50 \times 13.7}{120 + 50} = \frac{2772.0 + 685.0}{170} = \frac{3457.0}{170} = 20.34$$

c)

$$\text{Combined mean} = 21.74$$

$$\frac{80 \times 23.3 + 20 \times 15.5}{80 + 20} = \frac{1864.0 + 310.0}{100} = \frac{2174.0}{100} = 21.74$$

d)

$$\text{Combined mean} = 22.8$$

$$\frac{50 \times 23.9 + 50 \times 21.7}{50 + 50} = \frac{1195.0 + 1085.0}{100} = \frac{2280.0}{100} = 22.8$$

Weighted mean Total from mean

Exercise 4.20 (Find Missing Frequencies (Grouped))**.** Find the values of a and b.

a)

$$n = 100, \quad \bar{x} \approx 43.5$$

Class	Frequency
$20 \leq x < 30$	15
$30 \leq x < 40$	25
$40 \leq x < 50$	35
$50 \leq x < 60$	a
$60 \leq x < 70$	b

b)

$$n = 100, \quad \bar{x} \approx 43.5$$

Class	Frequency
$20 \leq x < 30$	15
$30 \leq x < 40$	25
$40 \leq x < 50$	35
$50 \leq x < 60$	a
$60 \leq x < 70$	b

c)

$$n = 100, \quad \bar{x} \approx 43.5$$

Class	Frequency
$20 \leq x < 30$	15
$30 \leq x < 40$	25
$40 \leq x < 50$	35
$50 \leq x < 60$	a
$60 \leq x < 70$	b

d)

$$n = 150, \quad \bar{x} \approx 48$$

Class	Frequency
$20 \leq x < 30$	13
$30 \leq x < 40$	15
$40 \leq x < 50$	40
$50 \leq x < 60$	a
$60 \leq x < 70$	b

View Solution

a)

$$a = 10, \; b = 15$$

Step 1: $a+b = 100-75 = 25$Step 2: $\Sigma fx = 43.5\times100 = 4350$Known $\Sigma fx = 2825 \quad \Rightarrow 5$

b)

$$a = 10, \; b = 15$$

Step 1: $a+b = 100-75 = 25$Step 2: $\Sigma fx = 43.5\times100 = 4350$Known $\Sigma fx = 2825 \quad \Rightarrow 5$

c)

$$a = 10, \; b = 15$$

Step 1: $a+b = 100-75 = 25$Step 2: $\Sigma fx = 43.5\times100 = 4350$Known $\Sigma fx = 2825 \quad \Rightarrow 5$

d)

$$a = 78, \; b = 4$$

Step 1: $a+b = 150-68 = 82$Step 2: $\Sigma fx = 48\times150 = 7200$Known $\Sigma fx = 2650 \quad \Rightarrow 55a$

Simultaneous equations Mean formula

∑ × π ÷ √ ◆ ∞ ± ∫ ≠ Δ

4.6 Part D — Cumulative Frequency

 Key Points

Measure	Position on CF axis
Median (Q_2)	$\frac{n}{2}$
Lower quartile (Q_1)	$\frac{n}{4}$
Upper quartile (Q_3)	$\frac{3n}{4}$

IQR $= Q_3 - Q_1$

4.6.1 Easy

Exercise 4.21 (Build Cumulative Frequency Table)**.** Complete the cumulative frequency table.

a)

Class	f
$28 \leq x < 37$	5
$37 \leq x < 46$	11
$46 \leq x < 55$	13
$55 \leq x < 64$	25
$64 \leq x < 73$	24
$73 \leq x < 82$	25

View Solution

a)

x	Cum. Freq
$x < 37$	5
$x < 46$	16
$x < 55$	29
$x < 64$	54
$x < 73$	78
$x < 82$	103

Running totals: $5, 16, 29, 54, 78, 103$

Running totals

Exercise 4.22 (Read Cumulative Frequency Table)**.** Use the table to answer the questions.

a)

x	Cum. Freq
$x < 39$	12
$x < 54$	30
$x < 69$	34
$x < 84$	54
$x < 99$	62

(a) Items ≤ 54?(b) Items > 54?

View Solution

a)

$$\text{(a) } 30 \quad \text{(b) } 32$$

$$\text{(a) CF at } 54 = 30 \quad \text{(b) } 62 - 30 = 32$$

Reading cf table Subtraction

∑ × π ÷ √ ◆ ∞ ± ∫ ≠ Δ

4.6.2 Medium

Exercise 4.23 (IQR from Cumulative Frequency)**.** Use the cumulative frequency graph to estimate the IQR.

a)

Complete the CF table.

b)

Estimate the median from the graph.

c)

Estimate the IQR.

d)

How many items are above 95?

View Solution

a)

x	Cum. Freq
$x < 35$	18
$x < 54$	23
$x < 73$	32
$x < 92$	55
$x < 111$	69
$x < 130$	76
$x < 149$	88

Running totals: $18, 23, 32, 55, 69, 76, 88$

b)

Median ≈ 82.9

Read at $\frac{n}{2} = \frac{88}{2} = 44$ on CF axis Median ≈ 82.9

c)

IQR $= 56.7$

$Q_1 \approx 50.2$ (read at 22.0) $Q_3 \approx 106.9$ (read at 66.0) IQR $= 106.9 - 50.2 = 56.7$

d)

≈ 31 items

Read CF at $95 \approx 57$ Above: $88 - 57 = 31$

Interpolation Quartile location

∑ × π ÷ √ ◆ ∞ ± ∫ ≠ Δ

4.6.3 Challenging

Exercise 4.24 (Estimate Frequency Above a Value)**.** Estimate the number of items above the given value.

a)

x	CF
$x < 32$	25
$x < 50$	45
$x < 68$	52
$x < 86$	69
$x < 104$	74
$x < 122$	88

How many items above 93?

View Solution

a)

$$\approx 17$$

Read CF at $93 \approx 71 \quad 88 - 71 = 17$

Interpolation Reading cf graph

Exercise 4.25 (Find Missing CF Values A and B)**.** Find the values of A and B in the cumulative frequency table.

a)

Find A and B.

x	Cum. Freq
$x < 21$	9
$x < 28$	A
$x < 35$	42
$x < 42$	B
$x < 49$	73

View Solution

a)

$$A = 29,\ B = 55$$

$$A = 9 + 20 = 29 \quad B = 42 + 13 = 55$$

Running totals Algebraic reasoning

Exercise 4.26 (Cumulative Frequency — Full Analysis)**.** Use the frequency table and cumulative frequency graph to answer the questions.

a)

Complete the cumulative frequency table.

b) Estimate the median from the graph.

c) Estimate Q_1 and Q_3.

d) Calculate the IQR.

e) Estimate how many items are above 57.

 View Solution

a)

x	CF
$x < 39$	19
$x < 50$	36
$x < 61$	56
$x < 72$	62
$x < 83$	71

Running totals: $19, 36, 56, 62, 71$

b) Median ≈ 49.7

$$\text{Read at } \frac{n}{2} = \frac{71}{2} = 35 \quad \Rightarrow \text{Median} \approx 49.7$$

c) $Q_1 \approx 38.3, \quad Q_3 \approx 59.5$

$$Q_1\text{: read at } \frac{n}{4} = 17.8 \Rightarrow Q_1 \approx 38.3 Q_3\text{: read at } \frac{3n}{4} = 53.2 \Rightarrow Q_3 \approx 59.5$$

d) IQR $= 21.2$

$$\text{IQR} = Q_3 - Q_1 = 59.5 - 38.3 = 21.2$$

e) ≈ 22 items

$$\text{Read CF at } 57 \approx 49 \quad 71 - 49 = 22$$

Cumulative frequency Interpolation Quartile location Iqr calculation Reading cf graph

∑ × π ÷ √ ◆ ∞ ± ∫ ≠ Δ

4.7 Part E — Box Plots and IQR

Key Points

$$\text{IQR} = Q_3 - Q_1$$

Outlier rule: A value is an outlier if it is $> Q_3 + 1.5 \times \text{IQR}$ or $< Q_1 - 1.5 \times \text{IQR}$

4.7.1 Easy

Exercise 4.27 (IQR from Raw Data)**.** Find the interquartile range.

a) $4,\ 9,\ 9,\ 10,\ 12,\ 12,\ 13,\ 17,\ 19,\ 22,\ 23,\ 24,\ 26,\ 29,\ 30$

View Solution

a) IQR $= 14$

$$Q_1 = 10,\ Q_3 = 24 \quad \text{IQR} = 24 - 10 = 14$$

Quartile calculation Subtraction

∑ × π ÷ √ ◆ ∞ ± ∫ ≠ Δ

4.7.2 Medium

Exercise 4.28 (Draw Box Plot from 5-Number Summary)**.** Find the IQR and draw a box plot.

a)

$$\text{min} = 10,\ Q_1 = 13,\ \text{med} = 16,\ Q_3 = 20,\ \text{max} = 24$$

View Solution

a)

$$\text{IQR} = 7$$

$$\text{IQR} = Q_3 - Q_1 = 20 - 13 = 7$$

Five number summary Box plot construction

Exercise 4.29 (Find Unknown Values from Box Plot Conditions)**.** Find the unknown values using the given conditions.

a)

$$h,\ 5,\ 5,\ 7,\ j,\ 15,\ k,\ k\text{Median} = 8,\ \text{Mode} = 20,\ \text{Range} = 17$$

View Solution

a)

$$h = 3,\ j = 9,\ k = 20$$

$k = 20$ (mode, appears twice) $h = k - 17 = 3$ (range) $j = 2 \times 8 - 7 = 9$ (median)

Simultaneous conditions Median rule Mode rule Range rule

Exercise 4.30 (Read a Box Plot)**.** Use the box plot to answer the questions.

a) Write down the minimum value.

b) Write down Q_1.

c) Write down the median.

d) Write down Q_3.

e) Calculate the IQR.

View Solution

a) Min $= 15$

Read the left whisker end: 15

b) $Q_1 = 21$

Read the left edge of the box: 21

c) Median $= 27$

Read the line inside the box: 27

d) $Q_3 = 35$

Read the right edge of the box: 35

e) IQR $= 14$

$$\text{IQR} = Q_3 - Q_1 = 35 - 21 = 14$$

Reading box plot Quartile identification Iqr calculation

∑ × π ÷ √ ◆ ∞ ± ∫ ≠ Δ

4.7.3 Challenging

Exercise 4.31 (Identify Outliers (1.5 × IQR Rule))**.** Identify outliers using the 1.5 × IQR rule and draw a box plot.

a)

$$14,\ 22,\ 24,\ 30,\ 31,\ 37,\ 48,\ 48,\ 65,\ 67,\ 107$$

View Solution

a)

$Q_1 = 24,\ Q_3 = 65,\ \text{IQR} = 41$Upper fence $= 126.5107 > 126.5 \Rightarrow$ outlier

Step 1: $Q_1 = 24,\ Q_3 = 65$Step 2: IQR $= Q_3 - Q_1 = 65 - 24 = 41$Step 3: Upper fence $= Q$

Outlier rule Quartile calculation

∑ × π ÷ √ ◆ ∞ ± ∫ ≠ Δ

4.8 Test Your Knowledge

5 Unit 5: Linear Equations

5.1 Learning Targets

LT1

One-Step Equations

Solve one-step equations using inverse operations — addition, subtraction, multiplication, and division.

Inverse operations Add & subtract Multiply & divide

LT2

Two-Step Equations

Solve two-step equations including those with brackets and fractions by applying two inverse operations in sequence.

Two inverse steps Brackets Fractions

LT3

Unknowns on Both Sides

Solve equations where the unknown appears on both sides by collecting like terms first.

Collect like terms Rearrange Balance method

LT4

Expand and Solve

Expand single and double brackets then solve the resulting linear equation.

Single brackets Double brackets Simplify then solve

LT5

Algebraic Fractions

Solve equations involving algebraic fractions by clearing denominators or using cross-multiplication.

Clear fractions Cross-multiply LCD method

LT6

Form and Solve from Context

Translate word problems into equations and solve them — including number riddles, ages, geometry, and formula problems.

Number problems Age problems Geometry & formulas

5.2 Formula Card

Linear Equations — Solving Strategies

The Balance Method

 Golden Rule Whatever you do to one side, you **must** do to the other side.

One-step

$$x + 5 = 12 \;\Rightarrow\; x = 12 - 5 = 7$$
$$3x = 21 \;\Rightarrow\; x = 21 \div 3 = 7$$

Two-step — undo addition/subtraction *first*, then multiplication/division

$$2x + 3 = 11 \;\Rightarrow\; 2x = 8 \;\Rightarrow\; x = 4$$

Unknowns on Both Sides

 Strategy Move all x terms to one side, all numbers to the other.

Method:

$$5x + 2 = 2x + 8$$
$$5x - 2x = 8 - 2$$
$$3x = 6 \;\Rightarrow\; x = 2$$

Expanding Brackets

⚠ **Remember** The number outside multiplies **every** term inside the bracket.

Single bracket:

$$3(2x+5) = 27 \;\Rightarrow\; 6x+15 = 27 \;\Rightarrow\; x = 2$$

Two brackets:

$$\begin{gathered} 2(x+3) + 4(x-1) = 18 \\ 2x + 6 + 4x - 4 = 18 \\ 6x + 2 = 18 \;\Rightarrow\; x = \frac{8}{3} \end{gathered}$$

Algebraic Fractions

 Key Move Multiply every term by the denominator (or LCM) to clear fractions first.

Single fraction:

$$\frac{2x+1}{3} = 5 \;\Rightarrow\; 2x+1 = 15 \;\Rightarrow\; x = 7$$

Fractions on both sides — cross multiply:

$$\frac{x+1}{2} = \frac{x+3}{4} \;\Rightarrow\; 4(x+1) = 2(x+3) \;\Rightarrow\; x = 1$$

Adding fractions — use LCM:

$$\frac{x}{2} + \frac{x}{3} = 5 \;\xrightarrow{\times 6}\; 3x + 2x = 30 \;\Rightarrow\; x = 6$$

Forming Equations from Words

Step-by-step:

1. Define your variable — *let* $x = \ldots$
2. Translate each condition into algebra
3. Form the equation
4. Solve and check your answer makes sense

Example: Three consecutive integers sum to 51.

$$x + (x+1) + (x+2) = 51 \;\Rightarrow\; 3x + 3 = 51 \;\Rightarrow\; x = 16$$

Numbers: 16, 17, 18. ✓

5.3 Practice Problems

5.3.1 Easy

Exercise 5.1.

a) $t - 9 = -14$

b) $t - 4 = 0$

c) $y - 8 = -9$

d) $t + 11 = 19$

e) $x + 6 = 12$

f) $x - 12 = -10$

g) $y - 5 = 17$

h) $p - 12 = 20$

i) $y + 10 = 19$

j) $m - 5 = 13$

k) $p - 4 = -15$

l) $y + 10 = 12$

View Solutions

a) $t - 9 = -14$

$$t = -5$$

b) $t - 4 = 0$

$$t = 4$$

c) $y - 8 = -9$

$$y = -1$$

d) $t + 11 = 19$

$$t = 8$$

e) $x + 6 = 12$

$$x = 6$$

f) $x - 12 = -10$

$$x = 2$$

g) $y - 5 = 17$

$$y = 22$$

h) $p - 12 = 20$

$$p = 32$$

i) $y + 10 = 19$

$$y = 9$$

j) $m - 5 = 13$

$$m = 18$$

k) $p - 4 = -15$

$$p = -11$$

l) $y + 10 = 12$

$$y = 2$$

Exercise 5.2.

a) $6n = -36$

b) $7y = -49$

c) $8x = -56$

d) $7n = 63$

e) $2n = 26$

f) $10m = -70$

g) $3m = 21$

h) $12n = 108$

i) $11n = -44$

j) $3t = -27$

k) $5t = 70$

l) $3n = -9$

View Solutions

a) $6n = -36$

$$n = -6$$

b) $7y = -49$

$$y = -7$$

c) $8x = -56$

$$x = -7$$

d) $7n = 63$

$$n = 9$$

e) $2n = 26$

$$n = 13$$

f) $10m = -70$

$$m = -7$$

g) $3m = 21$

$$m = 7$$

h) $12n = 108$

$$n = 9$$

i) $11n = -44$

$$n = -4$$

j) $3t = -27$

$$t = -9$$

k) $5t = 70$

$$t = 14$$

l) $3n = -9$

$$n = -3$$

Exercise 5.3.

a) $\frac{x}{8} = -2$

b) $\frac{m}{7} = -5$

c) $\frac{n}{7} = -4$

d) $\frac{t}{6} = 12$

e) $\frac{t}{3} = 9$

f) $\frac{t}{4} = 7$

g) $\frac{t}{5} = -5$

h) $\frac{m}{8} = -2$

View Solutions

a) $\frac{x}{8} = -2$

$$x = -16$$

b) $\frac{m}{7} = -5$

$$m = -35$$

c) $\frac{n}{7} = -4$

$$n = -28$$

d) $\frac{t}{6} = 12$

$$t = 72$$

e) $\frac{t}{3} = 9$

$$t = 27$$

f) $\frac{t}{4} = 7$

$$t = 28$$

g) $\frac{t}{5} = -5$

$$t = -25$$

h) $\frac{m}{8} = -2$

$$m = -16$$

Exercise 5.4.

a) $10t - 3 = -73$

b) $2y + 2 = -10$

c) $7y - 4 = 10$

d) $9t - 6 = -42$

e) $10y + 7 = 107$

f) $8m + 1 = -31$

g) $9p - 8 = -53$

h) $4y + 3 = 19$

View Solutions

a) $10t - 3 = -73$

$$\begin{aligned} 10t &= -70 \\ t &= -7 \end{aligned}$$

b) $2y + 2 = -10$

$$\begin{aligned} 2y &= -12 \\ y &= -6 \end{aligned}$$

c) $7y - 4 = 10$

$$\begin{aligned} 7y &= 14 \\ y &= 2 \end{aligned}$$

d) $9t - 6 = -42$

$$\begin{aligned} 9t &= -36 \\ t &= -4 \end{aligned}$$

e) $10y + 7 = 107$

$$\begin{aligned} 10y &= 100 \\ y &= 10 \end{aligned}$$

f) $8m + 1 = -31$

$$\begin{aligned} 8m &= -32 \\ m &= -4 \end{aligned}$$

g) $9p - 8 = -53$

$$\begin{aligned} 9p &= -45 \\ p &= -5 \end{aligned}$$

h) $4y + 3 = 19$

$$\begin{aligned} 4y &= 16 \\ y &= 4 \end{aligned}$$

∑ × π ÷ √ ◆ ∞ ± ∫ ≠ Δ

5.3.2 Medium

Exercise 5.5.

a) $\dfrac{m+6}{6} = 2$

b) $\dfrac{n-8}{6} = 1$

c) $\dfrac{n+2}{8} = -1$

d) $\dfrac{t+8}{7} = 2$

e) $\dfrac{x+1}{8} = 1$

f) $\dfrac{p+2}{2} = -4$

View Solutions

a) $\dfrac{m+6}{6} = 2$

$$\begin{aligned} m+6 &= 12 \\ m &= 6 \end{aligned}$$

b) $\dfrac{n-8}{6} = 1$

$$\begin{aligned} n-8 &= 6 \\ n &= 14 \end{aligned}$$

c) $\dfrac{n+2}{8} = -1$

$$\begin{aligned} n+2 &= -8 \\ n &= -10 \end{aligned}$$

d) $\dfrac{t+8}{7} = 2$

$$\begin{aligned} t+8 &= 14 \\ t &= 6 \end{aligned}$$

e) $\dfrac{x+1}{8} = 1$

$$\begin{aligned} x+1 &= 8 \\ x &= 7 \end{aligned}$$

f) $\dfrac{p+2}{2} = -4$

$$\begin{aligned} p+2 &= -8 \\ p &= -10 \end{aligned}$$

Exercise 5.6.

a) $4(x+1) = 0$

b) $3(x-6) = -36$

c) $5(t-6) = 15$

d) $3(y+7) = 48$

e) $4(y+8) = 76$

f) $3(m-2) = -3$

View Solutions

a) $4(x+1) = 0$

$$\begin{aligned} x+1 &= 0 \\ x &= -1 \end{aligned}$$

b) $3(x-6) = -36$

$$\begin{aligned} x - 6 &= -12 \\ x &= -6 \end{aligned}$$

c) $5(t-6) = 15$

$$\begin{aligned} t - 6 &= 3 \\ t &= 9 \end{aligned}$$

d) $3(y+7) = 48$

$$\begin{aligned} y + 7 &= 16 \\ y &= 9 \end{aligned}$$

e) $4(y+8) = 76$

$$\begin{aligned} y + 8 &= 19 \\ y &= 11 \end{aligned}$$

f) $3(m-2) = -3$

$$\begin{aligned} m - 2 &= -1 \\ m &= 1 \end{aligned}$$

Exercise 5.7.

a) $7m + 6 = 8m$

b) $5x - 8 = 4x - 6$

c) $10x + 8 = 4x + 2$

d) $3x - 3 = x - 15$

e) $7x + 6 = 2x + 1$

f) $9n + 7 = 4n - 13$

View Solutions

a) $7m + 6 = 8m$

$$\begin{aligned} -m + 6 &= 0 \\ -m &= -6 \\ m &= 6 \end{aligned}$$

b) $5x - 8 = 4x - 6$

$$\begin{aligned} x - 8 &= -6 \\ x &= 2 \end{aligned}$$

c) $10x + 8 = 4x + 2$

$$\begin{aligned} 6x + 8 &= 2 \\ 6x &= -6 \\ x &= -1 \end{aligned}$$

d) $3x - 3 = x - 15$

$$\begin{aligned} 2x - 3 &= -15 \\ 2x &= -12 \\ x &= -6 \end{aligned}$$

e) $7x + 6 = 2x + 1$

$$\begin{aligned} 5x + 6 &= 1 \\ 5x &= -5 \\ x &= -1 \end{aligned}$$

f) $9n + 7 = 4n - 13$

$$\begin{aligned} 5n + 7 &= -13 \\ 5n &= -20 \\ n &= -4 \end{aligned}$$

Exercise 5.8.

a) $5(t+6) = 3(t+12)$

b) $8(m-6) = 7(m-7)$

c) $3(y+1) = 6(y-2)$

d) $4(y-3) = 5(y-1)$

View Solutions

a) $5(t+6) = 3(t+12)$

$$\begin{aligned} 2t + 30 &= 36 \\ 2t &= 6 \\ t &= 3 \end{aligned}$$

b) $8(m-6) = 7(m-7)$

$$\begin{aligned} m - 48 &= -49 \\ m &= -1 \end{aligned}$$

c) $3(y+1) = 6(y-2)$

$$\begin{aligned} -3y + 3 &= -12 \\ -3y &= -15 \\ y &= 5 \end{aligned}$$

d) $4(y-3) = 5(y-1)$

$$\begin{aligned} -y - 12 &= -5 \\ -y &= 7 \\ y &= -7 \end{aligned}$$

Exercise 5.9.

a) $6(2t-6) = -24$

b) $5(5y+7) = 35$

c) $2(3m+4) = -28$

d) $4(5m+1) = 144$

e) $6(5t-4) = -24$

f) $3(2n-7) = 3$

View Solutions

a) $6(2t-6) = -24$

$$\begin{aligned} 12t - 36 &= -24 \\ 12t &= 12 \\ t &= 1 \end{aligned}$$

b) $5(5y+7) = 35$

$$\begin{aligned} 25y + 35 &= 35 \\ 25y &= 0 \\ y &= 0 \end{aligned}$$

c) $2(3m+4) = -28$

$$\begin{aligned} 6m + 8 &= -28 \\ 6m &= -36 \\ m &= -6 \end{aligned}$$

d) $4(5m+1) = 144$

$$\begin{aligned} 20m + 4 &= 144 \\ 20m &= 140 \\ m &= 7 \end{aligned}$$

e) $6(5t-4) = -24$

$$\begin{aligned} 30t - 24 &= -24 \\ 30t &= 0 \\ t &= 0 \end{aligned}$$

f) $3(2n-7)=3$

$$\begin{aligned} 6n-21&=3\\ 6n&=24\\ n&=4 \end{aligned}$$

Exercise 5.10.

a) $\dfrac{5p+7}{2}=16$

b) $\dfrac{4x+4}{4}=1$

c) $\dfrac{4m+2}{5}=2$

d) $\dfrac{4m-6}{5}=-6$

e) $\dfrac{6x-8}{4}=7$

f) $\dfrac{2p+8}{3}=0$

View Solutions

a) $\dfrac{5p+7}{2}=16$

$$\begin{aligned} 5p+7&=32\\ 5p&=25\\ p&=5 \end{aligned}$$

b) $\dfrac{4x+4}{4}=1$

$$\begin{aligned} 4x+4&=4\\ 4x&=0\\ x&=0 \end{aligned}$$

c) $\dfrac{4m+2}{5} = 2$

$$\begin{aligned} 4m + 2 &= 10 \\ 4m &= 8 \\ m &= 2 \end{aligned}$$

d) $\dfrac{4m-6}{5} = -6$

$$\begin{aligned} 4m - 6 &= -30 \\ 4m &= -24 \\ m &= -6 \end{aligned}$$

e) $\dfrac{6x-8}{4} = 7$

$$\begin{aligned} 6x - 8 &= 28 \\ 6x &= 36 \\ x &= 6 \end{aligned}$$

f) $\dfrac{2p+8}{3} = 0$

$$\begin{aligned} 2p + 8 &= 0 \\ 2p &= -8 \\ p &= -4 \end{aligned}$$

Exercise 5.11.

Sara buys tickets. Adult tickets cost \$7 each and child tickets cost \$4 each. The total cost is \$29. Sara bought 3 adult tickets.

(i) Write a formula for the total cost T.

(ii) How many child tickets were bought?

View Solution

$$\begin{aligned} \text{(i) } T &= 7a + 4c \\ \text{(ii) } 21 + 4c &= 29 \\ 4c &= 8 \\ c &= 2 \end{aligned}$$

Answer: $T = 7a + 4c,\ c = 2$

Exercise 5.12.

The sum of four consecutive integers is 70.

(i) Let the first integer be x. Write an equation.

(ii) Find the numbers.

View Solution

$$\begin{aligned} \text{(i) } x + (x+1) + (x+2) + (x+3) &= 70 \\ \text{(ii) } 4x + 6 &= 70 \\ x &= 16 \\ \text{Numbers: } 16, 17, 18, 19 \end{aligned}$$

Answer: $x = 16,$ numbers: $16, 17, 18, 19$

Exercise 5.13.

Ibrahim is x years old. Yusuf is 6 years older than Ibrahim. The sum of their ages is 54.

(i) Form an equation in x.

(ii) Find the ages of Ibrahim and Yusuf.

View Solution

$$\text{(i) } x + (x + 6) = 54$$
$$\text{(ii) } 2x + 6 = 54$$
$$2x = 48$$
$$x = 24$$
Ibrahim = 24, Yusuf = 30

Answer: $x = 24,\ Ibrahim = 24,\ Yusuf = 30$

Exercise 5.14.

The angles of a triangle are $(x + 13)^\circ$, $(2x + 14)^\circ$ and $(3x + 45)^\circ$.

- Angles in a triangle sum to 180°.

(i) Form an equation in x.

(ii) Find x and each angle.

View Solution

$$\text{(i) } 6x + 72 = 180$$
$$\text{(ii) } x = 18$$
$$\text{Angles: } 31^\circ,\ 50^\circ,\ 99^\circ$$

Answer: $x = 18,\ 31^\circ,\ 50^\circ,\ 99^\circ$

Exercise 5.15.

The lengths, in cm, of the sides of the triangle are $3(x + 2)$, $2x - 2$ and $2x + 7$.

(i) Write an expression for the perimeter of the triangle.

(ii) The perimeter is 32 cm. Find x.

View Solution

$$\text{(i) } (3(x+2)) + (2x-2) + (2x+7) = 7x + 11 \text{ cm}$$
$$\text{(ii) } 7x + 11 = 32$$
$$7x = 21$$
$$x = 3$$

Answer: $(i)\ 7x + 11$ cm, $\quad (ii)\ x = 3$

Exercise 5.16.

4(2x + 3)4x - 34x + 1Diagram NOT accurately drawn

The lengths, in cm, of the sides of the triangle are $4(2x+3)$, $4x-3$ and $4x+1$.

(i) Write an expression for the perimeter of the triangle.

(ii) The perimeter is 58 cm. Find x.

View Solution

$$\text{(i) } (4(2x+3)) + (4x-3) + (4x+1) = 16x + 10 \text{ cm}$$
$$\text{(ii) } 16x + 10 = 58$$
$$16x = 48$$
$$x = 3$$

Answer: $(i)\ 16x + 10$ cm, $\quad (ii)\ x = 3$

Exercise 5.17.

(2x + 16)°(3x + 13)°(3x + 47)°Diagram NOT accurately drawn

The angles of a triangle are $(2x+16)^\circ$, $(3x+13)^\circ$ and $(3x+47)^\circ$.

- Angles in a triangle sum to 180°.

(i) Form an equation in x.

(ii) Find x and each angle.

View Solution

$$\text{(i) } 8x + 76 = 180$$
$$\text{(ii) } x = 13$$
$$\text{Angles: } 42^\circ,\ 52^\circ,\ 86^\circ$$

Answer: $x = 13,\ 42^\circ,\ 52^\circ,\ 86^\circ$

Exercise 5.18.

The angles of a triangle are $(3x + 22)^\circ$, $(2x + 24)^\circ$ and $(x + 74)^\circ$.

- Angles in a triangle sum to 180°.

(i) Form an equation in x.

(ii) Find x and each angle.

View Solution

$$\text{(i) } 6x + 120 = 180$$
$$\text{(ii) } x = 10$$
$$\text{Angles: } 52^\circ,\ 44^\circ,\ 84^\circ$$

Answer: $x = 10,\ 52^\circ,\ 44^\circ,\ 84^\circ$

Exercise 5.19.

4x + 6 cmx + 4 cmDiagram NOT accurately drawn

A rectangle has length $(4x + 6)$ cm and width $(x + 4)$ cm.

(i) Write an expression for the perimeter.

(ii) The perimeter is 90 cm. Find x.

View Solution

$$\text{(i) } 2(4x+6+x+4) = 10x+20 \text{ cm}$$
$$\text{(ii) } 10x+20 = 90$$
$$10x = 70$$
$$x = 7$$

Answer: $(i)\ 10x+20$ cm, $\quad (ii)\ x=7$

Exercise 5.20.

A rectangle has length $(2x+8)$ cm and width $(x+6)$ cm.

(i) Write an expression for the perimeter.

(ii) The perimeter is 64 cm. Find x.

View Solution

$$\text{(i) } 2(2x+8+x+6) = 6x+28 \text{ cm}$$
$$\text{(ii) } 6x+28 = 64$$
$$6x = 36$$
$$x = 6$$

Answer: $(i)\ 6x+28$ cm, $\quad (ii)\ x=6$

Exercise 5.21.

2x - 4x + 2x + 5Diagram NOT accurately drawn

An isosceles triangle has equal sides $(2x-4)$ cm and $(x+2)$ cm. The base is $(x+5)$ cm.

(i) Find x.

(ii) Find the perimeter of the triangle.

View Solution

$$\text{(i) } 2x - 4 = x + 2$$
$$x = 6$$
$$\text{(ii) Perimeter} = 27 \text{ cm}$$

Answer: $x = 6,\ P = 27$ cm

Exercise 5.22.

An isosceles triangle has equal sides $(3x + 5)$ cm and $(x + 17)$ cm. The base is $(2x + 2)$ cm.

(i) Find x.

(ii) Find the perimeter of the triangle.

View Solution

$$\text{(i) } 3x + 5 = x + 17$$
$$2x = 12$$
$$x = 6$$
$$\text{(ii) Perimeter} = 60 \text{ cm}$$

Answer: $x = 6,\ P = 60$ cm

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5.3.3 Challenging

Exercise 5.23.

a) $2(4x + 2) + 3(x + 2) = -34$

b) $2(y + 4) + 3(4y - 5) = 49$

c) $2(y + 4) + 4(3y - 3) = 66$

d) $4(2t - 2) + 5(2t + 4) = 102$

View Solutions

a) $2(4x+2)+3(x+2)=-34$

$$\begin{aligned} 8x+4+3x+6 &= -34 \\ 11x+10 &= -34 \\ 11x &= -44 \\ x &= -4 \end{aligned}$$

b) $2(y+4)+3(4y-5)=49$

$$\begin{aligned} 2y+8+12y-15 &= 49 \\ 14y-7 &= 49 \\ 14y &= 56 \\ y &= 4 \end{aligned}$$

c) $2(y+4)+4(3y-3)=66$

$$\begin{aligned} 2y+8+12y-12 &= 66 \\ 14y-4 &= 66 \\ 14y &= 70 \\ y &= 5 \end{aligned}$$

d) $4(2t-2)+5(2t+4)=102$

$$\begin{aligned} 8t-8+10t+20 &= 102 \\ 18t+12 &= 102 \\ 18t &= 90 \\ t &= 5 \end{aligned}$$

Exercise 5.24.

a) $(n+3)^2=(n+4)(n+1)$

b) $(3y+4)^2=(3y+5)(3y+2)$

c) $(2y+4)^2=(2y+6)(2y+3)$

d) $(2m+2)^2 = (2m+1)(2m+4)$

View Solutions

a) $(n+3)^2 = (n+4)(n+1)$

Expand both sides, $1n^2$ cancels:

$$\begin{aligned} 6n+9 &= 5n+4 \\ 1n &= -5 \\ n &= -5 \end{aligned}$$

b) $(3y+4)^2 = (3y+5)(3y+2)$

Expand both sides, $9y^2$ cancels:

$$\begin{aligned} 24y+16 &= 21y+10 \\ 3y &= -6 \\ y &= -2 \end{aligned}$$

c) $(2y+4)^2 = (2y+6)(2y+3)$

Expand both sides, $4y^2$ cancels:

$$\begin{aligned} 16y+16 &= 18y+18 \\ -2y &= 2 \\ y &= -1 \end{aligned}$$

d) $(2m+2)^2 = (2m+1)(2m+4)$

Expand both sides, $4m^2$ cancels:

$$\begin{aligned} 8m+4 &= 10m+4 \\ -2m &= 0 \\ m &= 0 \end{aligned}$$

Exercise 5.25.

a) $5(3y+2)+4(3y-1) = -2(3y+5)$

b) $5(3p+1)+2(3p+1) = -2(2p+6)$

c) $4(3y+1)+4(y-2)=-2(y+5)$

d) $3(2n-1)+3(3n-5)=-1(n-3)$

View Solutions

a) $5(3y+2)+4(3y-1)=-2(3y+5)$

$$\begin{aligned} 15y+10+12y-4 &= -6y-10 \\ 27y+6 &= -6y-10 \\ 33y &= -16 \\ y &= -\frac{16}{33} \end{aligned}$$

b) $5(3p+1)+2(3p+1)=-2(2p+6)$

$$\begin{aligned} 15p+5+6p+2 &= -4p-12 \\ 21p+7 &= -4p-12 \\ 25p &= -19 \\ p &= -\frac{19}{25} \end{aligned}$$

c) $4(3y+1)+4(y-2)=-2(y+5)$

$$\begin{aligned} 12y+4+4y-8 &= -2y-10 \\ 16y-4 &= -2y-10 \\ 18y &= -6 \\ y &= -\frac{1}{3} \end{aligned}$$

d) $3(2n-1)+3(3n-5)=-1(n-3)$

$$\begin{aligned} 6n-3+9n-15 &= -n+3 \\ 15n-18 &= -n+3 \\ 16n &= 21 \\ n &= \frac{21}{16} \end{aligned}$$

Exercise 5.26.

a) $\dfrac{5y+2}{3} = \dfrac{2y+26}{4}$

b) $\dfrac{4p+2}{3} = \dfrac{3p+1}{3}$

c) $\dfrac{4n-2}{2} = \dfrac{n-4}{4}$

d) $\dfrac{3x+2}{2} = \dfrac{2x}{4}$

View Solutions

a) $\dfrac{5y+2}{3} = \dfrac{2y+26}{4}$

$$\begin{aligned} \text{Cross multiply: } 4(5y+2) &= 3(2y+26) \\ 20y+8 &= 6y+78 \\ 14y &= 70 \\ y &= 5 \end{aligned}$$

b) $\dfrac{4p+2}{3} = \dfrac{3p+1}{3}$

$$\begin{aligned} \text{Cross multiply: } 3(4p+2) &= 3(3p+1) \\ 12p+6 &= 9p+3 \\ 3p &= -3 \\ p &= -1 \end{aligned}$$

c) $\dfrac{4n-2}{2} = \dfrac{n-4}{4}$

$$\begin{aligned} \text{Cross multiply: } 4(4n-2) &= 2(n-4) \\ 16n-8 &= 2n-8 \\ 14n &= 0 \\ n &= 0 \end{aligned}$$

d) $\dfrac{3x+2}{2} = \dfrac{2x}{4}$

$$\begin{aligned} \text{Cross multiply: } 4(3x+2) &= 2(2x) \\ 12x+8 &= 4x \\ 8x &= -8 \\ x &= -1 \end{aligned}$$

Exercise 5.27.

a) $\dfrac{2p-1}{2} + \dfrac{2p+5}{2} = -10$

b) $\dfrac{2n+5}{2} + \dfrac{n-1}{2} = 17$

c) $\dfrac{m-1}{2} + \dfrac{3m}{5} = -6$

d) $\dfrac{p-1}{5} + \dfrac{2p+3}{5} = -2$

View Solutions

a) $\dfrac{2p-1}{2} + \dfrac{2p+5}{2} = -10$

$$\begin{aligned} \text{Multiply by } 2: 2p-1+2p+5 &= -20 \\ 4p+4 &= -20 \\ 4p &= -24 \\ p &= -6 \end{aligned}$$

b) $\dfrac{2n+5}{2} + \dfrac{n-1}{2} = 17$

$$\begin{aligned} \text{Multiply by } 2: 2n+5+n-1 &= 34 \\ 3n+4 &= 34 \\ 3n &= 30 \\ n &= 10 \end{aligned}$$

c) $\dfrac{m-1}{2} + \dfrac{3m}{5} = -6$

$$\begin{aligned} \text{Multiply by } 10: 5m - 5 + 6m &= -60 \\ 11m - 5 &= -60 \\ 11m &= -55 \\ m &= -5 \end{aligned}$$

d) $\dfrac{p-1}{5} + \dfrac{2p+3}{5} = -2$

$$\begin{aligned} \text{Multiply by } 5: p - 1 + 2p + 3 &= -10 \\ 3p + 2 &= -10 \\ 3p &= -12 \\ p &= -4 \end{aligned}$$

Exercise 5.28.

a) $\dfrac{t-2}{3} - \dfrac{2t+2}{5} = \dfrac{3t-1}{3} + \dfrac{t-19}{3}$

b) $\dfrac{2p}{4} - \dfrac{2p-4}{3} = \dfrac{3p}{2} + \dfrac{p-32}{3}$

c) $\dfrac{y-2}{4} - \dfrac{3y-3}{4} = \dfrac{2y-4}{2} + \dfrac{y+16}{4}$

View Solutions

a) $\dfrac{t-2}{3} - \dfrac{2t+2}{5} = \dfrac{3t-1}{3} + \dfrac{t-19}{3}$

$$\begin{aligned} \text{Multiply by } 15: 5t - 10 - 6t + 6 &= 15t - 5 + 5t - 95 \\ -21t &= -84 \\ t &= 4 \end{aligned}$$

b) $\dfrac{2p}{4} - \dfrac{2p-4}{3} = \dfrac{3p}{2} + \dfrac{p-32}{3}$

$$\begin{aligned} \text{Multiply by } 12: 6p - 8p - 16 &= 18p + 4p - 128 \\ -24p &= -144 \\ p &= 6 \end{aligned}$$

c) $\dfrac{y-2}{4} - \dfrac{3y-3}{4} = \dfrac{2y-4}{2} + \dfrac{y+16}{4}$

$$\text{Multiply by } 4: y - 2 - 3y - 3 = 4y - 8 + y + 16$$
$$-7y = 7$$
$$y = -1$$

Exercise 5.29.

ABCD(6x - 23)°(3x + 59)°(3x + 13)°

ABCD is a trapezium.

- BC is parallel to AD.
- Angle A = $(6x - 23)°$, Angle B = $(3x + 59)°$, Angle D = $(3x + 13)°$.

(i) Find x.

(ii) Find the size of the largest angle in the trapezium.

View Solution

$$\text{(i) Co-interior: } (6x - 23) + (3x + 59) = 180$$
$$9x + 36 = 180$$
$$x = 16$$
$$\text{(ii) A=}73°,\ \text{B=}107°,\ \text{C=}119°,\ \text{D=}61°$$
$$\text{Largest = } 119°$$

Answer: $x = 16,$ largest $= 119°$

Exercise 5.30.

ABCD(5x - 26)°(3x + 54)°(2x + 16)°

ABCD is a trapezium.

- BC is parallel to AD.
- Angle A = $(5x - 26)°$, Angle B = $(3x + 54)°$, Angle D = $(2x + 16)°$.

(i) Find x.

(ii) Find the size of the largest angle in the trapezium.

 View Solution

$$\text{(i) Co-interior: } (5x-26)+(3x+54)=180$$
$$8x+28=180$$
$$x=19$$
$$\text{(ii) A=}69^\circ,\ \text{B=}111^\circ,\ \text{C=}126^\circ,\ \text{D=}54^\circ$$
$$\text{Largest = }126^\circ$$

Answer: $x=19,\ \text{largest}=126^\circ$

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5.4 Test Your Knowledge

6 Unit 6: Quadratic Equations

6.1 Learning Targets

LT1

Factorising — Simple

Factorise $x^2 + bx + c$ using the sum-product method.

Sum-product Factor pairs Monic quadratic

LT2

Factorising — Harder

Factorise $ax^2 + bx + c$ where $a \neq 1$ using the ac-method.

ac-method Split middle term Factor by grouping

LT3

Special Forms

Recognise and factorise difference of two squares and perfect square trinomials.

Difference of squares Perfect square Pattern recognition

LT4

Solve by Factorising

Solve quadratic equations by factorising and applying the zero-product property.

Zero-product Set factors to zero Two solutions

LT5

Quadratic Formula & Discriminant

Apply $x = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ and use $\Delta = b^2 - 4ac$ to determine the nature of roots and find unknown parameters.

Quadratic formula Discriminant Nature of roots

LT6

Completing the Square

Write $ax^2 + bx + c$ in completed square form and use it to solve equations and identify the vertex.

Completed square form Vertex form Solve from CTS

LT7

Word Problems

Form and solve quadratic equations from real-world contexts including area, consecutive integers, and even/odd number problems.

Area problems Consecutive integers Form the equation

LT8

Geometry Problems

Apply quadratic equations to geometric problems involving right triangles, rectangles, L-shapes, and cuboids.

Right triangles Rectangles Composite shapes

LT9

Roots & Vieta's Formulas

Form equations from given roots and use $\alpha + \beta = -b/a$ and $\alpha\beta = c/a$ to find symmetric expressions.

Vieta's formulas Sum & product of roots Symmetric expressions

LT10

Algebraic Fraction Equations

Solve equations with algebraic fractions by multiplying through to clear denominators and form a quadratic.

Clear denominators Form quadratic Check solutions

LT11

Disguised Quadratics

Recognise and solve $x^4 + bx^2 + c = 0$ and $(x + a)^2 + b(x + a) + c = 0$ using substitution.

Substitution Biquadratic Nested form

LT12

Surd Equations

Solve equations involving square roots by squaring both sides; identify and reject extraneous roots.

Square both sides Extraneous roots Check solutions

LT13

Indices → Quadratic

Solve exponential equations like $(b^{x+p})^{x+q} = b^k$ by equating exponents to form and solve a quadratic.

Equate exponents Form quadratic Exponential form

6.2 Formula Card

Quadratic Equations — Formula Card

Factorising (a = 1)

 Sum-Product Method Find two numbers that **multiply** to c and **add** to b.

Standard form: $x^2 + bx + c = (x + p)(x + q)$

where $p \times q = c$ and $p + q = b$

Examples:

$$x^2 + 5x + 6 = (x + 2)(x + 3) \quad [2 \times 3 = 6,\ 2 + 3 = 5]$$

$$x^2 - x - 12 = (x + 3)(x - 4) \quad [3 \times (-4) = -12,\ 3 + (-4) = -1]$$

Factorising (a 1) — ac Method

 ac Method Multiply $a \times c$, then find two numbers that multiply to ac and add to b.

Steps: For $ax^2 + bx + c$:

1. Find $p \times q = ac$ and $p + q = b$
2. Split the middle term: $ax^2 + px + qx + c$
3. Group and factorise

Example: $2x^2 + 7x + 3,\ ac = 6$, find $p, q : p \times q = 6,\ p + q = 7 \Rightarrow 1, 6$

$$2x^2 + x + 6x + 3 = x(2x + 1) + 3(2x + 1) = (x + 3)(2x + 1)$$

Special Forms

⚠ **Spot These First!** Always check for special forms before using other methods.

Difference of Two Squares:

$$a^2 - b^2 = (a + b)(a - b)$$

Perfect Square:

$$a^2 + 2ab + b^2 = (a + b)^2 \qquad a^2 - 2ab + b^2 = (a - b)^2$$

Examples:

$$x^2 - 25 = (x + 5)(x - 5)$$
$$x^2 + 8x + 16 = (x + 4)^2$$

Quadratic Formula

 Always works! Use when factorising is not possible.

For $ax^2 + bx + c = 0$:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

The Discriminant $\Delta = b^2 - 4ac$:

$\Delta > 0$	Two distinct real roots
$\Delta = 0$	One repeated root
$\Delta < 0$	No real roots

Example: $x^2 - 5x + 3 = 0$

$$x = \frac{5 \pm \sqrt{25 - 12}}{2} = \frac{5 \pm \sqrt{13}}{2}$$

Completing the Square

Monic ($a = 1$): $x^2 + bx + c = \left(x + \frac{b}{2}\right)^2 - \left(\frac{b}{2}\right)^2 + c$

Non-monic ($a \neq 1$): $ax^2 + bx + c = a\left(x + \frac{b}{2a}\right)^2 - \frac{b^2}{4a} + c$

To solve: Write in form $(x + p)^2 = q$, then $x = -p \pm \sqrt{q}$

Example: $x^2 + 6x + 2 = 0$

$$(x + 3)^2 - 9 + 2 = 0 \implies (x + 3)^2 = 7 \implies x = -3 \pm \sqrt{7}$$

6.3 Practice Problems

6.3.1 Easy

Exercise 6.1. *Factorise each expression.*

a) $n^2 + 3n + 2$

b) $y^2 + 8y + 16$

c) $x^2 + 11x + 18$

d) $x^2 + 17x + 70$

e) $y^2 + 3y + 2$

f) $x^2 + 13x + 36$

g) $m^2 + 13m + 36$

h) $n^2 + 12n + 32$

View Solutions

a) $n^2 + 3n + 2$

$$\text{Find } p, q : p \times q = 2,\ p + q = 3$$
$$p = 2,\ q = 1$$
$$n^2 + 3n + 2 = (n + 2)(n + 1)$$

b) $y^2 + 8y + 16$

$$\text{Find } p, q : p \times q = 16,\ p + q = 8$$
$$p = 4,\ q = 4$$
$$y^2 + 8y + 16 = (y + 4)(y + 4)$$

c) $x^2 + 11x + 18$

$$\text{Find } p, q : p \times q = 18,\ p + q = 11$$
$$p = 2,\ q = 9$$
$$x^2 + 11x + 18 = (x + 2)(x + 9)$$

d) $x^2 + 17x + 70$

$$\text{Find } p, q : p \times q = 70,\ p + q = 17$$
$$p = 10,\ q = 7$$
$$x^2 + 17x + 70 = (x + 10)(x + 7)$$

e) $y^2 + 3y + 2$

$$\text{Find } p, q : p \times q = 2,\ p + q = 3$$
$$p = 1,\ q = 2$$
$$y^2 + 3y + 2 = (y + 1)(y + 2)$$

f) $x^2 + 13x + 36$

$$\text{Find } p, q : p \times q = 36,\ p + q = 13$$
$$p = 4,\ q = 9$$
$$x^2 + 13x + 36 = (x + 4)(x + 9)$$

g) $m^2 + 13m + 36$

$$\text{Find } p, q : p \times q = 36,\ p + q = 13$$
$$p = 9,\ q = 4$$
$$m^2 + 13m + 36 = (m + 9)(m + 4)$$

h) $n^2 + 12n + 32$

$$\text{Find } p, q : p \times q = 32,\ p + q = 12$$
$$p = 4,\ q = 8$$
$$n^2 + 12n + 32 = (n + 4)(n + 8)$$

Exercise 6.2. *Factorise each expression.*

a) $m^2 - 2m - 3$

b) $y^2 + y - 30$

c) $x^2 - 2x - 24$

d) $x^2 - 5x - 14$

e) $x^2 + 5x - 50$

f) $x^2 - x - 72$

g) $n^2 + 5n - 14$

h) $y^2 + 4y - 60$

View Solutions

a) $m^2 - 2m - 3$

$$\text{Find } p, q : p \times q = -3,\ p + q = -2$$
$$p = 1,\ q = -3$$
$$m^2 - 2m - 3 = (m + 1)(m - 3)$$

b) $y^2 + y - 30$

$$\text{Find } p, q : p \times q = -30,\ p + q = 1$$
$$p = 6,\ q = -5$$
$$y^2 + y - 30 = (y + 6)(y - 5)$$

c) $x^2 - 2x - 24$

$$\text{Find } p, q : p \times q = -24,\ p + q = -2$$
$$p = 4,\ q = -6$$
$$x^2 - 2x - 24 = (x + 4)(x - 6)$$

d) $x^2 - 5x - 14$

$$\text{Find } p, q : p \times q = -14,\ p + q = -5$$
$$p = 2,\ q = -7$$
$$x^2 - 5x - 14 = (x + 2)(x - 7)$$

e) $x^2 + 5x - 50$

$$\text{Find } p, q : p \times q = -50,\ p + q = 5$$
$$p = 10,\ q = -5$$
$$x^2 + 5x - 50 = (x + 10)(x - 5)$$

f) $x^2 - x - 72$

$$\text{Find } p, q : p \times q = -72,\ p + q = -1$$
$$p = 8,\ q = -9$$
$$x^2 - x - 72 = (x + 8)(x - 9)$$

g) $n^2 + 5n - 14$

$$\text{Find } p, q : p \times q = -14,\ p + q = 5$$
$$p = 7,\ q = -2$$
$$n^2 + 5n - 14 = (n + 7)(n - 2)$$

h) $y^2 + 4y - 60$

$$\text{Find } p, q : p \times q = -60,\ p + q = 4$$
$$p = 10,\ q = -6$$
$$y^2 + 4y - 60 = (y + 10)(y - 6)$$

Exercise 6.3. *Factorise using the difference of two squares.*

a) $x^2 - 1$

b) $y^2 - 121$

c) $4x^2 - 100$

d) $4m^2 - 49$

e) $9x^2 - 25$

f) $n^2 - 121$

g) $n^2 - 4$

h) $4m^2 - 9$

View Solutions

a) $x^2 - 1$

$$x^2 - 1^2 = (x + 1)(x - 1)$$

b) $y^2 - 121$

$$y^2 - 11^2 = (y + 11)(y - 11)$$

c) $4x^2 - 100$

$$(2x)^2 - (10)^2 = (2x + 10)(2x - 10)$$

d) $4m^2 - 49$

$$(2m)^2 - (7)^2 = (2m + 7)(2m - 7)$$

e) $9x^2 - 25$

$$(3x)^2 - (5)^2 = (3x + 5)(3x - 5)$$

f) $n^2 - 121$

$$n^2 - 11^2 = (n + 11)(n - 11)$$

g) $n^2 - 4$

$$n^2 - 2^2 = (n + 2)(n - 2)$$

h) $4m^2 - 9$

$$(2m)^2 - (3)^2 = (2m + 3)(2m - 3)$$

Exercise 6.4. *Recognise and factorise each perfect square.*

a) $y^2 - 2y + 1$

b) $n^2 + 18n + 81$

c) $y^2 + 18y + 81$

d) $x^2 + 12x + 36$

e) $m^2 - 20m + 100$

f) $n^2 + 2n + 1$

View Solutions

a) $y^2 - 2y + 1$

$$(y-1)^2 \quad \text{[perfect square]}$$

b) $n^2 + 18n + 81$

$$(n+9)^2 \quad \text{[perfect square]}$$

c) $y^2 + 18y + 81$

$$(y+9)^2 \quad \text{[perfect square]}$$

d) $x^2 + 12x + 36$

$$(x+6)^2 \quad \text{[perfect square]}$$

e) $m^2 - 20m + 100$

$$(m-10)^2 \quad \text{[perfect square]}$$

f) $n^2 + 2n + 1$

$$(n+1)^2 \quad \text{[perfect square]}$$

Exercise 6.5. *Solve each quadratic equation.*

a) $x^2 - 1 = 0$

b) $x^2 - 7x + 6 = 0$

c) $y^2 + y - 42 = 0$

d) $y^2 + 3y - 28 = 0$

e) $y^2 + 3y - 10 = 0$

f) $n^2 + 5n + 4 = 0$

g) $m^2 + 14m + 48 = 0$

h) $y^2 - 6y + 5 = 0$

View Solutions

a) $x^2 - 1 = 0$

$$\begin{aligned} \text{Find } p, q : p \times q = -1,\ p + q = 0 \\ p = 1,\ q = -1 \\ (x+1)(x-1) = 0 \\ x = -1 \text{ or } x = 1 \end{aligned}$$

b) $x^2 - 7x + 6 = 0$

$$\begin{aligned} \text{Find } p, q : p \times q = 6,\ p + q = -7 \\ p = -1,\ q = -6 \\ (x-1)(x-6) = 0 \\ x = 1 \text{ or } x = 6 \end{aligned}$$

c) $y^2 + y - 42 = 0$

$$\begin{aligned} \text{Find } p, q : p \times q = -42,\ p + q = 1 \\ p = 7,\ q = -6 \\ (y+7)(y-6) = 0 \\ y = -7 \text{ or } y = 6 \end{aligned}$$

d) $y^2 + 3y - 28 = 0$

$$\begin{aligned} \text{Find } p, q : p \times q = -28,\ p + q = 3 \\ p = -4,\ q = 7 \\ (y-4)(y+7) = 0 \\ y = 4 \text{ or } y = -7 \end{aligned}$$

e) $y^2 + 3y - 10 = 0$

$$\text{Find } p, q : p \times q = -10,\ p + q = 3$$
$$p = 5,\ q = -2$$
$$(y+5)(y-2) = 0$$
$$y = -5 \text{ or } y = 2$$

f) $n^2 + 5n + 4 = 0$

$$\text{Find } p, q : p \times q = 4,\ p + q = 5$$
$$p = 1,\ q = 4$$
$$(n+1)(n+4) = 0$$
$$n = -1 \text{ or } n = -4$$

g) $m^2 + 14m + 48 = 0$

$$\text{Find } p, q : p \times q = 48,\ p + q = 14$$
$$p = 6,\ q = 8$$
$$(m+6)(m+8) = 0$$
$$m = -6 \text{ or } m = -8$$

h) $y^2 - 6y + 5 = 0$

$$\text{Find } p, q : p \times q = 5,\ p + q = -6$$
$$p = -5,\ q = -1$$
$$(y-5)(y-1) = 0$$
$$y = 5 \text{ or } y = 1$$

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6.3.2 Medium

Exercise 6.6. *Factorise each expression.*

a) $y^2 - 3y + 2$

b) $y^2 - 7y + 10$

c) $n^2 - 9n + 14$

d) $y^2 - 14y + 48$

e) $y^2 - 12y + 36$

f) $y^2 - 7y + 10$

View Solutions

a) $y^2 - 3y + 2$

$$\begin{aligned} \text{Find } p, q : p \times q = 2,\ p + q = -3 \\ p = -2,\ q = -1 \\ y^2 - 3y + 2 = (y-2)(y-1) \end{aligned}$$

b) $y^2 - 7y + 10$

$$\begin{aligned} \text{Find } p, q : p \times q = 10,\ p + q = -7 \\ p = -5,\ q = -2 \\ y^2 - 7y + 10 = (y-5)(y-2) \end{aligned}$$

c) $n^2 - 9n + 14$

$$\begin{aligned} \text{Find } p, q : p \times q = 14,\ p + q = -9 \\ p = -2,\ q = -7 \\ n^2 - 9n + 14 = (n-2)(n-7) \end{aligned}$$

d) $y^2 - 14y + 48$

$$\begin{aligned} \text{Find } p, q : p \times q = 48,\ p + q = -14 \\ p = -8,\ q = -6 \\ y^2 - 14y + 48 = (y-8)(y-6) \end{aligned}$$

e) $y^2 - 12y + 36$

$$\text{Find } p, q : p \times q = 36,\ p + q = -12$$
$$p = -6,\ q = -6$$
$$y^2 - 12y + 36 = (y-6)(y-6)$$

f) $y^2 - 7y + 10$

$$\text{Find } p, q : p \times q = 10,\ p + q = -7$$
$$p = -5,\ q = -2$$
$$y^2 - 7y + 10 = (y-5)(y-2)$$

Exercise 6.7. *Factorise each expression.*

a) $3m^2 + 14m + 8$

b) $4y^2 + 30y + 36$

c) $4x^2 + 9x + 2$

d) $4x^2 + 16x + 12$

e) $3n^2 + 23n + 30$

f) $3m^2 + 18m + 24$

View Solutions

a) $3m^2 + 14m + 8$

$$ac = 3 \times 8 = 24$$
$$\text{Find } p, q : p \times q = 24,\ p + q = 14$$
$$p = 2,\ q = 12$$
$$3m^2 + 2m + 12m + 8$$
$$m(3m+2) + \text{group}$$
$$3m^2 + 14m + 8 = (3m+2)(m+4)$$

b) $4y^2 + 30y + 36$

$$ac = 4 \times 36 = 144$$
$$\text{Find } p, q : p \times q = 144,\ p + q = 30$$
$$p = 6,\ q = 24$$
$$4y^2 + 6y + 24y + 36$$
$$y(4y + 3) + \text{group}$$
$$4y^2 + 30y + 36 = (4y + 6)(y + 6)$$

c) $4x^2 + 9x + 2$

$$ac = 4 \times 2 = 8$$
$$\text{Find } p, q : p \times q = 8,\ p + q = 9$$
$$p = 1,\ q = 8$$
$$4x^2 + x + 8x + 2$$
$$x(4x + 1) + \text{group}$$
$$4x^2 + 9x + 2 = (4x + 1)(x + 2)$$

d) $4x^2 + 16x + 12$

$$ac = 4 \times 12 = 48$$
$$\text{Find } p, q : p \times q = 48,\ p + q = 16$$
$$p = 4,\ q = 12$$
$$4x^2 + 4x + 12x + 12$$
$$x(4x + 1) + \text{group}$$
$$4x^2 + 16x + 12 = (4x + 4)(x + 3)$$

e) $3n^2 + 23n + 30$

$$ac = 3 \times 30 = 90$$
$$\text{Find } p, q : p \times q = 90,\ p + q = 23$$
$$p = 5,\ q = 18$$
$$3n^2 + 5n + 18n + 30$$
$$n(3n + 5) + \text{group}$$
$$3n^2 + 23n + 30 = (3n + 5)(n + 6)$$

f) $3m^2 + 18m + 24$

$$\begin{aligned} ac &= 3 \times 24 = 72 \\ \text{Find } p, q : p \times q &= 72,\ p + q = 18 \\ p &= 6,\ q = 12 \\ 3m^2 + 6m + 12m + 24 \\ m(3m + 2) &+ \text{group} \\ 3m^2 + 18m + 24 &= (3m + 6)(m + 4) \end{aligned}$$

Exercise 6.8. *Solve each quadratic equation.*

a) $2y^2 - 13y + 6 = 0$

b) $4x^2 - 27x + 18 = 0$

c) $4y^2 - 20y - 24 = 0$

d) $2x^2 - 8x + 6 = 0$

e) $4m^2 - 11m + 6 = 0$

f) $2m^2 - 6m - 8 = 0$

View Solutions

a) $2y^2 - 13y + 6 = 0$

$$\begin{aligned} ac &= 2 \times 6 = 12 \\ p &= -12,\ q = -1 \\ (2y - 1)(y - 6) &= 0 \\ y &= \frac{1}{2} \text{ or } y = 6 \end{aligned}$$

b) $4x^2 - 27x + 18 = 0$

$$
\begin{aligned}
ac &= 4 \times 18 = 72 \\
p &= -24,\ q = -3 \\
(4x-3)(x-6) &= 0 \\
x &= \frac{3}{4} \text{ or } x = 6
\end{aligned}
$$

c) $4y^2 - 20y - 24 = 0$

$$
\begin{aligned}
ac &= 4 \times -24 = -96 \\
p &= -24,\ q = 4 \\
(4y+4)(y-6) &= 0 \\
y &= -1 \text{ or } y = 6
\end{aligned}
$$

d) $2x^2 - 8x + 6 = 0$

$$
\begin{aligned}
ac &= 2 \times 6 = 12 \\
p &= -6,\ q = -2 \\
(2x-6)(x-1) &= 0 \\
x &= 3 \text{ or } x = 1
\end{aligned}
$$

e) $4m^2 - 11m + 6 = 0$

$$
\begin{aligned}
ac &= 4 \times 6 = 24 \\
p &= -8,\ q = -3 \\
(4m-3)(m-2) &= 0 \\
m &= \frac{3}{4} \text{ or } m = 2
\end{aligned}
$$

f) $2m^2 - 6m - 8 = 0$

$$
\begin{aligned}
ac &= 2 \times -8 = -16 \\
p &= -8,\ q = 2 \\
(2m+2)(m-4) &= 0 \\
m &= -1 \text{ or } m = 4
\end{aligned}
$$

Exercise 6.9. *Solve each equation.*

a) $m^2 - 64 = 0$

b) $4x^2 - 64 = 0$

c) $m^2 + 16m + 64 = 0$

d) $4x^2 - 64 = 0$

e) $y^2 - 25 = 0$

f) $y^2 + 14y + 49 = 0$

View Solutions

a) $m^2 - 64 = 0$

$$(m+8)(m-8) = 0$$
$$m = 8 \text{ or } m = -8$$

b) $4x^2 - 64 = 0$

$$(2x+8)(2x-8) = 0$$
$$x = 4 \text{ or } x = -4$$

c) $m^2 + 16m + 64 = 0$

$$(m+8)^2 = 0$$
$$m = -8 \text{ (repeated root)}$$

d) $4x^2 - 64 = 0$

$$(2x+8)(2x-8) = 0$$
$$x = 4 \text{ or } x = -4$$

e) $y^2 - 25 = 0$

$$(y+5)(y-5) = 0$$
$$y = 5 \text{ or } y = -5$$

f) $y^2 + 14y + 49 = 0$

$$(y+7)^2 = 0$$
$$y = -7 \text{ (repeated root)}$$

Exercise 6.10. *Solve using the quadratic formula.*

a) $x^2 + 7x + 6 = 0$

b) $y^2 + 15y + 56 = 0$

c) $m^2 - 2m - 15 = 0$

d) $y^2 - 2y - 8 = 0$

e) $n^2 + 4n - 32 = 0$

f) $m^2 - 7m + 6 = 0$

View Solutions

a) $x^2 + 7x + 6 = 0$

$$x = \frac{-(7) \pm \sqrt{(7)^2 - 4(1)(6)}}{2}$$
$$x = \frac{-7 \pm 5}{2}$$
$$x = -1 \text{ or } x = -6$$

b) $y^2 + 15y + 56 = 0$

$$y = \frac{-(15) \pm \sqrt{(15)^2 - 4(1)(56)}}{2}$$
$$y = \frac{-15 \pm 1}{2}$$
$$y = -7 \text{ or } y = -8$$

c) $m^2 - 2m - 15 = 0$

$$m = \frac{-(-2) \pm \sqrt{(-2)^2 - 4(1)(-15)}}{2}$$
$$m = \frac{2 \pm 8}{2}$$
$$m = 5 \text{ or } m = -3$$

d) $y^2 - 2y - 8 = 0$

$$y = \frac{-(-2) \pm \sqrt{(-2)^2 - 4(1)(-8)}}{2}$$
$$y = \frac{2 \pm 6}{2}$$
$$y = 4 \text{ or } y = -2$$

e) $n^2 + 4n - 32 = 0$

$$n = \frac{-(4) \pm \sqrt{(4)^2 - 4(1)(-32)}}{2}$$
$$n = \frac{-4 \pm 12}{2}$$
$$n = 4 \text{ or } n = -8$$

f) $m^2 - 7m + 6 = 0$

$$m = \frac{-(-7) \pm \sqrt{(-7)^2 - 4(1)(6)}}{2}$$
$$m = \frac{7 \pm 5}{2}$$
$$m = 6 \text{ or } m = 1$$

Exercise 6.11. *Write each expression in completed square form.*

a) $n^2 + 2n - 6$

b) $x^2 + 4x$

c) $x^2 - 10x + 4$

d) $y^2 - 8y + 7$

e) $y^2 + 6y - 2$

f) $y^2 + y - 8$

View Solutions

a) $n^2 + 2n - 6$

$$(n+1)^2 - 7$$

b) $x^2 + 4x$

$$(x+2)^2 - 4$$

c) $x^2 - 10x + 4$

$$\left(x - \frac{10}{2}\right)^2 - \left(\frac{10}{2}\right)^2 + 4$$
$$(x-5)^2 - 25 + 4$$
$$(x-5)^2 - 21$$

d) $y^2 - 8y + 7$

$$\left(y - \frac{8}{2}\right)^2 - \left(\frac{8}{2}\right)^2 + 7$$
$$(y-4)^2 - 16 + 7$$
$$(y-4)^2 - 9$$

e) $y^2 + 6y - 2$

$$(y+3)^2 - 11$$

f) $y^2 + y - 8$

$$(y + \frac{1}{2})^2 - \frac{33}{4}$$

Exercise 6.12.

A rectangle has length $(x+3)$ cm and width $(x+2)$ cm. The area is 42 cm².

(i) Form a quadratic equation in x.

(ii) Solve the equation, giving exact answers.

(iii) Find the dimensions of the rectangle.

View Solution

$$\begin{aligned}
\text{(i) } (x+3)(x+2) &= 42 \\
x^2+5x+6 &= 42 \\
x^2+5x-36 &= 0 \\
\text{(ii) } x &= \frac{-(5) \pm \sqrt{(5)^2-4(1)(-36)}}{2} \\
x &= \frac{-5 \pm 13}{2} \\
x &= 4 \text{ or } x = -9 \\
\text{(iii) Taking positive root: } x &= 4 \\
\text{Length } &= 4+3 = 7 \text{ cm, Width } = 4+2 = 6 \text{ cm}
\end{aligned}$$

Answer: $x = 4$

Exercise 6.13.

A rectangle has length $(x+3)$ cm and width $(x+4)$ cm. The area is 30 cm².

(i) Form a quadratic equation in x.

(ii) Solve the equation, giving exact answers.

(iii) Find the dimensions of the rectangle.

View Solution

$$\text{(i) } (x+3)(x+4) = 30$$
$$x^2 + 7x + 12 = 30$$
$$x^2 + 7x - 18 = 0$$
$$\text{(ii) } x = \frac{-(7) \pm \sqrt{(7)^2 - 4(1)(-18)}}{2}$$
$$x = \frac{-7 \pm 11}{2}$$
$$x = 2 \text{ or } x = -9$$
$$\text{(iii) Taking positive root: } x = 2$$
$$\text{Length } = 2 + 3 = 5 \text{ cm, Width } = 2 + 4 = 6 \text{ cm}$$

Answer: $x = 2$

Exercise 6.14.

A rectangle has length $(x+3)$ cm and width $(x+6)$ cm. The area is 130 cm².

(i) Form a quadratic equation in x.

(ii) Solve the equation, giving exact answers.

(iii) Find the dimensions of the rectangle.

View Solution

$$\text{(i) } (x+3)(x+6) = 130$$
$$x^2 + 9x + 18 = 130$$
$$x^2 + 9x - 112 = 0$$
$$\text{(ii) } x = \frac{-(9) \pm \sqrt{(9)^2 - 4(1)(-112)}}{2}$$
$$x = \frac{-9 \pm 23}{2}$$
$$x = 7 \text{ or } x = -16$$
$$\text{(iii) Taking positive root: } x = 7$$
$$\text{Length } = 7 + 3 = 10 \text{ cm, Width } = 7 + 6 = 13 \text{ cm}$$

Answer: $x = 7$

Exercise 6.15.

The product of two consecutive positive integers is 240.

(i) Form a quadratic equation in n.

(ii) Solve the equation exactly.

(iii) Find the two integers.

View Solution

$$\text{(i) } n(n+1) = 240$$
$$n^2 + n - 240 = 0$$
$$\text{(ii) } n = \frac{-(1) \pm \sqrt{(1)^2 - 4(1)(-240)}}{2}$$
$$n = \frac{-1 \pm 31}{2}$$
$$n = 15 \text{ or } n = -16$$
$$\text{(iii) Since integers are positive, } n = 15$$
$$\text{Integers: } 15 \text{ and } 16$$

Answer: $Integers: 15 and 16$

Exercise 6.16.

The product of two consecutive positive integers is 90.

(i) Form a quadratic equation in n.

(ii) Solve the equation exactly.

(iii) Find the two integers.

View Solution

$$\text{(i) } n(n+1) = 90$$
$$n^2 + n - 90 = 0$$
$$\text{(ii) } n = \frac{-(1) \pm \sqrt{(1)^2 - 4(1)(-90)}}{2}$$
$$n = \frac{-1 \pm 19}{2}$$
$$n = 9 \text{ or } n = -10$$
$$\text{(iii) Since integers are positive, } n = 9$$
$$\text{Integers: } 9 \text{ and } 10$$

Answer: $Integers: 9 and 10$

Exercise 6.17.

The product of two consecutive positive integers is 240.

(i) Form a quadratic equation in n.

(ii) Solve the equation exactly.

(iii) Find the two integers.

View Solution

$$\text{(i) } n(n+1) = 240$$
$$n^2 + n - 240 = 0$$
$$\text{(ii) } n = \frac{-(1) \pm \sqrt{(1)^2 - 4(1)(-240)}}{2}$$
$$n = \frac{-1 \pm 31}{2}$$
$$n = 15 \text{ or } n = -16$$
$$\text{(iii) Since integers are positive, } n = 15$$
$$\text{Integers: } 15 \text{ and } 16$$

Answer: $Integers: 15 and 16$

Exercise 6.18.

A rectangle has perimeter 32 cm and area 28 cm². Let the length be x cm, so the width is $(16 - x)$ cm.

(i) Show that $x^2 - 16x + 28 = 0$.

(ii) Solve the equation exactly.

(iii) Find the dimensions of the rectangle.

View Solution

$$\text{(i) } x(16 - x) = 28$$
$$x^2 - 16x + 28 = 0$$
$$\text{(ii) } x = \frac{-(-16) \pm \sqrt{(-16)^2 - 4(1)(28)}}{2}$$
$$x = \frac{16 \pm 12}{2}$$
$$x = 14 \text{ or } x = 2$$
$$\text{(iii) } x = 14 \text{ or } x = 2$$
$$\text{Length } = 14 \text{ cm, Width } = 2 \text{ cm}$$

Answer: $Length = 14cm, Width = 2cm$

Exercise 6.19.

A rectangle has perimeter 32 cm and area 55 cm². Let the length be x cm, so the width is $(16 - x)$ cm.

(i) Show that $x^2 - 16x + 55 = 0$.

(ii) Solve the equation exactly.

(iii) Find the dimensions of the rectangle.

View Solution

$$\text{(i) } x(16 - x) = 55$$
$$x^2 - 16x + 55 = 0$$
$$\text{(ii) } x = \frac{-(-16) \pm \sqrt{(-16)^2 - 4(1)(55)}}{2}$$
$$x = \frac{16 \pm 6}{2}$$
$$x = 11 \text{ or } x = 5$$
$$\text{(iii) } x = 11 \text{ or } x = 5$$
$$\text{Length } = 11 \text{ cm, Width } = 5 \text{ cm}$$

Answer: $Length = 11cm, Width = 5cm$

Exercise 6.20. *Find the discriminant and state the nature of the roots.*

a) $3y^2 - 5y = 0$

b) $m^2 - 7m + 5 = 0$

c) $y^2 + 8y + 6 = 0$

d) $n^2 + 5n - 5 = 0$

e) $3x^2 + 2x + 4 = 0$

f) $m^2 + 8m - 2 = 0$

View Solutions

a) $3y^2 - 5y = 0$

$$\Delta = b^2 - 4ac = (-5)^2 - 4(3)(0)$$
$$\Delta = 25 - 0 = 25$$
$$\Delta > 0 \Rightarrow \text{Two distinct real roots}$$

b) $m^2 - 7m + 5 = 0$

$$\Delta = b^2 - 4ac = (-7)^2 - 4(1)(5)$$
$$\Delta = 49 - 20 = 29$$
$$\Delta > 0 \Rightarrow \text{Two distinct real roots}$$

c) $y^2 + 8y + 6 = 0$

$$\Delta = b^2 - 4ac = (8)^2 - 4(1)(6)$$
$$\Delta = 64 - 24 = 40$$
$$\Delta > 0 \Rightarrow \text{Two distinct real roots}$$

d) $n^2 + 5n - 5 = 0$

$$\Delta = b^2 - 4ac = (5)^2 - 4(1)(-5)$$
$$\Delta = 25 + 20 = 45$$
$$\Delta > 0 \Rightarrow \text{Two distinct real roots}$$

e) $3x^2 + 2x + 4 = 0$

$$\Delta = b^2 - 4ac = (2)^2 - 4(3)(4)$$
$$\Delta = 4 - 48 = -44$$
$$\Delta < 0 \Rightarrow \text{No real roots}$$

f) $m^2 + 8m - 2 = 0$

$$\Delta = b^2 - 4ac = (8)^2 - 4(1)(-2)$$
$$\Delta = 64 + 8 = 72$$
$$\Delta > 0 \Rightarrow \text{Two distinct real roots}$$

Exercise 6.21.

The product of two consecutive positive even integers is 624.

(i) Form a quadratic equation in n.

(ii) Solve the equation exactly.

(iii) Find the two integers.

View Solution

$$\text{(i) } n(n+2) = 624$$
$$n^2 + 2n - 624 = 0$$
$$\text{(ii) } n = \frac{-(2) \pm \sqrt{(2)^2 - 4(1)(-624)}}{2}$$
$$n = \frac{-2 \pm 50}{2}$$
$$n = 24 \text{ or } n = -26$$
$$\text{Since integers are positive even, } n = 24$$
$$\text{Integers: } 24 \text{ and } 26$$

Answer: $Integers : 24 and 26$

Exercise 6.22.

The product of two consecutive positive even integers is 48.

(i) Form a quadratic equation in n.

(ii) Solve the equation exactly.

(iii) Find the two integers.

View Solution

$$\text{(i) } n(n+2) = 48$$
$$n^2 + 2n - 48 = 0$$
$$\text{(ii) } n = \frac{-(2) \pm \sqrt{(2)^2 - 4(1)(-48)}}{2}$$
$$n = \frac{-2 \pm 14}{2}$$
$$n = 6 \text{ or } n = -8$$
$$\text{Since integers are positive even, } n = 6$$
$$\text{Integers: } 6 \text{ and } 8$$

Answer: $Integers : 6and8$

Exercise 6.23.

The product of two consecutive positive even integers is 528.

(i) Form a quadratic equation in n.

(ii) Solve the equation exactly.

(iii) Find the two integers.

View Solution

$$\text{(i) } n(n+2) = 528$$
$$n^2 + 2n - 528 = 0$$
$$\text{(ii) } n = \frac{-(2) \pm \sqrt{(2)^2 - 4(1)(-528)}}{2}$$
$$n = \frac{-2 \pm 46}{2}$$
$$n = 22 \text{ or } n = -24$$
$$\text{Since integers are positive even, } n = 22$$
$$\text{Integers: } 22 \text{ and } 24$$

Answer: $Integers : 22and24$

Exercise 6.24. *Form a quadratic equation with roots α and β. Give your answer in the form $y^2 + py + q = 0$.*

a) $\alpha = -1,\ \beta = -2$

b) $\alpha = -2,\ \beta = 5$

c) $\alpha = 2,\ \beta = 3$

d) $\alpha = 1,\ \beta = -4$

View Solutions

a) $\alpha = -1,\ \beta = -2$

$$\begin{aligned} \text{Sum of roots: } \alpha + \beta &= -1 + (-2) = -3 \\ \text{Product of roots: } \alpha\beta &= (-1) \times (-2) = 2 \\ \text{Equation: } y^2 - (\alpha + \beta)y + \alpha\beta &= 0 \\ y^2 + 3y + 2 &= 0 \end{aligned}$$

b) $\alpha = -2,\ \beta = 5$

$$\begin{aligned} \text{Sum of roots: } \alpha + \beta &= -2 + (5) = 3 \\ \text{Product of roots: } \alpha\beta &= (-2) \times (5) = -10 \\ \text{Equation: } y^2 - (\alpha + \beta)y + \alpha\beta &= 0 \\ y^2 - 3y - 10 &= 0 \end{aligned}$$

c) $\alpha = 2,\ \beta = 3$

$$\begin{aligned} \text{Sum of roots: } \alpha + \beta &= 2 + (3) = 5 \\ \text{Product of roots: } \alpha\beta &= (2) \times (3) = 6 \\ \text{Equation: } x^2 - (\alpha + \beta)x + \alpha\beta &= 0 \\ x^2 - 5x + 6 &= 0 \end{aligned}$$

d) $\alpha = 1,\ \beta = -4$

$$\text{Sum of roots: } \alpha + \beta = 1 + (-4) = -3$$
$$\text{Product of roots: } \alpha\beta = (1) \times (-4) = -4$$
$$\text{Equation: } m^2 - (\alpha + \beta)m + \alpha\beta = 0$$
$$m^2 + 3m - 4 = 0$$

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6.3.3 Challenging

Exercise 6.25. *Factorise each expression.*

a) $4n^2 - 12n + 8$

b) $2n^2 - 12n + 10$

c) $4m^2 - 21m + 20$

d) $3m^2 - 8m + 4$

View Solutions

a) $4n^2 - 12n + 8$

$$ac = 4 \times 8 = 32$$
$$\text{Find } p, q : p \times q = 32,\ p + q = -12$$
$$p = -8,\ q = -4$$
$$4n^2 - 12n + 8 = (4n - 4)(n - 2)$$

b) $2n^2 - 12n + 10$

$$ac = 2 \times 10 = 20$$
$$\text{Find } p, q : p \times q = 20,\ p + q = -12$$
$$p = -10,\ q = -2$$
$$2n^2 - 12n + 10 = (2n - 2)(n - 5)$$

c) $4m^2 - 21m + 20$

$$ac = 4 \times 20 = 80$$
$$\text{Find } p, q : p \times q = 80,\ p + q = -21$$
$$p = -16,\ q = -5$$
$$4m^2 - 21m + 20 = (4m - 5)(m - 4)$$

d) $3m^2 - 8m + 4$

$$ac = 3 \times 4 = 12$$
$$\text{Find } p, q : p \times q = 12,\ p + q = -8$$
$$p = -6,\ q = -2$$
$$3m^2 - 8m + 4 = (3m - 2)(m - 2)$$

Exercise 6.26. *Solve using the quadratic formula. Leave answers in surd form.*

a) $y^2 - 2y - 2 = 0$

b) $n^2 - 7n + 5 = 0$

c) $m^2 - 7m + 3 = 0$

d) $x^2 - 7x + 2 = 0$

View Solutions

a) $y^2 - 2y - 2 = 0$

$$y = \frac{-(-2) \pm \sqrt{(-2)^2 - 4(1)(-2)}}{2}$$
$$y = \frac{2 \pm \sqrt{12}}{2}$$
$$y = \frac{2 \pm 2\sqrt{3}}{2}$$
$$y = 1 \pm \sqrt{3}$$

b) $n^2 - 7n + 5 = 0$

$$n = \frac{-(-7) \pm \sqrt{(-7)^2 - 4(1)(5)}}{2}$$

$$n = \frac{7 \pm \sqrt{29}}{2}$$

c) $m^2 - 7m + 3 = 0$

$$m = \frac{-(-7) \pm \sqrt{(-7)^2 - 4(1)(3)}}{2}$$

$$m = \frac{7 \pm \sqrt{37}}{2}$$

d) $x^2 - 7x + 2 = 0$

$$x = \frac{-(-7) \pm \sqrt{(-7)^2 - 4(1)(2)}}{2}$$

$$x = \frac{7 \pm \sqrt{41}}{2}$$

Exercise 6.27. *Use the discriminant to determine the nature of roots.*

a) $3x^2 - x + 7 = 0$

b) $3x^2 - x + 1 = 0$

c) $2n^2 + 4n + 6 = 0$

d) $n^2 + n + 4 = 0$

View Solutions

a) $3x^2 - x + 7 = 0$

$$x = \frac{-(-1) \pm \sqrt{(-1)^2 - 4(3)(7)}}{6}$$

$$x = \frac{1 \pm 0}{6}$$

Discriminant $= -83 < 0$

No real solutions

b) $3x^2 - x + 1 = 0$

$$x = \frac{-(-1) \pm \sqrt{(-1)^2 - 4(3)(1)}}{6}$$

$$x = \frac{1 \pm 0}{6}$$

Discriminant $= -11 < 0$

No real solutions

c) $2n^2 + 4n + 6 = 0$

$$n = \frac{-(4) \pm \sqrt{(4)^2 - 4(2)(6)}}{4}$$

$$n = \frac{-4 \pm 0}{4}$$

Discriminant $= -32 < 0$

No real solutions

d) $n^2 + n + 4 = 0$

$$n = \frac{-(1) \pm \sqrt{(1)^2 - 4(1)(4)}}{2}$$

$$n = \frac{-1 \pm 0}{2}$$

Discriminant $= -15 < 0$

No real solutions

Exercise 6.28. *Write in the form* $a(x+p)^2+q$.

a) $3m^2+m-4$

b) $-2x^2+x+3$

c) $-2x^2+x+4$

d) $2y^2-5y+5$

View Solutions

a) $3m^2+m-4$

$$3\left[m^2+\frac{1}{3}m\right]-4$$
$$3\left(m+\frac{1}{6}\right)^2-\frac{1}{12}-4$$
$$3(m+\frac{1}{6})^2-\frac{49}{12}$$

b) $-2x^2+x+3$

$$-2\left[x^2+\frac{1}{-2}x\right]+3$$
$$-2\left(x+\frac{1}{-4}\right)^2+\frac{1}{8}+3$$
$$-2(x-\frac{1}{4})^2+\frac{25}{8}$$

c) $-2x^2+x+4$

$$-2\left[x^2+\frac{1}{-2}x\right]+4$$
$$-2\left(x+\frac{1}{-4}\right)^2+\frac{1}{8}+4$$
$$-2(x-\frac{1}{4})^2+\frac{33}{8}$$

d) $2y^2 - 5y + 5$

$$2\left[y^2 + \frac{-5}{2}y\right] + 5$$
$$2\left(y + \frac{-5}{4}\right)^2 - \frac{25}{8} + 5$$
$$2(y - \frac{5}{4})^2 + \frac{15}{8}$$

Exercise 6.29. *Solve by completing the square.*

a) $n^2 + 2n - 2 = 0$

b) $n^2 + 8n + 7 = 0$

c) $n^2 + 2n - 4 = 0$

d) $n^2 - 4n - 7 = 0$

View Solutions

a) $n^2 + 2n - 2 = 0$

$$n^2 + 2n - 2 = 0$$
$$(n+1)^2 - 3 = 0$$
$$(n+1)^2 = 3$$
$$n + 1 = \pm\sqrt{3}$$
$$n = -1 + \sqrt{3} \text{ or } n = -1 - \sqrt{3}$$

b) $n^2 + 8n + 7 = 0$

$$n^2 + 8n + 7 = 0$$
$$(n+4)^2 - 9 = 0$$
$$(n+4)^2 = 9$$
$$n + 4 = \pm 3$$
$$n = -1 \text{ or } n = -7$$

c) $n^2 + 2n - 4 = 0$

$$\begin{aligned} n^2 + 2n - 4 &= 0 \\ (n+1)^2 - 5 &= 0 \\ (n+1)^2 &= 5 \\ n + 1 &= \pm\sqrt{5} \\ n = -1 + \sqrt{5} \text{ or } n &= -1 - \sqrt{5} \end{aligned}$$

d) $n^2 - 4n - 7 = 0$

$$\begin{aligned} n^2 - 4n - 7 &= 0 \\ (n-2)^2 - 11 &= 0 \\ (n-2)^2 &= 11 \\ n - 2 &= \pm\sqrt{11} \\ n = 2 + \sqrt{11} \text{ or } n &= 2 - \sqrt{11} \end{aligned}$$

Exercise 6.30. *Solve by completing the square.*

a) $2y^2 - 4y = 0$

b) $2x^2 + x - 1 = 0$

c) $2m^2 - 5m - 1 = 0$

View Solutions

a) $2y^2 - 4y = 0$

$$\begin{aligned} \text{Divide by } 2: y^2 - 2y + 0 &= 0 \\ \text{Complete the square:} \\ (y-1)^2 &= 1 \\ y - 1 &= \pm 1 \\ y = 2 \text{ or } y &= 0 \end{aligned}$$

b) $2x^2 + x - 1 = 0$

$$\text{Divide by } 2: x^2 + \frac{1}{2}x - \frac{1}{2} = 0$$

Complete the square:

$$\left(x + \frac{1}{4}\right)^2 = \frac{9}{16}$$

$$x + \frac{1}{4} = \pm\frac{3}{4}$$

$$x = \frac{1}{2} \text{ or } x = -1$$

c) $2m^2 - 5m - 1 = 0$

$$\text{Divide by } 2: m^2 - \frac{5}{2}m - \frac{1}{2} = 0$$

Complete the square:

$$\left(m - \frac{5}{4}\right)^2 = \frac{33}{16}$$

$$m - \frac{5}{4} = \pm\frac{\sqrt{33}}{4}$$

$$m = \frac{5}{4} + \frac{\sqrt{33}}{4} \text{ or } m = \frac{5}{4} - \frac{\sqrt{33}}{4}$$

Exercise 6.31.

A ball is thrown upward with initial velocity 15 m/s. Its height (m) after t seconds is $h = -5t^2 + 15t + 15$.

(i) What does the constant 15 represent in this context?

(ii) Find when the ball hits the ground. Give exact answers in surd form.

(iii) Find the time to 2 d.p. and interpret your answer.

View Solution

(i) The constant term 15 is the initial height (height at $t = 0$).

(ii) Set $h = 0$:

$$-5t^2 + 15t + 15 = 0$$

$$5t^2 - 15t - 15 = 0$$

$$t = \frac{-(-15) \pm \sqrt{(-15)^2 - 4(5)(-15)}}{10}$$

$$t = \frac{15 \pm \sqrt{525}}{10}$$

$$t = \frac{15 \pm 5\sqrt{21}}{10}$$

$$t = \frac{3 \pm \sqrt{21}}{2}$$

(iii) Taking positive root: $t \approx 3.79$ s

This is the time when the ball hits the ground.

Answer: $t \approx 3.79$ s (time to hit ground)

Exercise 6.32.

A ball is thrown upward with initial velocity 15 m/s. Its height (m) after t seconds is $h = -5t^2 + 15t + 5$.

(i) What does the constant 5 represent in this context?

(ii) Find when the ball hits the ground. Give exact answers in surd form.

(iii) Find the time to 2 d.p. and interpret your answer.

View Solution

(i) The constant term 5 is the initial height (height at $t = 0$).

(ii) Set $h = 0$:

$$-5t^2 + 15t + 5 = 0$$

$$5t^2 - 15t - 5 = 0$$

$$t = \frac{-(-15) \pm \sqrt{(-15)^2 - 4(5)(-5)}}{10}$$

$$t = \frac{15 \pm \sqrt{325}}{10}$$

$$t = \frac{15 \pm 5\sqrt{13}}{10}$$

$$t = \frac{3 \pm \sqrt{13}}{2}$$

(iii) Taking positive root: $t \approx 3.30$ s

This is the time when the ball hits the ground.

Answer: $t \approx 3.30$ s (time to hit ground)

Exercise 6.33.

A right-angled triangle has legs x cm and $(x + 2)$ cm. The hypotenuse is 10 cm.

(i) Use Pythagoras' theorem to form a quadratic equation.

(ii) Solve the equation exactly.

(iii) Find the lengths of the two legs.

View Solution

$$\text{(i) Pythagoras: } x^2 + (x+2)^2 = 10^2$$
$$x^2 + x^2 + 4x + 4 = 100$$
$$2x^2 + 4x - 96 = 0$$
$$\text{(ii) } x = \frac{-(4) \pm \sqrt{(4)^2 - 4(2)(-96)}}{4}$$
$$x = \frac{-4 \pm 28}{4}$$
$$x = 6 \text{ or } x = -8$$
$$\text{(iii) } x = 6 \text{ cm}$$

Answer: $x = 6$ cm, legs: 6 and 8 cm

Exercise 6.34.

A right-angled triangle has legs x cm and $(x+4)$ cm. The hypotenuse is 20 cm.

(i) Use Pythagoras' theorem to form a quadratic equation.

(ii) Solve the equation exactly.

(iii) Find the lengths of the two legs.

View Solution

$$\text{(i) Pythagoras: } x^2 + (x+4)^2 = 20^2$$
$$x^2 + x^2 + 8x + 16 = 400$$
$$2x^2 + 8x - 384 = 0$$
$$\text{(ii) } x = \frac{-(8) \pm \sqrt{(8)^2 - 4(2)(-384)}}{4}$$
$$x = \frac{-8 \pm 56}{4}$$
$$x = 12 \text{ or } x = -16$$
$$\text{(iii) } x = 12 \text{ cm}$$

Answer: $x = 12$ cm, legs: 12 and 16 cm

Exercise 6.35. *Find the value(s) of k for which the equation has real roots.*

a) $n^2 - 6n + k = 0$

b) $m^2 + km + 9 = 0$

c) $x^2 + 4x + k = 0$

d) $y^2 + 6y + k = 0$

View Solutions

a) $n^2 - 6n + k = 0$

$$\begin{aligned} \text{Given: } n^2 - 6n + k &= 0 \\ \text{For real roots: } \Delta &\geq 0 \\ (-6)^2 - 4k &\geq 0 \\ 36 &\geq 4k \\ k &\leq 9 \end{aligned}$$

b) $m^2 + km + 9 = 0$

$$\begin{aligned} \text{Given: } m^2 + km + 9 &= 0 \\ \text{For real roots: } \Delta &\geq 0 \\ k^2 - 4(9) &\geq 0 \\ k^2 &\geq 36 \\ k \leq -6 \text{ or } k &\geq 6 \end{aligned}$$

c) $x^2 + 4x + k = 0$

$$\begin{aligned} \text{Given: } x^2 + 4x + k &= 0 \\ \text{For real roots: } \Delta &\geq 0 \\ (4)^2 - 4k &\geq 0 \\ 16 &\geq 4k \\ k &\leq 4 \end{aligned}$$

d) $y^2 + 6y + k = 0$

$$\begin{aligned}\text{Given: } y^2 + 6y + k &= 0 \\ \text{For real roots: } \Delta &\geq 0 \\ (6)^2 - 4k &\geq 0 \\ 36 &\geq 4k \\ k &\leq 9\end{aligned}$$

Exercise 6.36.

Two consecutive positive odd integers: the product of the larger and twice the smaller is 30.

(i) Form a quadratic equation in n.

(ii) Solve the equation exactly.

(iii) Find the two integers.

View Solution

$$\begin{aligned}text(i)(n+2)(2n) &= 30 \\ 2n^2 + 4n - 30 &= 0 \\ text(ii)n &= \frac{-(4) \pm \sqrt{(4)^2 - 4(2)(-30)}}{4} \\ n &= \frac{-4 \pm 16}{4} \\ n = 3 \text{ or } n &= -5 \\ textSinceintegersarepositiveodd, n &= 3 \\ textIntegers: &3 \\ textand5 &\end{aligned}$$

Answer: $Integers: 3and5$

Exercise 6.37.

Two consecutive positive odd integers: the product of the larger and twice the smaller is 646.

(i) Form a quadratic equation in n.

(ii) Solve the equation exactly.

(iii) Find the two integers.

View Solution

$$\begin{aligned} text(i)(n+2)(2n) &= 646 \\ 2n^2 + 4n - 646 &= 0 \\ text(ii)n &= \frac{-(4) \pm \sqrt{(4)^2 - 4(2)(-646)}}{4} \\ n &= \frac{-4 \pm 72}{4} \\ n = 17 \text{ or } n &= -19 \\ textSinceintegersarepositiveodd, n &= 17 \\ textIntegers: &17 \\ textand &19 \end{aligned}$$

Answer: $Integers: 17and19$

Exercise 6.38.

The product of two consecutive positive odd integers is 15.

(i) Form a quadratic equation in n.

(ii) Solve the equation exactly.

(iii) Find the two integers.

View Solution

$$\begin{aligned} text(i)n(n+2) &= 15 \\ n^2+2n-15 &= 0 \\ text(ii)n &= \frac{-(2)\pm\sqrt{(2)^2-4(1)(-15)}}{2} \\ n &= \frac{-2\pm 8}{2} \\ n = 3 \text{ or } n &= -5 \\ textSinceintegersarepositiveodd, n &= 3 \\ textIntegers: &3 \\ textand5 \end{aligned}$$

Answer: $Integers: 3and5$

Exercise 6.39. *The equations below have roots α and β. Use Vieta's formulas to find the given expression.*

a) $2m^2+6m-5=0$

b) $x^2+6x+6=0$

c) $x^2+2x-3=0$

d) $y^2+y-2=0$

View Solutions

a) $2m^2+6m-5=0$

$$\text{From } 2m^2 + 6m - 5 = 0 :$$
$$\alpha + \beta = \frac{-b}{a} = \frac{-6}{2} = -3$$
$$\alpha\beta = \frac{c}{a} = \frac{-5}{2} = -\frac{5}{2}$$
$$\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta} = \frac{-3}{-\frac{5}{2}} = \frac{6}{5}$$

b) $x^2 + 6x + 6 = 0$

$$\text{From } x^2 + 6x + 6 = 0 :$$
$$\alpha + \beta = \frac{-b}{a} = \frac{-6}{1} = -6$$
$$\alpha\beta = \frac{c}{a} = \frac{6}{1} = 6$$
$$\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta} = \frac{-6}{6} = -1$$

c) $x^2 + 2x - 3 = 0$

$$\text{From } x^2 + 2x - 3 = 0 :$$
$$\alpha + \beta = \frac{-b}{a} = \frac{-2}{1} = -2$$
$$\alpha\beta = \frac{c}{a} = \frac{-3}{1} = -3$$
$$(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta = (-2)^2 - 4(-3) = 16$$

d) $y^2 + y - 2 = 0$

$$\text{From } y^2 + y - 2 = 0:$$
$$\alpha + \beta = \frac{-b}{a} = \frac{-1}{1} = -1$$
$$\alpha\beta = \frac{c}{a} = \frac{-2}{1} = -2$$
$$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta = (-1)^2 - 2(-2) = 5$$

Exercise 6.40.

x + 5x - 22x - 1

Shown is a right-angled triangle. All sides are in cm.

(a) Show that $x^2 - 5x - 14 = 0$.

(b) Find x.

(c) Find the area of the triangle.

View Solution

$$\text{(a) Pythagoras: } (x+5)^2 + (x-2)^2 = (2x-1)^2$$
$$x^2 - 5x - 14 = 0$$
$$x^2 - 5x - 14 = 0$$
$$\text{(b) } x = \frac{-(-5) \pm \sqrt{(-5)^2 - 4(1)(-14)}}{2}$$
$$x = \frac{5 \pm 9}{2}$$
$$x = 7 \text{ or } x = -2$$
$$\text{Taking positive root: } x = 7$$
$$\text{(c) Legs: } 12 \text{ cm and } 5 \text{ cm}$$
$$\text{Area } = \tfrac{1}{2} \times 12 \times 5 = 30 \text{ cm}^2$$

Answer: $x = 7,$ Area $= 30$ cm^2

Exercise 6.41.

x + 5x - 22x - 1

Shown is a right-angled triangle. All sides are in cm.

(a) Show that $x^2 - 5x - 14 = 0$.

(b) Find x.

(c) Find the area of the triangle.

View Solution

$$\text{(a) Pythagoras: } (x+5)^2 + (x-2)^2 = (2x-1)^2$$
$$x^2 - 5x - 14 = 0$$
$$x^2 - 5x - 14 = 0$$
$$\text{(b) } x = \frac{-(-5) \pm \sqrt{(-5)^2 - 4(1)(-14)}}{2}$$
$$x = \frac{5 \pm 9}{2}$$
$$x = 7 \text{ or } x = -2$$
$$\text{Taking positive root: } x = 7$$
$$\text{(c) Legs: } 12 \text{ cm and } 5 \text{ cm}$$
$$\text{Area } = \tfrac{1}{2} \times 12 \times 5 = 30 \text{ cm}^2$$

Answer: $x = 7,$ Area $= 30$ cm^2

Exercise 6.42.

x+33x−224

The area of the L-shaped region is 234 cm². The big rectangle is $(3x-2)$ cm by $(x+3)$ cm with a 2 cm $\times$ 4 cm rectangle removed.

(a) Show that $3x^2 + 7x - 248 = 0$.

(b) Solve to find x.

View Solution

$$\text{(a) Area: } (3x-2)(x+3)-8=234$$
$$3x^2+7x-6-8=234$$
$$3x^2+7x-248=0$$
$$3x^2+7x-248=0$$
$$\text{(b) } x=\frac{-(7)\pm\sqrt{(7)^2-4(3)(-248)}}{6}$$
$$x=\frac{-7\pm 55}{6}$$
$$x=8 \text{ or } x=-\frac{31}{3}$$
$$\text{Taking positive root: } x=8$$

Answer: $x=8$

Exercise 6.43.

x+73x−221

The area of the L-shaped region is 206 cm². The big rectangle is $(3x-2)$ cm by $(x+7)$ cm with a 2 cm $\times$ 1 cm rectangle removed.

(a) Show that $3x^2+19x-222=0$.

(b) Solve to find x.

View Solution

$$\text{(a) Area: } (3x-2)(x+7)-2=206$$
$$3x^2+19x-14-2=206$$
$$3x^2+19x-222=0$$
$$3x^2+19x-222=0$$
$$\text{(b) } x=\frac{-(19)\pm\sqrt{(19)^2-4(3)(-222)}}{6}$$
$$x=\frac{-19\pm 55}{6}$$
$$x=6 \text{ or } x=-\frac{37}{3}$$
$$\text{Taking positive root: } x=6$$

Answer: $x=6$

Exercise 6.44.

A cuboid has dimensions x cm, x cm and $(x+5)$ cm. Its surface area is 544 cm².

(a) Show that $3x^2+10x-272=0$.

(b) Find x.

(c) Find the volume of the cuboid.

View Solution

$$\text{(a) Surface area: } 2x^2 + 4x(x+5) = 544$$
$$6x^2 + 20x - 544 = 0$$
$$3x^2 + 10x - 272 = 0$$
$$\text{(b) } x = \frac{-(10) \pm \sqrt{(10)^2 - 4(3)(-272)}}{6}$$
$$x = \frac{-10 \pm 58}{6}$$
$$x = 8 \text{ or } x = -\frac{34}{3}$$
$$x = 8 \text{ cm}$$
$$\text{(c) Volume } = 8^2 \cdot (13) = 832 \text{ cm}^3$$

Answer: $x = 8,\ V = 832$ cm^3

Exercise 6.45.

A cuboid has dimensions x cm, x cm and $(x+6)$ cm. Its surface area is 840 cm².

(a) Show that $x^2 + 4x - 140 = 0$.

(b) Find x.

(c) Find the volume of the cuboid.

View Solution

$$\text{(a) Surface area: } 2x^2 + 4x(x+6) = 840$$
$$6x^2 + 24x - 840 = 0$$
$$x^2 + 4x - 140 = 0$$
$$\text{(b) } x = \frac{-(4) \pm \sqrt{(4)^2 - 4(1)(-140)}}{2}$$
$$x = \frac{-4 \pm 24}{2}$$
$$x = 10 \text{ or } x = -14$$
$$x = 10 \text{ cm}$$
$$\text{(c) Volume } = 10^2 \cdot (16) = 1600 \text{ cm}^3$$

Answer: $x = 10,\ V = 1600$ cm^3

Exercise 6.46.

x2x−228

A cuboid has dimensions x cm, $(2x - 22)$ cm and 8 cm. Its volume is 416 cm³.

(a) Show that $2x^2 - 22x - 52 = 0$.

(b) Find x.

View Solution

$$\text{(a) Volume: } 8 \times x \times (2x - 22) = 416$$
$$8x(2x - 22) = 416$$
$$2x^2 - 22x - 52 = 0$$
$$\text{(b) } x = \frac{-(-22) \pm \sqrt{(-22)^2 - 4(2)(-52)}}{4}$$
$$x = \frac{22 \pm 30}{4}$$
$$x = 13 \text{ or } x = -2$$
$$\text{Taking positive root: } x = 13 \text{ cm}$$

Answer: $x = 13$

Exercise 6.47.

x2x−2215

A cuboid has dimensions x cm, $(2x - 22)$ cm and 15 cm. Its volume is 3780 cm³.

(a) Show that $2x^2 - 22x - 252 = 0$.

(b) Find x.

View Solution

$$\begin{aligned}
\text{(a) Volume: } 15 \times x \times (2x - 22) &= 3780 \\
15x(2x - 22) &= 3780 \\
2x^2 - 22x - 252 &= 0 \\
\text{(b) } x &= \frac{-(-22) \pm \sqrt{(-22)^2 - 4(2)(-252)}}{4} \\
x &= \frac{22 \pm 50}{4} \\
x = 18 \text{ or } x &= -7 \\
\text{Taking positive root: } x &= 18 \text{ cm}
\end{aligned}$$

Answer: $x = 18$

Exercise 6.48. *Solve. State any values of x that must be excluded.*

a) $\dfrac{4}{x} + \dfrac{4}{x-3} = 2$

b) $\dfrac{5}{x} + \dfrac{10}{x-6} = 3$

c) $\dfrac{6}{x} + \dfrac{12}{x-5} = 3$

d) $\dfrac{6}{x} + \dfrac{12}{x-5} = 3$

View Solutions

a) $\dfrac{4}{x} + \dfrac{4}{x-3} = 2$

$$\begin{aligned} &\text{Multiply through by } x(x-3): \\ 4(x-3) + 4x &= 2x(x-3) \\ 2x^2 - 14x + 12 &= 0 \\ (x-6)(x-1) &= 0 \\ x = 6 \text{ or } x &= 1 \end{aligned}$$

b) $\dfrac{5}{x} + \dfrac{10}{x-6} = 3$

$$\begin{aligned} &\text{Multiply through by } x(x-6): \\ 5(x-6) + 10x &= 3x(x-6) \\ 3x^2 - 33x + 30 &= 0 \\ (x-10)(x-1) &= 0 \\ x = 10 \text{ or } x &= 1 \end{aligned}$$

c) $\dfrac{6}{x} + \dfrac{12}{x-5} = 3$

$$\begin{aligned} &\text{Multiply through by } x(x-5): \\ 6(x-5) + 12x &= 3x(x-5) \\ 3x^2 - 33x + 30 &= 0 \\ (x-10)(x-1) &= 0 \\ x = 10 \text{ or } x &= 1 \end{aligned}$$

d) $\dfrac{6}{x} + \dfrac{12}{x-5} = 3$

$$\begin{aligned} &\text{Multiply through by } x(x-5): \\ 6(x-5) + 12x &= 3x(x-5) \\ 3x^2 - 33x + 30 &= 0 \\ (x-10)(x-1) &= 0 \\ x = 10 \text{ or } x &= 1 \end{aligned}$$

Exercise 6.49. *Solve. State any values of x that must be excluded.*

a) $\dfrac{-4}{x+5} + \dfrac{2}{x-1} = -2$

b) $\dfrac{-3}{x+1} - \dfrac{7}{x-3} = 2$

c) $\dfrac{6}{x+1} - \dfrac{1}{x-1} = 1$

d) $\dfrac{8}{x+4} + \dfrac{8}{x-2} = 2$

View Solutions

a) $\dfrac{-4}{x+5} + \dfrac{2}{x-1} = -2$

$$\begin{aligned} \text{Multiply by } (x+5)(x-1): \\ -4(x-1) + 2(x+5) &= -2(x+5)(x-1) \\ -2x^2 - 6x - 4 &= 0 \\ (x+2)(x+1) &= 0 \\ x = -2 \text{ or } x &= -1 \end{aligned}$$

b) $\dfrac{-3}{x+1} - \dfrac{7}{x-3} = 2$

$$\begin{aligned} \text{Multiply by } (x+1)(x-3): \\ -3(x-3) - 7(x+1) &= 2(x+1)(x-3) \\ 2x^2 + 6x - 8 &= 0 \\ (x+4)(x-1) &= 0 \\ x = -4 \text{ or } x &= 1 \end{aligned}$$

c) $\dfrac{6}{x+1} - \dfrac{1}{x-1} = 1$

$$\begin{aligned} \text{Multiply by } (x+1)(x-1): \\ 6(x-1) - 1(x+1) &= 1(x+1)(x-1) \\ x^2 - 5x + 6 &= 0 \\ (x-2)(x-3) &= 0 \\ x = 2 \text{ or } x &= 3 \end{aligned}$$

d) $\dfrac{8}{x+4} + \dfrac{8}{x-2} = 2$

$$\begin{aligned} \text{Multiply by } (x+4)(x-2): \\ 8(x-2) + 8(x+4) &= 2(x+4)(x-2) \\ 2x^2 - 12x - 32 &= 0 \\ (x-8)(x+2) &= 0 \\ x = 8 \text{ or } x &= -2 \end{aligned}$$

Exercise 6.50. *Solve using the substitution $u = x^2$.*

a) $x^4 - 10x^2 + 16 = 0$

b) $x^4 - 6x^2 + 5 = 0$

c) $x^4 - 5x^2 + 4 = 0$

d) $x^4 - 5x^2 + 4 = 0$

View Solutions

a) $x^4 - 10x^2 + 16 = 0$

$$\begin{aligned} \text{Let } u = x^2: \quad u^2 - 10u + 16 &= 0 \\ (u-2)(u-8) &= 0 \\ u = 2 \text{ or } u &= 8 \\ x^2 = 2 \Rightarrow x &= \pm\sqrt{2} \\ x^2 = 8 \Rightarrow x &= \pm\sqrt{8} \end{aligned}$$

b) $x^4 - 6x^2 + 5 = 0$

$$\begin{aligned} \text{Let } u = x^2: \quad u^2 - 6u + 5 &= 0 \\ (u-1)(u-5) &= 0 \\ u = 1 \text{ or } u &= 5 \\ x^2 = 1 &\Rightarrow x = \pm 1 \\ x^2 = 5 &\Rightarrow x = \pm\sqrt{5} \end{aligned}$$

c) $x^4 - 5x^2 + 4 = 0$

$$\begin{aligned} \text{Let } u = x^2: \quad u^2 - 5u + 4 &= 0 \\ (u-4)(u-1) &= 0 \\ u = 4 \text{ or } u &= 1 \\ x^2 = 4 &\Rightarrow x = \pm 2 \\ x^2 = 1 &\Rightarrow x = \pm 1 \end{aligned}$$

d) $x^4 - 5x^2 + 4 = 0$

$$\begin{aligned} \text{Let } u = x^2: \quad u^2 - 5u + 4 &= 0 \\ (u-4)(u-1) &= 0 \\ u = 4 \text{ or } u &= 1 \\ x^2 = 4 &\Rightarrow x = \pm 2 \\ x^2 = 1 &\Rightarrow x = \pm 1 \end{aligned}$$

Exercise 6.51. *Solve using an appropriate substitution.*

a) $(x+5)^2 + 8(x+5) + 12 = 0$

b) $(x+2)^2 - 5(x+2) - 6 = 0$

c) $(x+1)^2 + 7(x+1) + 10 = 0$

d) $(x+4)^2 + 5(x+4) + 4 = 0$

View Solutions

a) $(x+5)^2 + 8(x+5) + 12 = 0$

$$\begin{aligned} \text{Let } u = (x+5): \quad u^2 + 8u + 12 &= 0 \\ (u+2)(u+6) &= 0 \\ u = -2 \Rightarrow x+5 = -2 &\Rightarrow x = -7 \\ u = -6 \Rightarrow x+5 = -6 &\Rightarrow x = -11 \end{aligned}$$

b) $(x+2)^2 - 5(x+2) - 6 = 0$

$$\begin{aligned} \text{Let } u = (x+2): \quad u^2 - 5u - 6 &= 0 \\ (u+1)(u-6) &= 0 \\ u = -1 \Rightarrow x+2 = -1 &\Rightarrow x = -3 \\ u = 6 \Rightarrow x+2 = 6 &\Rightarrow x = 4 \end{aligned}$$

c) $(x+1)^2 + 7(x+1) + 10 = 0$

$$\begin{aligned} \text{Let } u = (x+1): \quad u^2 + 7u + 10 &= 0 \\ (u+2)(u+5) &= 0 \\ u = -2 \Rightarrow x+1 = -2 &\Rightarrow x = -3 \\ u = -5 \Rightarrow x+1 = -5 &\Rightarrow x = -6 \end{aligned}$$

d) $(x+4)^2 + 5(x+4) + 4 = 0$

$$\begin{aligned} \text{Let } u = (x+4): \quad u^2 + 5u + 4 &= 0 \\ (u+1)(u+4) &= 0 \\ u = -1 \Rightarrow x+4 = -1 &\Rightarrow x = -5 \\ u = -4 \Rightarrow x+4 = -4 &\Rightarrow x = -8 \end{aligned}$$

Exercise 6.52. *Solve. Check for extraneous roots.*

a) $\sqrt{x+74} = x+2$

b) $\sqrt{x+29} = x-1$

c) $\sqrt{x+41} = x - 1$

d) $\sqrt{x+8} = x + 2$

View Solutions

a) $\sqrt{x+74} = x + 2$

$$\begin{aligned} \text{Square both sides: } x + 74 &= (x+2)^2 \\ x^2 + 3x - 70 &= 0 \\ (x-7)(x+10) &= 0 \\ x = 7 \text{ or } x &= -10 \\ \text{Check both in original equation:} \\ \text{Check } x = -10 : 8 = 8 &\neq -8 \text{ (extraneous)} \\ \text{Valid: } x &= 7 \end{aligned}$$

b) $\sqrt{x+29} = x - 1$

$$\begin{aligned} \text{Square both sides: } x + 29 &= (x-1)^2 \\ x^2 - 3x - 28 &= 0 \\ (x-7)(x+4) &= 0 \\ x = 7 \text{ or } x &= -4 \\ \text{Check both in original equation:} \\ \text{Check } x = -4 : 5 = 5 &\neq -5 \text{ (extraneous)} \\ \text{Valid: } x &= 7 \end{aligned}$$

c) $\sqrt{x+41} = x - 1$

$$\begin{aligned} \text{Square both sides: } x + 41 &= (x-1)^2 \\ x^2 - 3x - 40 &= 0 \\ (x-8)(x+5) &= 0 \\ x = 8 \text{ or } x &= -5 \\ \text{Check both in original equation:} \\ \text{Check } x = -5 : 6 = 6 &\neq -6 \text{ (extraneous)} \\ \text{Valid: } x &= 8 \end{aligned}$$

d) $\sqrt{x+8} = x+2$

$$\begin{aligned}
\text{Square both sides: } x+8 &= (x+2)^2 \\
x^2+3x-4 &= 0 \\
(x-1)(x+4) &= 0 \\
x = 1 \text{ or } x &= -4 \\
\text{Check both in original equation:} & \\
\text{Check } x=-4: 2 = 2 \neq -2 \text{ (extraneous)} & \\
\text{Valid: } x &= 1
\end{aligned}$$

Exercise 6.53. *Solve. Check for extraneous roots.*

a) $\sqrt{x+3} - \sqrt{7-x} = 2$

b) $\sqrt{x+5} - \sqrt{8-x} = 1$

c) $\sqrt{x+1} - \sqrt{4-x} = 1$

View Solutions

a) $\sqrt{x+3} - \sqrt{7-x} = 2$

$$\begin{aligned}
\text{Isolate: } \sqrt{x+3} &= 2+\sqrt{7-x} \\
\text{Square: } x+3 &= 2^2+2(2)\sqrt{7-x}+(7-x) \\
2(2)\sqrt{7-x} &= 2x-8 \\
\text{Square again: } 4(4)(7-x) &= (2x-8)^2 \\
4x^2-16x-48 &= 0 \\
x &= 6 \\
\text{(check both roots in original)} &
\end{aligned}$$

b) $\sqrt{x+5} - \sqrt{8-x} = 1$

$$\text{Isolate: } \sqrt{x+5} = 1 + \sqrt{8-x}$$
$$\text{Square: } x+5 = 1^2 + 2(1)\sqrt{8-x} + (8-x)$$
$$2(1)\sqrt{8-x} = 2x-4$$
$$\text{Square again: } 4(1)(8-x) = (2x-4)^2$$
$$4x^2 - 12x - 16 = 0$$
$$x = 4$$
(check both roots in original)

c) $\sqrt{x+1} - \sqrt{4-x} = 1$

$$\text{Isolate: } \sqrt{x+1} = 1 + \sqrt{4-x}$$
$$\text{Square: } x+1 = 1^2 + 2(1)\sqrt{4-x} + (4-x)$$
$$2(1)\sqrt{4-x} = 2x-4$$
$$\text{Square again: } 4(1)(4-x) = (2x-4)^2$$
$$4x^2 - 12x + 0 = 0$$
$$x = 3$$
(check both roots in original)

Exercise 6.54. *Solve. (Hint: use the index law $(a^m)^n = a^{mn}$, equate exponents, then solve the resulting quadratic.)*

a) $\left(2^{x-2}\right)^{x-2} = 16$

b) $\left(2^{x}\right)^{x-4} = 32$

c) $\left(2^{x-3}\right)^{x-3} = 16$

d) $\left(2^{x-4}\right)^{x} = 32$

View Solutions

a) $\left(2^{x-2}\right)^{x-2} = 16$

$$\text{Use index law: } (a^m)^n = a^{mn}$$
$$2^{(x-2)(x-2)} = 2^4$$
$$\text{Equate exponents: } (x-2)(x-2) = 4$$
$$x^2 - 4x + 4 = 4$$
$$x^2 - 4x = 0$$
$$x = \frac{-(-4) \pm \sqrt{(-4)^2 - 4(1)(0)}}{2}$$
$$x = \frac{4 \pm 4}{2}$$
$$x = 4 \text{ or } x = 0$$

b) $(2^x)^{x-4} = 32$

$$\text{Use index law: } (a^m)^n = a^{mn}$$
$$2^{(x)(x-4)} = 2^5$$
$$\text{Equate exponents: } (x)(x-4) = 5$$
$$x^2 - 4x = 5$$
$$x^2 - 4x - 5 = 0$$
$$x = \frac{-(-4) \pm \sqrt{(-4)^2 - 4(1)(-5)}}{2}$$
$$x = \frac{4 \pm 6}{2}$$
$$x = 5 \text{ or } x = -1$$

c) $(2^{x-3})^{x-3} = 16$

$$\text{Use index law: } (a^m)^n = a^{mn}$$
$$2^{(x-3)(x-3)} = 2^4$$
$$\text{Equate exponents: } (x-3)(x-3) = 4$$
$$x^2 - 6x + 9 = 4$$
$$x^2 - 6x + 5 = 0$$
$$x = \frac{-(-6) \pm \sqrt{(-6)^2 - 4(1)(5)}}{2}$$
$$x = \frac{6 \pm 4}{2}$$
$$x = 5 \text{ or } x = 1$$

d) $\left(2^{x-4}\right)^x = 32$

$$\text{Use index law: } (a^m)^n = a^{mn}$$
$$2^{(x-4)(x)} = 2^5$$
$$\text{Equate exponents: } (x-4)(x) = 5$$
$$x^2 - 4x = 5$$
$$x^2 - 4x - 5 = 0$$
$$x = \frac{-(-4) \pm \sqrt{(-4)^2 - 4(1)(-5)}}{2}$$
$$x = \frac{4 \pm 6}{2}$$
$$x = 5 \text{ or } x = -1$$

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6.4 Test Your Knowledge

7 Unit 7: Linear & Quadratic Functions

7.1 Learning Targets

LT1

Relations & Functions

Represent relations as sets, tables, and arrow diagrams, and determine whether a relation is a function using the vertical line test.

Sets & tables Arrow diagrams Vertical line test

LT2

Representations

Move fluently between tables, graphs, and equations for linear functions.

Table to graph Graph to equation Multiple representations

LT3

Domain & Range

Identify the domain and range of a relation from a set, table, or graph.

Input & output Set notation From graph

LT4

Linear Functions

Identify, evaluate, and interpret linear functions in real-world contexts.

Evaluate $f(x)$ Identify linear Interpret context

LT5

Slope & Rate of Change

Calculate slope from two points or a table and interpret it as a rate of change in context.

Slope formula From a table Rate of change

LT6

Linear Equations & Graphs

Write and graph linear equations; find equations of parallel and perpendicular lines.

Slope-intercept Parallel lines Perpendicular lines

LT7

Linear Modelling

Form, solve, and interpret linear models for real-world situations.

Form the model Solve & predict Interpret results

LT8

Quadratic Functions

Identify quadratic functions, evaluate them, and convert between vertex, standard, and factored forms.

Vertex form Standard form Factored form

LT9

Key Features of Graphs

Determine the vertex, roots, and axis of symmetry of a parabola from its equation or graph.

Vertex Roots / x-intercepts Axis of symmetry

LT12

Domain & Range — Quadratic

Determine and interpret the domain and range of a quadratic function from its equation or graph.

Domain: all reals Range from vertex Opens up or down

LT13

Quadratic Modelling

Model and interpret real-world situations using quadratic functions, including projectile motion and revenue problems.

Projectile motion Revenue problems Interpret vertex

7.2 Formula Card

Linear & Quadratic Functions — Key Facts

Relations & Functions

 Key Test A relation is a **function** if every input (x-value) maps to **exactly one** output (y-value). Use the vertical line test on a graph.

Function notation: $f(x) = mx + c$ means "the output when the input is x"

Domain: the set of all valid **inputs** (x-values)

Range: the set of all valid **outputs** (y-values)

Examples:

$$f(x) = 2x + 1 \quad \Rightarrow \quad f(3) = 2(3) + 1 = 7$$

Linear Functions

 Slope Formula

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{\Delta y}{\Delta x}$$

Slope-intercept form: $y = mx + c$, where m = slope, c = y-intercept

Parallel lines share the **same slope**: $m_1 = m_2$

Perpendicular lines: slopes are **negative reciprocals**: $m_1 \times m_2 = -1$

Examples:

$$\text{Through } (2, 5) \text{ with slope } 3: \quad y - 5 = 3(x - 2) \quad \Rightarrow \quad y = 3x - 1$$

$$\text{Perpendicular to } y = 2x + 1: \quad m_\perp = -\tfrac{1}{2}$$

Quadratic Functions — Three Forms

⚠ **Conversion Tip** Each form reveals different information. Convert using expansion or completing the square.

Form	Equation	Reveals
Standard	$f(x) = ax^2 + bx + c$	y-intercept $(0, c)$
Vertex	$f(x) = a(x-h)^2 + k$	Vertex (h, k)
Factored	$f(x) = a(x - r_1)(x - r_2)$	Roots $x = r_1, r_2$

Vertex formula (from standard form):

$$x_{\text{vertex}} = \frac{-b}{2a}$$

Axis of symmetry: $x = \dfrac{-b}{2a} = h$

Converting Between Forms

Vertex → Standard (expand):

$$2(x-3)^2 + 1 = 2(x^2 - 6x + 9) + 1 = 2x^2 - 12x + 19$$

Vertex → Factored (find roots first):

$$(x-3)^2 - 4 = 0 \Rightarrow x = 3 \pm 2 \Rightarrow f(x) = (x-1)(x-5)$$

Standard → Vertex (complete the square):

$$x^2 - 6x + 5 = (x-3)^2 - 9 + 5 = (x-3)^2 - 4$$

Factored → Standard (expand):

$$(x-1)(x-5) = x^2 - 6x + 5$$

Domain & Range

 Quick Rule Every quadratic has domain $x \in \mathbb{R}$. The range depends on whether the parabola opens up or down.

Function type	Domain	Range
Linear $f(x) = mx + c$	$x \in \mathbb{R}$	$y \in \mathbb{R}$
Quadratic, $a > 0$ (opens up)	$x \in \mathbb{R}$	$y \geq k$
Quadratic, $a < 0$ (opens down)	$x \in \mathbb{R}$	$y \leq k$

7.3 Practice Problems

7.3.1 Easy

Exercise 7.1. *Analyse the arrow diagram.*

The arrow diagram shows a relation.

Domain Range 1 6 3 4 2 6

(a) State the domain and range of this relation.

(b) Is this relation a function? Justify your answer.

View Solution

Check: does each domain element have exactly one arrow?

No — one input maps to more than one output.

Function definition Domain & range Arrow diagrams

Exercise 7.2. *Analyse the relation.*

Consider the relation: $\{\,(2,\ -1), (3,\ 4), (-4,\ -1), (5,\ -3), (5,\ -2)\,\}$

(a) State the domain and range.

(b) Is this a function? Explain.

View Solution

Domain: $\{-4, 2, 3, 5\}$

Range: $\{-3, -2, -1, 4\}$

Function? No — one x-value maps to two outputs.

Function definition Domain & range Arrow diagrams

Exercise 7.3. *Find the equation of the linear function shown in the table.*

a)

x	-2	-1	3	4
y	3	0	-12	-15

b)

x	-3	-2	0	3
y	-3	-2	0	3

c)

x	-3	0	1	3
y	-9	0	3	9

d)

x	-3	0	2	3
y	-5	4	10	13

View Solutions

a)

$$\text{Slope: } m = \frac{\Delta y}{\Delta x} = \frac{-3}{1} = -3$$
$$\text{Substitute } (-2, 3) : 3 = -3(-2) + c$$
$$c = -3$$
$$y = -3x - 3$$

b)

$$\text{Slope: } m = \frac{\Delta y}{\Delta x} = \frac{1}{1} = 1$$
$$\text{Substitute } (-3, -3) : -3 = 1(-3) + c$$
$$c = 0$$
$$y = x$$

c)

$$\text{Slope: } m = \frac{\Delta y}{\Delta x} = \frac{9}{3} = 3$$
$$\text{Substitute } (-3, -9) : -9 = 3(-3) + c$$
$$c = 0$$
$$y = 3x$$

d)

$$\text{Slope: } m = \frac{\Delta y}{\Delta x} = \frac{9}{3} = 3$$
$$\text{Substitute } (-3, -5) : -5 = 3(-3) + c$$
$$c = 4$$
$$y = 3x + 4$$

Table of values Linear equations Representations

Exercise 7.4. *Complete the table of values.*

a) $y = -2x + 2$

x	-1	0	1	4
y	$\square$	2	0	$\square$

b) $y = -x$

x	-2	-1	3	4
y	2	1	$\square$	$\square$

c) $y = 2x + 2$

x	-1	0	2	3
y	0	$\square$	6	$\square$

d) $y = -x - 4$

x	0	1	3	4
y	-4	$\square$	$\square$	-8

 View Solutions

a)

$$x = -1 : \quad y = -2(-1) + 2 = 4$$
$$x = 4 : \quad y = -2(4) + 2 = -6$$

b)

$$x = 3 : \quad y = -1(3) + 0 = -3$$
$$x = 4 : \quad y = -1(4) + 0 = -4$$

c)

$$x = 0 : \quad y = 2(0) + 2 = 2$$
$$x = 3 : \quad y = 2(3) + 2 = 8$$

d)

$$x = 1: \quad y = -1(1) - 4 = -5$$
$$x = 3: \quad y = -1(3) - 4 = -7$$

Table of values Linear equations Representations

Exercise 7.5. *State the domain and range of each relation.*

a) $\{(1,\ -1), (3,\ 2), (0,\ -2), (-1,\ 4), (4,\ 0)\}$

b) $\{(2,\ 2), (1,\ -1), (-2,\ 4), (3,\ -3), (-4,\ -3)\}$

c) $\{(2,\ -3), (3,\ 4), (0,\ 4), (-4,\ -3), (4,\ 0)\}$

d) $\{(1,\ -3), (0,\ 2), (-2,\ 3), (-4,\ 0), (4,\ -2)\}$

View Solutions

a)

Domain (all x-values): $\{-1, 0, 1, 3, 4\}$

Range (all y-values): $\{-2, -1, 0, 2, 4\}$

b)

Domain (all x-values): $\{-4, -2, 1, 2, 3\}$

Range (all y-values): $\{-3, -1, 2, 4\}$

c)

Domain (all x-values): $\{-4, 0, 2, 3, 4\}$

Range (all y-values): $\{-3, 0, 4\}$

d)

Domain (all x-values): $\{-4, -2, 0, 1, 4\}$

Range (all y-values): $\{-3, -2, 0, 2, 3\}$

Domain Range Set notation

Exercise 7.6. *State the domain and range of each function.*

a) $f(x) = 3x - 1$

b) $f(x) = 3x + 3$

c) $f(x) = 2x - 2$

d) $f(x) = -2x$

View Solutions

a)

Linear functions are defined for all real numbers.

Domain: $x \in \mathbb{R}$

Range: $y \in \mathbb{R}$

b)

Linear functions are defined for all real numbers.

Domain: $x \in \mathbb{R}$

Range: $y \in \mathbb{R}$

c)

Linear functions are defined for all real numbers.

Domain: $x \in \mathbb{R}$

Range: $y \in \mathbb{R}$

d)

Linear functions are defined for all real numbers.

Domain: $x \in \mathbb{R}$

Range: $y \in \mathbb{R}$

Domain Range Set notation

Exercise 7.7. *Identify which expression represents a linear function.*

a) A: $|t| + 1$ B: $\frac{1}{t} + 4$ C: $2t^2 + t + 1$ D: $-2t + 3$

b) A: $\frac{1}{y} + 1$ B: $|y| + 5$ C: $-2y - 5$ D: $2y^2 - 3y + 3$

c) A: $2n - 3$ B: $3n^2 - 2n + 4$ C: $\frac{1}{n} + 2$ D: $|n| + 3$

d) A: $2m^2 - 2m - 1$ B: $|m| + 3$ C: $\frac{1}{m} + 1$ D: $m + 3$

View Solutions

a)

A linear function has degree 1 (form $f(x) = mx + c$).

Answer: D — $f(x) = -2t + 3$ has degree 1. ✓

b)

A linear function has degree 1 (form $f(x) = mx + c$).

Answer: C — $f(x) = -2y - 5$ has degree 1. ✓

c)

A linear function has degree 1 (form $f(x) = mx + c$).

Answer: A — $f(x) = 2n - 3$ has degree 1. ✓

d)

A linear function has degree 1 (form $f(x) = mx + c$).

Answer: D — $f(x) = m + 3$ has degree 1. ✓

Linear functions Slope y-intercept

Exercise 7.8. *Evaluate the function at each given value.*

a) $f(x) = 2x - 3$ Find: $f(-3), f(4), f(2)$

b) $f(x) = 3x - 6$ Find: $f(0), f(-3), f(3)$

c) $f(x) = -x - 5$ Find: $f(0), f(-1), f(2)$

d) $f(x) = x - 3$ Find: $f(-1), f(2), f(1)$

View Solutions

a)

$$f(-3) = 2 \cdot (-3) - 3 = -9$$
$$f(4) = 2 \cdot (4) - 3 = 5$$
$$f(2) = 2 \cdot (2) - 3 = 1$$

b)

$$f(0) = 3 \cdot (0) - 6 = -6$$
$$f(-3) = 3 \cdot (-3) - 6 = -15$$
$$f(3) = 3 \cdot (3) - 6 = 3$$

c)

$$f(0) = -1 \cdot (0) - 5 = -5$$
$$f(-1) = -1 \cdot (-1) - 5 = -4$$
$$f(2) = -1 \cdot (2) - 5 = -7$$

d)

$$f(-1) = 1 \cdot (-1) - 3 = -4$$
$$f(2) = 1 \cdot (2) - 3 = -1$$
$$f(1) = 1 \cdot (1) - 3 = -2$$

Linear functions Slope y-intercept

Exercise 7.9. *Find the slope of the line through each pair of points.*

a) $(3,\ 2)$ and $(1,\ -1)$

b) $(-3,\ 2)$ and $(-4,\ 5)$

c) $(2,\ 3)$ and $(6,\ -1)$

d) $(-3,\ -4)$ and $(-2,\ -2)$

e) $(-3,\ -1)$ and $(-5,\ -2)$

f) $(3,\ -2)$ and $(4,\ -4)$

View Solutions

a)

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{-1-(2)}{1-(3)} = \frac{-3}{-2} = \frac{3}{2}$$

b)

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{5-(2)}{-4-(-3)} = \frac{3}{-1} = -3$$

c)

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{-1-(3)}{6-(2)} = \frac{-4}{4} = -1$$

d)

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{-2-(-4)}{-2-(-3)} = \frac{2}{1} = 2$$

e)

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{-2-(-1)}{-5-(-3)} = \frac{-1}{-2} = \frac{1}{2}$$

f)

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{-4-(-2)}{4-(3)} = \frac{-2}{1} = -2$$

Rate of change Slope formula Interpretation

Exercise 7.10. *Find the rate of change from the table.*

a)

x	-2	1	4	5
y	-1	5	11	13

b)

x	-2	-1	3	5
y	0	-2	-10	-14

c)

x	-2	0	1	4
y	4	2	1	-2

d)

x	-1	0	4	5
y	0	2	10	12

View Solutions

a)

$$\text{Rate of change} = \frac{\Delta y}{\Delta x} = \frac{6}{3} = 2$$

Constant rate → linear, slope $m = 2$

b)

$$\text{Rate of change} = \frac{\Delta y}{\Delta x} = \frac{-2}{1} = -2$$

Constant rate → linear, slope $m = -2$

c)

$$\text{Rate of change} = \frac{\Delta y}{\Delta x} = \frac{-2}{2} = -1$$

Constant rate → linear, slope $m = -1$

d)

$$\text{Rate of change} = \frac{\Delta y}{\Delta x} = \frac{2}{1} = 2$$

Constant rate → linear, slope $m = 2$

Rate of change Slope formula Interpretation

Exercise 7.11. *Identify which expression represents a quadratic function.*

a) A: $3t^2 - 3t - 3$ B: $3t^3 - 1t^2 + 3$ C: 2 D: $-2t + 4$

b) A: $2p^3 + 2p^2 + 3$ B: $-p^2 + 3p + 4$ C: 2 D: $-p + 4$

c) A: $3n^3 - 1n^2 + 3$ B: $-2n^2 + 4n - 2$ C: $-n - 1$ D: 9

d) A: $-3x + 1$ B: $-2x^2 + 2x$ C: $2x^3x^2 + 2$ D: 8

View Solutions

a)

A quadratic has highest degree 2.

Answer: $A - f(x) = 3t^2 - 3t - 3$ has degree 2. ✓

b)

A quadratic has highest degree 2.

Answer: $B - f(x) = -p^2 + 3p + 4$ has degree 2. ✓

c)

A quadratic has highest degree 2.

Answer: $B - f(x) = -2n^2 + 4n - 2$ has degree 2. ✓

d)

A quadratic has highest degree 2.

Answer: $B - f(x) = -2x^2 + 2x$ has degree 2. ✓

Quadratic functions Form conversions Vertex form

Exercise 7.12. *Evaluate each quadratic function at the given values.*

a) $f(t) = -2t^2 - 3t - 2$ Find: $f(-2), f(0), f(-1)$

b) $f(n) = 2n^2 - 2n - 4$ Find: $f(0), f(-2), f(3)$

c) $f(n) = -n^2 - n - 3$ Find: $f(2), f(1), f(-1)$

d) $f(m) = -m^2 - 2m - 2$ Find: $f(-2), f(-1), f(0)$

View Solutions

a)

$$f(-2) = -2(-2)^2 + 6 - 2 = -4$$
$$f(0) = -2(0)^2 + 0 - 2 = -2$$
$$f(-1) = -2(-1)^2 + 3 - 2 = -1$$

b)

$$f(0) = 2(0)^2 + 0 - 4 = -4$$
$$f(-2) = 2(-2)^2 + 4 - 4 = 8$$
$$f(3) = 2(3)^2 - 6 - 4 = 8$$

c)

$$f(2) = -1(2)^2 - 2 - 3 = -9$$
$$f(1) = -1(1)^2 - 1 - 3 = -5$$
$$f(-1) = -1(-1)^2 + 1 - 3 = -3$$

d)

$$f(-2) = -1(-2)^2 + 4 - 2 = -2$$
$$f(-1) = -1(-1)^2 + 2 - 2 = -1$$
$$f(0) = -1(0)^2 + 0 - 2 = -2$$

Quadratic functions Form conversions Vertex form

Exercise 7.13. *State the equation of the axis of symmetry.*

a) $f(x) = -x^2 - 4x - 4$

b) $f(x) = 2x^2 - 20x + 55$

c) $f(x) = -x^2 + 8x - 13$

d) $f(x) = x^2 - 2x + 3$

e) $f(x) = -2x^2 - 4x - 4$

f) $f(x) = -x^2 + 2x + 5$

View Solutions

a)

$$x = \frac{-b}{2a} = \frac{-(-4)}{2(-1)} = -2$$

b)

$$x = \frac{-b}{2a} = \frac{-(-20)}{2(2)} = 5$$

c)

$$x = \frac{-b}{2a} = \frac{-(8)}{2(-1)} = 4$$

d)

$$x = \frac{-b}{2a} = \frac{-(-2)}{2(1)} = 1$$

e)

$$x = \frac{-b}{2a} = \frac{-(-4)}{2(-2)} = -1$$

f)

$$x = \frac{-b}{2a} = \frac{-(2)}{2(-1)} = 1$$

Vertex Axis of symmetry Roots

Exercise 7.14. *State whether each graph represents a function. Write Yes or No.*

a)

x y

b)

x y

c)

x y

d)

x y

 View Solutions

 (a) Yes — passes the Vertical Line Test
 (b) No — fails the Vertical Line Test
 (c) No — fails the Vertical Line Test
 (d) Yes — passes the Vertical Line Test

Function definition Domain & range Arrow diagrams

∑ × π ÷ √ ◆ ∞ ± ∫ ≠ Δ

7.3.2 Medium

Exercise 7.15. *Read and interpret the linear function.*

A phone plan costs $30 per month plus $3 per GB of data.

(a) Write an equation for T in terms of g.

(b) Interpret the slope and intercept in context.

(c) Find the value when $g = 5$.

 View Solution

$$\text{(a) } T = 3g + 30$$

$$\text{(b) Slope } = 3: \text{ cost per GB used}$$

$$\text{Intercept } = 30: \text{ monthly base cost}$$

$$\text{(c) When } g = 5: T = 3(5) + 30 = 45 \text{ \$}$$

Linear functions Slope y-intercept

Exercise 7.16. *Interpret the slope in context.*

A plumber charges a call-out fee of $39 plus $2 per hour. The equation is $C = 2h + 39$.

(a) Identify the slope and interpret it in context.

(b) What does the intercept represent?

View Solution

$$\text{(a) Slope } = 2 \text{ (C per hour)}$$

$$\text{cost per hour of work}$$

$$\text{(b) Intercept } = 39: \text{ fixed call-out fee}$$

Rate of change Slope formula Interpretation

Exercise 7.17. *Find intercepts and analyse the graph.*

Consider the linear function $y = -2x + 1$.

x y -4 -3 -2 -1 1 2 3 4 (-4,9) (0,1)

(a) Find the x-intercept and y-intercept.

(b) Use the graph to verify your intercepts.

View Solution

$$y\text{-int: set } x = 0 : y = 1$$

$$x\text{-int: set } y = 0 : 0 = -2x + 1 \Rightarrow x = \frac{1}{2}$$

$$\text{Plot the points: x-int at } (\frac{1}{2}, 0) \text{ and y-int at } (0,\ 1).$$

Graphing Intercepts Parallel & perpendicular

Exercise 7.18. *Write the equation of the line.*

A line passes through $(1,\ -7)$ and $(0,\ -5)$.

x y -4 -3 -2 -1 1 2 3 4 (-4,3) (0,-5)

(a) Calculate the slope of the line.

(b) Write the equation in the form $y = mx + c$.

View Solution

$$m = \frac{-5-(-7)}{0-(1)} = -2$$
$$y-(-7) = -2(x-(1))$$
$$y = -2x-5$$

Graphing Intercepts Parallel & perpendicular

Exercise 7.19. *Analyse the parabola.*

The parabola is given by $f(x) = 2(x-3)^2 - 1$.

x y x=3 (3, -1) 2.3 3.7

(a) State the vertex and axis of symmetry.

(b) Does the parabola open upward or downward? Why?

(c) Write the equation in standard form $f(x) = ax^2 + bx + c$.

View Solution

(a) Vertex: $(3,\ -1)$, axis: $x = 3$

(b) Opens upward since $a = 2$ > 0

(c) Standard form: $f(x) = 2x^2 - 12x + 17$

Quadratic functions Form conversions Vertex form

Exercise 7.20. *Convert from vertex form to standard form.*

a) $f(x) = (x+2)^2 + 6$

b) $f(x) = -2(x-2)^2 - 6$

c) $f(x) = 2(x+1)^2 - 3$

d) $f(x) = 2(x-1)^2 - 2$

View Solutions

a)

$$\text{Expand the bracket:}$$
$$= 1(x^2 + 4x + 4) + 6$$
$$= x^2 + 4x + 10$$

b)

$$\text{Expand the bracket:}$$
$$= -2(x^2 - 4x + 4) - 6$$
$$= -2x^2 + 8x - 14$$

c)

$$\text{Expand the bracket:}$$
$$= 2(x^2 + 2x + 1) - 3$$
$$= 2x^2 + 4x - 1$$

d)

$$\text{Expand the bracket:}$$
$$= 2(x^2 - 2x + 1) - 2$$
$$= 2x^2 - 4x$$

Quadratic functions Form conversions Vertex form

Exercise 7.21. *Expand to standard form.*

a) $f(x) = (x - 1)(x - 5)$

b) $f(x) = -(x - 1)(x - 5)$

c) $f(x) = (x + 1)(x + 4)$

d) $f(x) = 2(x + 5)(x - 4)$

View Solutions

a)

$$\begin{aligned} & \text{Expand} \\ &= 1(x^2 - 6x + 5) \\ &= x^2 - 6x + 5 \end{aligned}$$

b)

$$\begin{aligned} & \text{Expand} \\ &= -1(x^2 - 6x + 5) \\ &= -x^2 + 6x - 5 \end{aligned}$$

c)

$$\begin{aligned} & \text{Expand} \\ &= 1(x^2 + 5x + 4) \\ &= x^2 + 5x + 4 \end{aligned}$$

d)

$$\begin{aligned} & \text{Expand} \\ &= 2(x^2 + x - 20) \\ &= 2x^2 + 2x - 40 \end{aligned}$$

Quadratic functions Form conversions Vertex form

Exercise 7.22. *Find the vertex of each quadratic function.*

a) $f(x) = 3x^2 + 6x + 2$

b) $f(x) = 2x^2 + 12x + 22$

c) $f(x) = x^2 - 2x + 5$

d) $f(x) = -2x^2 + 2$

e) $f(x) = x^2 - 4x + 3$

f) $f(x) = 2x^2 + 2$

View Solutions

a)

$$x_{\text{vertex}} = \frac{-b}{2a} = \frac{-(6)}{2(3)} = -1$$

$$y = 3(-1)^2 - 6 + 2 = -1$$

$$\text{Vertex: } x = -1,\ y = -1$$

b)

$$x_{\text{vertex}} = \frac{-b}{2a} = \frac{-(12)}{2(2)} = -3$$

$$y = 2(-3)^2 - 36 + 22 = 4$$

$$\text{Vertex: } x = -3,\ y = 4$$

c)

$$x_{\text{vertex}} = \frac{-b}{2a} = \frac{-(-2)}{2(1)} = 1$$

$$y = 1(1)^2 - 2 + 5 = 4$$

$$\text{Vertex: } x = 1,\ y = 4$$

d)

$$x_{\text{vertex}} = \frac{-b}{2a} = \frac{-(0)}{2(-2)} = 0$$

$$y = -2(0)^2 + 2 = 2$$

$$\text{Vertex: } x = 0,\ y = 2$$

e)

$$x_{\text{vertex}} = \frac{-b}{2a} = \frac{-(-4)}{2(1)} = 2$$

$$y = 1(2)^2 - 8 + 3 = -1$$

$$\text{Vertex: } x = 2,\ y = -1$$

f)

$$x_{\text{vertex}} = \frac{-b}{2a} = \frac{-(0)}{2(2)} = 0$$

$$y = 2(0)^2 + 2 = 2$$

$$\text{Vertex: } x = 0,\ y = 2$$

Vertex Axis of symmetry Roots

Exercise 7.23. *Read key features from the graph.*

The graph of a quadratic function is shown.

x y x=2 (2, -4) 0 4

(a) State the roots of the function.

(b) State the coordinates of the vertex.

(c) Write the equation of the axis of symmetry.

View Solution

$$\text{Roots: } x = 0 \text{ and } x = 4$$

$$\text{Vertex: } (2,\ -4)$$

$$\text{Axis of symmetry: } x = 2$$

Vertex Axis of symmetry Roots

Exercise 7.24. *State the domain and range of each quadratic function.*

a) $f(x) = 2(x-3)^2 + 2$

b) $f(x) = -(x-4)^2 + 2$

c) $f(x) = -(x+3)^2 - 1$

d) $f(x) = (x+3)^2 - 2$

View Solutions

a)

Domain: all real numbers

$x \in \mathbb{R}$

Parabola opens upward; vertex at $(3,\ 2)$

Range: $y \geq 2$

b)

Domain: all real numbers

$x \in \mathbb{R}$

Parabola opens downward; vertex at $(4,\ 2)$

Range: $y \leq 2$

c)

Domain: all real numbers

$x \in \mathbb{R}$

Parabola opens downward; vertex at $(-3,\ -1)$

Range: $y \leq -1$

d)

Domain: all real numbers

$x \in \mathbb{R}$

Parabola opens upward; vertex at $(-3,\ -2)$

Range: $y \geq -2$

Domain & range Vertex Quadratic functions

Exercise 7.25. *State the domain and range from the graph.*

The graph of a quadratic function is shown.

x y (-2, -1)

(a) State the domain of the function.

(b) State the range of the function.

 View Solution

(a) Domain: $x \in \mathbb{R}$

(b) Vertex at $(-2,\ -1)$; opens upward

Range: $y \geq -1$

Domain & range Vertex Quadratic functions

Exercise 7.26. *Read the graph and find the equation.*

The graph of a quadratic function is shown.

x y

(a) State the roots (x-intercepts) of the function.

(b) State the coordinates of the vertex.

(c) Write the equation of the axis of symmetry.

(d) Write the equation of the quadratic function.

 View Solution

(a) Roots: $x = -1$ and $x = 1$

(b) Vertex: $(0,\ -2)$

(c) Axis of symmetry: $x = 0$

(d) Equation: $2(x+1)(x-1)$ or $2x^2 - 2$

Vertex Axis of symmetry Roots

∑ × π ÷ √ ◆ ∞ ± ∫ ≠ Δ

7.3.3 Challenging

Exercise 7.27. *Find the equation of each line described.*

a) Parallel to $y = 3x + 3$, through $(-2,\ -3)$

b) Parallel to $y = 3x - 4$, through $(3,\ -2)$

c) Parallel to $y = -2x + 4$, through $(-3,\ 2)$

d) Parallel to $y = -x + 1$, through $(-3,\ 1)$

View Solutions

a)

$$\text{Parallel lines have equal slopes: } m = 3$$
$$\text{Through } (-2,\ -3) : -3 = 3(-2) + c$$
$$c = 3$$
$$y = 3x + 3$$

b)

$$\text{Parallel lines have equal slopes: } m = 3$$
$$\text{Through } (3,\ -2) : -2 = 3(3) + c$$
$$c = -11$$
$$y = 3x - 11$$

c)

$$\text{Parallel lines have equal slopes: } m = -2$$
$$\text{Through } (-3,\ 2) : 2 = -2(-3) + c$$
$$c = -4$$
$$y = -2x - 4$$

d)

$$\text{Parallel lines have equal slopes: } m = -1$$
$$\text{Through } (-3,\ 1) : 1 = -1(-3) + c$$
$$c = -2$$
$$y = -x - 2$$

Graphing Intercepts Parallel & perpendicular

Exercise 7.28. *Find the equation of each line described.*

a) Perpendicular to $y = -x - 4$, through $(2,\ -2)$

b) Perpendicular to $y = 2x + 4$, through $(0,\ 0)$

c) Perpendicular to $y = -x + 1$, through $(-1,\ 0)$

d) Perpendicular to $y = 2x - 2$, through $(0,\ 1)$

View Solutions

a)

$$\text{Perpendicular slope: } m_\perp = \frac{-1}{-1} = 1$$
$$\text{Through } (2,\ -2): -2 = 1(2) + c$$
$$c = -4$$
$$y = x - 4$$

b)

$$\text{Perpendicular slope: } m_\perp = \frac{-1}{2} = -\frac{1}{2}$$
$$\text{Through } (0,\ 0): 0 = -\frac{1}{2}(0) + c$$
$$c = 0$$
$$y = -\frac{1}{2}x$$

c)

$$\text{Perpendicular slope: } m_\perp = \frac{-1}{-1} = 1$$
$$\text{Through } (-1,\ 0): 0 = 1(-1) + c$$
$$c = 1$$
$$y = x + 1$$

d)

$$\text{Perpendicular slope: } m_\perp = \frac{-1}{2} = -\frac{1}{2}$$

$$\text{Through } (0,\ 1) : 1 = -\frac{1}{2}(0) + c$$

$$c = 1$$

$$y = -\frac{1}{2}x + 1$$

Graphing Intercepts Parallel & perpendicular

Exercise 7.29. *Form and solve a linear model.*

A phone plan charges $10 per month plus $2 per GB of data.

(a) Write an equation for total monthly bill T after using g GB.

(b) Find the bill for 12 GB usage.

(c) How many GB can be used for $90?

View Solution

(a) $T = 2g + 10$

(b) $T = 2(12) + 10 = 34$ $

(c) $90 = 2g + 10 \Rightarrow g = 40$ GB

Linear modelling Real-world context Interpretation

Exercise 7.30. *Convert from vertex form to factored form.*

a) $f(x) = (x+1)^2 - 1$

b) $f(x) = (x-2)^2 - 4$

c) $f(x) = (x+3)^2 - 4$

d) $f(x) = (x-4)^2 - 4$

View Solutions

a)

$$\text{Set } f(x) = 0:$$
$$(x - (-1))^2 = 1$$
$$x + 1 = \pm 1$$
$$x = -2 \text{ or } x = 0$$
$$\therefore f(x) = (x + 2)(x)$$

b)

$$\text{Set } f(x) = 0:$$
$$(x - (2))^2 = 4$$
$$x - 2 = \pm 2$$
$$x = 0 \text{ or } x = 4$$
$$\therefore f(x) = (x)(x - 4)$$

c)

$$\text{Set } f(x) = 0:$$
$$(x - (-3))^2 = 4$$
$$x + 3 = \pm 2$$
$$x = -5 \text{ or } x = -1$$
$$\therefore f(x) = (x + 5)(x + 1)$$

d)

$$\text{Set } f(x) = 0:$$
$$(x - (4))^2 = 4$$
$$x - 4 = \pm 2$$
$$x = 2 \text{ or } x = 6$$
$$\therefore f(x) = (x - 2)(x - 6)$$

Quadratic functions Form conversions Vertex form

Exercise 7.31. *Convert to vertex form by completing the square.*

a) $f(x) = -x^2 + 2x - 7$

b) $f(x) = x^2 - 4$

c) $f(x) = -x^2 - 6$

d) $f(x) = x^2 - 4x + 3$

View Solutions

a)

$$\begin{aligned} \text{Factor out } -1: \\ = -1\left(x^2 - 2x\right) - 7 \\ = -1\left(x - 1\right)^2 + 1 - 7 \\ = -(x-1)^2 - 6 \end{aligned}$$

b)

$$\begin{aligned} \text{Complete the square:} \\ = (x+0)^2 - 0 - 4 \\ = (x+0)^2 - 4 \\ = (x)^2 - 4 \end{aligned}$$

c)

$$\begin{aligned} \text{Factor out } -1: \\ = -1\left(x^2 + 0x\right) - 6 \\ = -1\left(x + 0\right)^2 + 0 - 6 \\ = -(x)^2 - 6 \end{aligned}$$

d)

$$\begin{aligned} \text{Complete the square:} \\ = (x-2)^2 - 4 + 3 \\ = (x-2)^2 - 1 \\ = (x-2)^2 - 1 \end{aligned}$$

Quadratic functions Form conversions Vertex form

Exercise 7.32. *Analyse the projectile model.*

A ball is thrown upward. Its height (m) after t seconds is $h(t) = -5t^2 + 15t$.

t y x=3/2 (3/2, 11) 0 3

(a) What is the initial height of the ball?

(b) Find the maximum height and when it occurs.

(c) Set up the equation to find when the ball hits the ground.

View Solution

$$\text{(a) } h(0) = 0 \text{ m — initial height}$$

$$\text{(b) Vertex: } t = \frac{-15}{2(-5)} = \frac{3}{2} \text{ s}$$

$$\text{Max height } = 11 \text{ m}$$

$$\text{(c) Set } h = 0 : -5t^2 + 15t = 0$$

Quadratic modelling Projectile motion Revenue

Exercise 7.33. *Analyse the revenue model.*

A company sells items at price p dollars. Demand is $100 - 2p$ units.

(a) Write an equation for revenue R in terms of p.

(b) Find the price that maximises revenue and state the maximum revenue.

View Solution

$$\text{(a) Demand } = 100 - 2p$$

$$R = p(100 - 2p) = 100p - 2p^2$$

$$\text{(b) Vertex: } p = \frac{100}{2 \times 2} = 25$$

$$R_{\text{max}} = 1250$$

Quadratic modelling Projectile motion Revenue

Exercise 7.34. *Find the intersection of the two functions.*

$f(x) = x^2 - 9x + 6$ and $g(x) = -2x - 4$

(a) Set $f(x) = g(x)$ and form a quadratic equation.

(b) Solve to find the x-coordinates of the intersection points.

(c) State the coordinates of the intersection points.

View Solution

$$\textbf{(a)} \quad x^2 - 9x + 6 = -2x - 4 \implies x^2 - 7x + 10 = 0$$

$$\textbf{(b)} \quad (x-2)(x-5) = 0 \implies x = 2 \text{ or } x = 5$$

$$\textbf{(c)} \quad (2,\ -8) \text{ and } (5,\ -14)$$

Vertex Axis of symmetry Roots

∑ × π ÷ √ ◆ ∞ ± ∫ ≠ Δ

7.4 Test Your Knowledge

Part II

Geometry

8 Unit 1: Trigonometry

8.1 Learning Targets

LT1

Label Right-Angled Triangles

Identify and correctly label the hypotenuse, opposite side, and adjacent side relative to a given angle in a right-angled triangle.

Hypotenuse Opposite & adjacent Angle reference

LT2

Recall the Trigonometric Ratios

Define and recall sin θ, cos θ, and tan θ as ratios of sides using SOH-CAH-TOA, and identify the correct ratio for a given problem.

SOH-CAH-TOA Ratio selection Fraction form

LT3

Find a Missing Side Length

Use a trigonometric ratio to set up and solve an equation to find an unknown side length, given one side and one acute angle.

Equation set-up Multiply & divide Calculator use

LT4

Find a Missing Angle

Use inverse trigonometric functions ($\sin^{-1}$, $\cos^{-1}$, $\tan^{-1}$) to find an unknown angle when two sides of a right-angled triangle are known.

Inverse trig Two-side ratio Degree rounding

LT5

Solve Elevation & Depression Problems

Interpret diagrams and written descriptions involving angles of elevation and depression, and apply trigonometry to find distances and heights.

Elevation & depression Diagram reading Real-world modelling

LT6

Solve Multi-step Problems

Combine two or more trigonometric steps, or integrate trigonometry with other geometric knowledge, to solve complex problems.

Multi-step reasoning Combined geometry Exact & rounded answers

8.2 Formula Card

Trigonometry — Right-Angled Triangles

Labelling the Triangle

 Key Rule The labels **opposite**, **adjacent**, and **hypotenuse** depend on which angle **θ** you are working from. The **hypotenuse** is always opposite the right angle — it never changes.

Three sides relative to angle θ

- **Hypotenuse (H)** — longest side, always opposite the right angle
- **Opposite (O)** — the side directly across from angle θ
- **Adjacent (A)** — the side next to θ (not the hypotenuse)

SOH-CAH-TOA

 Memory Trick — **S**ome **O**ld **H**ouses **C**an **A**lways **H**ide **T**heir **O**ld **A**ge

The three ratios

$$\sin\theta = \frac{O}{H} \qquad \cos\theta = \frac{A}{H} \qquad \tan\theta = \frac{O}{A}$$

Choosing the right ratio — identify which two sides are involved (known + unknown), then pick the ratio that contains both of them.

Known & Unknown	Ratio to use
Opp and Hyp	**sin**
Adj and Hyp	**cos**
Opp and Adj	**tan**

Finding a Missing Side

⚠ **Common Mistake** Check whether the unknown is on the **top** or **bottom** before rearranging. $x = H \times \sin\theta$ is very different from $x = \dfrac{O}{\sin\theta}$.

Unknown on top — multiply

$$\sin\theta = \frac{x}{H} \implies x = H \times \sin\theta$$

$$\cos\theta = \frac{x}{H} \implies x = H \times \cos\theta$$

$$\tan\theta = \frac{x}{A} \implies x = A \times \tan\theta$$

Unknown on bottom — divide

$$\sin\theta = \frac{O}{x} \implies x = \frac{O}{\sin\theta}$$

$$\cos\theta = \frac{A}{x} \implies x = \frac{A}{\cos\theta}$$

$$\tan\theta = \frac{O}{x} \implies x = \frac{O}{\tan\theta}$$

Example — find x given hypotenuse = 10 cm, θ = 30°:

$$\sin 30^\circ = \frac{x}{10} \implies x = 10 \times \sin 30^\circ = 5 \text{ cm}$$

Finding a Missing Angle

 Key Idea Use the **inverse** trig function to undo the ratio. On your calculator: press **SHIFT** (or **2nd**) then **sin / cos / tan**.

The three inverse functions

$$\theta = \sin^{-1}\left(\frac{O}{H}\right)$$

$$\theta = \cos^{-1}\left(\frac{A}{H}\right)$$

$$\theta = \tan^{-1}\left(\frac{O}{A}\right)$$

Example — find θ given opposite = 5 cm, hypotenuse = 13 cm:

$$\sin\theta = \frac{5}{13} \implies \theta = \sin^{-1}\left(\frac{5}{13}\right) \approx 22.6^\circ$$

Angles of Elevation & Depression

 Key Fact Both angles are measured from the **horizontal**. They are **equal** (alternate angles) when the ground is level.

Elevation — angle measured *upward* from horizontal

$$\tan\theta = \frac{h}{d} \qquad h = d\tan\theta \qquad d = \frac{h}{\tan\theta}$$

Depression — angle measured *downward* from horizontal

$$\tan\theta = \frac{h}{d} \qquad \text{(same formula — alternate angles)}$$

Angle Sum Reminder

In any right-angled triangle the two acute angles sum to 90°:

$$\alpha + \beta = 90^\circ \implies \beta = 90^\circ - \alpha$$

8.3 Practice Problems

8.3.1 Easy

Exercise 8.1. *Label the hypotenuse, opposite, and adjacent sides relative to the marked angle.*

Triangle UDA has a right angle at U and angle θ at D.

(a) Which side is the hypotenuse?

(b) Which side is opposite to angle θ?

(c) Which side is adjacent to angle θ?

View Solution

a)

$$\text{Hyp} = \text{DA (opposite right angle at U)}$$

b)

$$\text{Opp} = \text{UA (across from } \theta \text{ at D)}$$

c)

$$\text{Adj} = \text{UD (next to } \theta\text{, not hyp)}$$

Hypotenuse identification Opposite side Adjacent side

Exercise 8.2. *Write the trigonometric ratio as a fraction.*

The diagram shows a right-angled triangle.

(a) Write sin $45°$ as a fraction.

View Solution

a)

$$\sin\theta = \frac{\text{Opp}}{\text{Hyp}} = \frac{8}{11.31}$$

SOH-CAH-TOA Fraction form Ratio identification

Exercise 8.3. *State which trigonometric ratio should be used and write the equation.*

In a right triangle, $\theta = 30°$, opposite=13 cm, adjacent=6 cm.

(a) State the correct ratio and write the equation.

View Solution

a)

$$\text{Opp and Adj} \Rightarrow \tan\theta = \frac{13}{6}$$

SOH-CAH-TOA Ratio selection Side identification

Exercise 8.4. *Find the length of the unknown side. Give your answer to 2 decimal places.*

Hypotenuse = 5 cm, θ = 55°.

(a) Find x.

View Solution

a)

$$\sin 55° = \frac{x}{5}$$
$$x = 5 \times \sin 55°$$
$$x = 4.1 \text{ cm}$$

SOH-CAH-TOA Equation set-up Calculator use

Exercise 8.5. *Find the length of the unknown side. Give your answer to 2 decimal places.*

Hypotenuse = 6 cm, θ = 25°.

(a) Find x.

View Solution

a)

$$\cos 25° = \frac{x}{6}$$
$$x = 6 \times \cos 25°$$
$$x = 5.44 \text{ cm}$$

SOH-CAH-TOA Equation set-up Calculator use

Exercise 8.6. *Find the length of the unknown side. Give your answer to 2 decimal places.*

Adjacent = 8 cm, θ = 40°.

(a) Find x.

 View Solution

a)

$$\tan 40^\circ = \frac{x}{8}$$
$$x = 8 \times \tan 40^\circ$$
$$x = 6.71 \text{ cm}$$

SOH-CAH-TOA Equation set-up Calculator use

∑ × π ÷ √ ◆ ∞ ± ∫ ≠ Δ

8.3.2 Medium

Exercise 8.7. *Find the length of the hypotenuse. Give your answer to 2 decimal places.*

Opposite = 14 cm, θ = 65°.

(a) Find x.

 View Solution

a)

$$\sin 65^\circ = \frac{14}{x}$$
$$x = \frac{14}{\sin 65^\circ}$$
$$x = 15.45 \text{ cm}$$

SOH-CAH-TOA Rearranging equations Calculator use

Exercise 8.8. *Find the length of the hypotenuse. Give your answer to 2 decimal places.*

Adjacent = 13 cm, θ = 25°.

(a) Find x.

View Solution

a)

$$\cos 25° = \frac{13}{x}$$
$$x = \frac{13}{\cos 25°}$$
$$x = 14.34 \text{ cm}$$

SOH-CAH-TOA Rearranging equations Calculator use

Exercise 8.9. *Find the length of the unknown side. Give your answer to 2 decimal places.*

Opposite = 15 cm, θ = 40°.

(a) Find x.

View Solution

a)

$$\tan 40° = \frac{15}{x}$$
$$x = \frac{15}{\tan 40°}$$
$$x = 17.88 \text{ cm}$$

SOH-CAH-TOA Rearranging equations Calculator use

Exercise 8.10. *Find the size of angle θ. Give your answer to 1 decimal place.*

Opposite = 9.97 cm, hypotenuse = 11 cm.

(a) Find angle θ.

View Solution

a)

$$\sin\theta = \frac{9.97}{11}$$
$$\theta = \sin^{-1}\left(\frac{9.97}{11}\right)$$
$$\theta = 65.0°$$

Inverse trig functions Two-side ratio Degree rounding

Exercise 8.11. *Find the size of angle θ. Give your answer to 1 decimal place.*

Adjacent = 9.19 cm, hypotenuse = 12 cm.

(a) Find angle θ.

View Solution

a)

$$\cos\theta = \frac{9.19}{12}$$
$$\theta = \cos^{-1}\left(\frac{9.19}{12}\right)$$
$$\theta = 40.0°$$

Inverse trig functions Two-side ratio Degree rounding

Exercise 8.12. *Find the size of angle θ. Give your answer to 1 decimal place.*

Opposite = 5.0 cm, adjacent = 5 cm.

(a) Find angle θ.

View Solution

a)

$$\tan\theta = \frac{5.0}{5}$$
$$\theta = \tan^{-1}\left(\frac{5.0}{5}\right)$$
$$\theta = 45.0°$$

Inverse trig functions Two-side ratio Degree rounding

Exercise 8.13. *Use trigonometry to find the unknown measurement. Give your answer to 2 decimal places.*

A person stands 50 m from a vertical tower. The angle of elevation to the top is 55°.

(a) Find the height of the tower.

View Solution

a)

$$\tan 55^\circ = \frac{h}{50}$$
$$h = 50 \times \tan 55^\circ$$
$$h = 71.41 \text{ m}$$

Elevation diagram tan ratio Real-world modelling

Exercise 8.14. *Use trigonometry to find the unknown measurement. Give your answer to 2 decimal places.*

An observer at the top of a 40 m cliff looks down at a boat with angle of depression 30°.

(a) Find the horizontal distance from the cliff base to the boat.

View Solution

a)

$$\tan 30^\circ = \frac{40}{d}$$
$$d = \frac{40}{\tan 30^\circ}$$
$$d = 69.28 \text{ m}$$

Depression diagram tan ratio Alternate angles

Exercise 8.15. *Show all working. Give your answer to 2 decimal places.*

A flagpole casts a shadow of 5.72 m when the angle of elevation to the sun is 65°.

(a) Find the height of the flagpole.

View Solution

a)

$$\tan 65^\circ = \frac{h}{5.72}$$
$$h = 5.72 \times \tan 65^\circ = 12.27 \text{ m}$$

tan ratio Shadow geometry Real-world modelling

Exercise 8.16. *Show all working. Give your answer to 2 decimal places.*

A hill is inclined at 12° to the horizontal, base at sea level.

(a) If you walk 150 m up the hill, what is your height above sea level?

(b) If you climb to 80 m above sea level, how far have you walked?

View Solution

a)

$$h = 150 \sin 12^\circ = 31.19 \text{ m}$$

b)

$$\sin 12^\circ = \frac{80}{d}$$
$$d = \frac{80}{\sin 12^\circ} = 384.78 \text{ m}$$

sin for height cos for horizontal Hill geometry

Exercise 8.17. *Show all working. Give your answer to 2 decimal places.*

A train climbs 6.4 m for every 150 m of track.

(a) Find the angle of incline.

View Solution

a)

$$\sin\theta = \frac{6.4}{150}$$
$$\theta = \sin^{-1}\left(\frac{6.4}{150}\right) = 2.4°$$

sin⁻¹ Gradient geometry Small angles

Exercise 8.18. *Show all working. Give your answer to 1 decimal place.*

A 65 m building. Point A is 113 m from its base.

(a) Find the angle of elevation from A to the top of the building.

(b) What is the angle of depression from the top to A?

View Solution

a)

$$\tan\theta = \frac{65}{113}$$
$$\theta = \tan^{-1}\left(\frac{65}{113}\right) = 29.9°$$

b)

$$\text{Depression angle} = \text{elevation angle} = 29.9°$$
$$\text{(Alternate interior angles with horizontal)}$$

Elevation angle Depression angle Alternate angles

Exercise 8.19. *Show all working. Give your answer to 2 decimal places.*

The handle is held 85 cm above the ground. The string is 10 m long at 55° to the horizontal.

Given: String = 10 m · Angle = 55° · Handle = 85 cm

(a) Calculate the height of the kite above the ground.

View Solution

a)

$$\text{Vertical} = 10 \sin 55° = 8.19 \text{ m}$$
$$\text{Handle} = 85 \text{ cm} = 0.85 \text{ m}$$
$$\text{Total} = 8.19 + 0.85 = 9.04 \text{ m}$$

sin ratio Height calculation Unit conversion

Exercise 8.20. *Show all working. Give angles to 1 decimal place.*

A ladder leans against a wall. Wall height = 5.6 m, base distance = 0.9 m. Safe angle: 70°–80° to the ground.

(a) Is the ladder safe? Show your working.

(b) Calculate the length of the ladder.

View Solution

a)

$$\tan\theta = \frac{5.6}{0.9}$$
$$\theta = \tan^{-1}\left(\frac{5.6}{0.9}\right) = 80.9°$$
$$80.9° \text{ is NOT safe (outside 70°–80°)}$$

b)

$$L^2 = 5.6^2 + 0.9^2 = 32.17$$
$$L = \sqrt{32.17} = 5.67 \text{ m}$$

tan ratio Angle calculation Safety constraint

Exercise 8.21. *Convert units where needed. Give your answer to 2 decimal places.*

A ramp: horizontal distance 2.0 m, height 20 cm.

(a) Calculate the size of angle y.

View Solution

a)

$$\text{Convert: } 20 \text{ cm} = 0.2 \text{ m}$$
$$\tan y = \frac{0.2}{2.0}$$
$$y = \tan^{-1}\left(\frac{0.2}{2.0}\right) = 5.7°$$

Unit conversion tan⁻¹ Ramp geometry

∑ × π ÷ √ ◆ ∞ ± ∫ ≠ Δ

8.3.3 Challenging

Exercise 8.22. *Draw a diagram, label all measurements, and find the answer. Give your answer to 2 decimal places.*

A tree of height 20 m is observed at an angle of elevation of 40° from the ground.

(a) Draw and label a diagram.

(b) Find the horizontal distance to 2 d.p.

View Solution

a)

$$\text{Vertical = height, horizontal = distance, } \theta \text{ at base}$$

b)

$$\tan 40° = \frac{20}{d}$$
$$d = \frac{20}{\tan 40°}$$
$$d = 23.84 \text{ m}$$

Problem decomposition Diagram construction tan ratio

Exercise 8.23. *Draw a diagram, label all measurements, and find the answer. Give your answer to 2 decimal places.*

A drone hovers 20 m above ground. Angle of depression to a landmark is 30°.

(a) Draw and label a diagram.

(b) Find the horizontal distance to 2 d.p.

View Solution

a)

Horizontal line at top; depression angle = interior angle

b)

$$\text{Depression angle} = 30^\circ \text{ (alternate angles)}$$
$$\tan 30^\circ = \frac{20}{d}$$
$$d = \frac{20}{\tan 30^\circ} = 34.64 \text{ m}$$

Problem decomposition Diagram construction Alternate angles

Exercise 8.24. *Show all working clearly. Give your answer to 2 decimal places.*

Two points A and B on opposite sides of a tower. From A: elevation 30°, distance 15 m. From B: elevation 35°.

(a) Find the height of the tower.

(b) Find the horizontal distance from B to the tower.

View Solution

a)

$$h = 15 \times \tan 30^\circ = 8.66 \text{ m}$$

b)

$$d_B = \frac{8.66}{\tan 35^\circ} = 12.37 \text{ m}$$

Multi-step reasoning Shared side Combined trig

Exercise 8.25. *Show all working clearly. Give your answer to 2 decimal places.*

A ship sails 12 km on bearing 060°.

(a) How far north of port is the ship?

(b) How far east of port is the ship?

View Solution

a)

$$N = 12\cos 60^\circ = 6 \text{ km}$$

b)

$$E = 12\sin 60^\circ = 10.39 \text{ km}$$

Bearing interpretation North reference Multi-step trig

Exercise 8.26. *Show all working clearly. Give your answer to 2 decimal places.*

Two right-angled triangles share side BD.

Given: AB=8 cm · BC=2 cm · ∠BCD=55°

(a) Calculate angle ABD (marked x).

View Solution

a)

$$\triangle BCD: \quad BD = BC \times \tan 55^\circ = 2 \times \tan 55^\circ = 2.86 \text{ cm}$$

$$\triangle ABD: \quad \sin(x) = \frac{2.86}{8}$$

$$x = \sin^{-1}\left(\frac{2.86}{8}\right) = 20.9^\circ$$

Multi-step reasoning Vertex labelling Combined trig

Exercise 8.27. *Show all working clearly. Give your answer to 2 decimal places.*

Right-angled triangles ABD and BCD share side BD.

Given: AB=16 cm · ∠ABD=70° · ∠ABD:∠DBC=5:2

(a) Work out the length of CD.

View Solution

a)

$$\angle DBC = \frac{2}{5} \times 70° = 28.0°$$
$$BD = 16 \sin 70° = 15.04 \text{ cm}$$
$$CD = BD \times \tan 28.0° = 15.04 \times \tan 28.0° = 8 \text{ cm}$$

Ratio splitting Multi-step trig Angle arithmetic

Exercise 8.28. *Show all working. Give your answer to 1 decimal place.*

Isosceles triangle FLG: $FL = LG = 32$ cm, $FG = 20$ cm.

(a) Work out angle LFG.

(b) Work out angle FLG.

View Solution

a)

Drop perpendicular from L to midpoint M of FG

$$FM = \frac{20}{2} = 10.0 \text{ cm}$$
$$\sin(\angle LFG) = \frac{10.0}{32}$$
$$\angle LFG = \sin^{-1}\left(\frac{10.0}{32}\right) = 18.2°$$

b)

$$\angle FLG = 180° - 2 \times 18.2° = 143.6°$$

Isosceles properties Perpendicular bisector sin application

Exercise 8.29. *Show all working. Give bearing to nearest degree.*

A helicopter flies 10 km due east, then 6 km due north.

(a) Find the direct distance.

(b) Calculate the bearing.

View Solution

a)

$$d^2 = 10^2 + 6^2 = 136$$
$$d = \sqrt{136} = 11.66 \text{ km}$$

b)

$$\tan\theta = \frac{10}{6}$$
$$\theta = \tan^{-1}\left(\frac{10}{6}\right) = 59.0°$$
$$\text{Bearing} = 059°$$

Pythagoras $\tan^{-1}$ for bearing Compass directions

Exercise 8.30. *Show all working. Give your answer to 2 decimal places.*

A boat approaches a 120 m cliff with a 25 m lighthouse on top. Angle of elevation to top of lighthouse is 20°.

Given: Cliff = 120 m · Lighthouse = 25 m · Elevation to lighthouse top = 20°

(a) Find the distance from the boat to the base of the cliff.

(b) Find the angle of elevation from the boat to the top of the cliff.

View Solution

a)

$$\tan 20° = \frac{145}{d}$$
$$d = \frac{145}{\tan 20°} = 398.38 \text{ m}$$

b)

$$\tan\alpha = \frac{120}{398.38}$$
$$\alpha = \tan^{-1}\left(\frac{120}{398.38}\right) = 16.8°$$

Multi-step trig Total height Elevation angles

Exercise 8.31. *Show all working. Give your answer to 2 decimal places.*

A chord of 5.3 cm subtends an angle of 34° at the centre.

Given: Chord = 5.3 cm · Angle at centre = 34°

(a) Find the radius of the circle.

View Solution

a)

$$\text{Half-chord} = \frac{5.3}{2} = 2.65 \text{ cm}$$
$$\sin 34^\circ = \frac{2.65}{r}$$
$$r = \frac{2.65}{\sin 34^\circ} = 4.74 \text{ cm}$$

Circle geometry sin ratio Right triangle in circle

Exercise 8.32. *Show all working. Give your answer to 2 decimal places.*

From 200 m away: elevation to flagpole bottom = 34°, to top = 38°.

(a) Find the height of the flagpole.

View Solution

a)

$$h_1 = 200 \tan 34^\circ = 134.9 \text{ m}$$
$$h_2 = 200 \tan 38^\circ = 156.26 \text{ m}$$
$$\text{Flagpole} = 156.26 - 134.9 = 21.36 \text{ m}$$

Multi-step trig Two-angle setup Subtraction

Exercise 8.33. *Show all working. Give your answer to 2 decimal places.*

Depression from top of 18 m cliff: first 3.3°, then 5.5°. How far does the boat move?

(a) How much closer must the boat move?

View Solution

a)

$$d_1 = \frac{18}{\tan 3.3^\circ} = 312.18 \text{ m}$$
$$d_2 = \frac{18}{\tan 5.5^\circ} = 186.94 \text{ m}$$
$$\text{Moved} = 312.18 - 186.94 = 125.24 \text{ m}$$

Two depression angles Distance difference Multi-step trig

Exercise 8.34. *Show all working. Give your answer to 2 decimal places.*

Helicopter at 90 km/h. Takes 15 s to go from overhead to elevation angle 65°.

(a) Find the height of the helicopter.

View Solution

a)

$$\text{Distance} = 90 \times \frac{15}{3600} = 0.38 \text{ km}$$
$$h = 0.38 \times \tan 65^\circ = 0.81 \text{ km}$$
$$h = 810 \text{ m}$$

Speed-distance-time tan ratio Unit conversion

∑ × π ÷ √ ◆ ∞ ± ∫ ≠ Δ

8.4 Test Your Knowledge